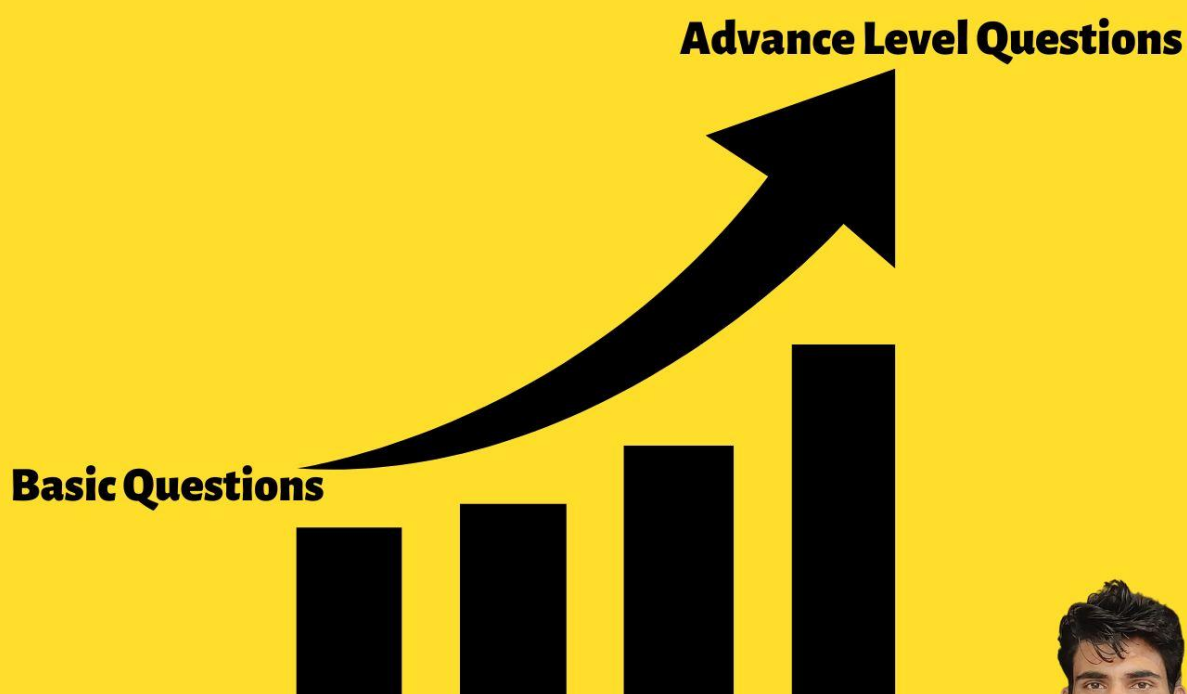


MATHS

GEOMETRY

**ALL TYPE WISE
QUESTIONS**



Maths with Gajanand



Line and angle

Point :- A Zero dimensional figure which has no length, no breadth, no thickness. It's called point.

OR

A Circle which does not have radius is called Point.

Line :- Line is a set of points having only length.

OR

Line is a set of infinite points that should be in one dimension, which can be extended on both the end points.

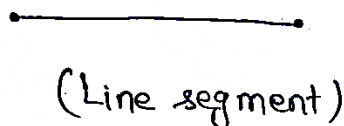
→ Minimum two points is required to draw a line.



Ray :- Ray has a fixed starting point but has not a fixed end Point.

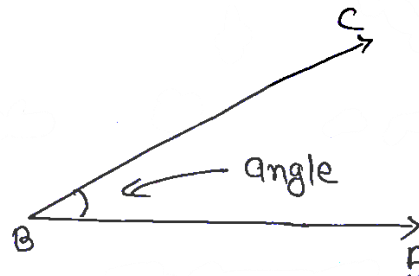


Line Segment :- A line segment is a part of a line with end points at both ends.



Angle :- Two non collinear rays which have one common starting point is called 'Angle'.

→ Rays making an angle are called "arms" and that Common Point is called "vertex".



AB, BC → arms

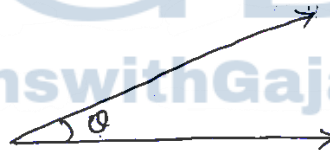
B → vertex.

Types of Angle :-

1. Acute-Angle :-

It is greater than 0 and smaller than 90°

$$0^\circ < \theta < 90^\circ$$



2. Right-Angle :-

The measurement of right angle is 90°



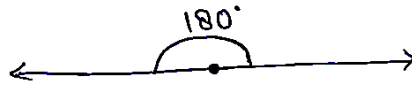
3. Obtuse-Angle :-

It is greater than 90° and smaller than 180°

$$90^\circ < \theta < 180^\circ$$

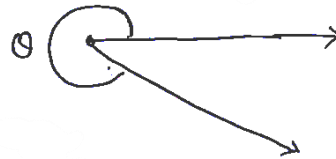


4. Straight Angle :- The measurement of straight angle is 180° .



5. Reflex Angle :- It is greater than 180° and less than 360° .

$$180^\circ < \theta < 360^\circ$$



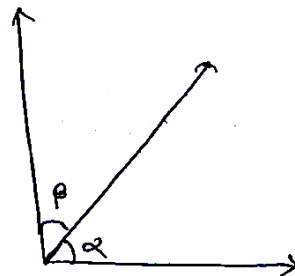
6. Complete - Angle :- The measurement of complete angle is 360° .



Relation between Angles :-

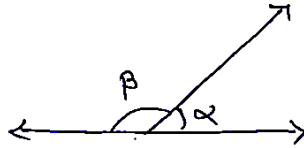
Complementary Angle :- Two angles whose sum is 90° called Complementary angle

$$\alpha + \beta = 90$$

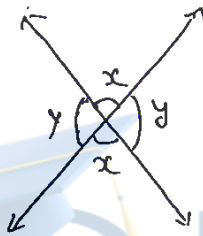


Supplementary Angle :- Two angles whose sum is 180° is called supplementary angle.

$$\alpha + \beta = 180^\circ$$

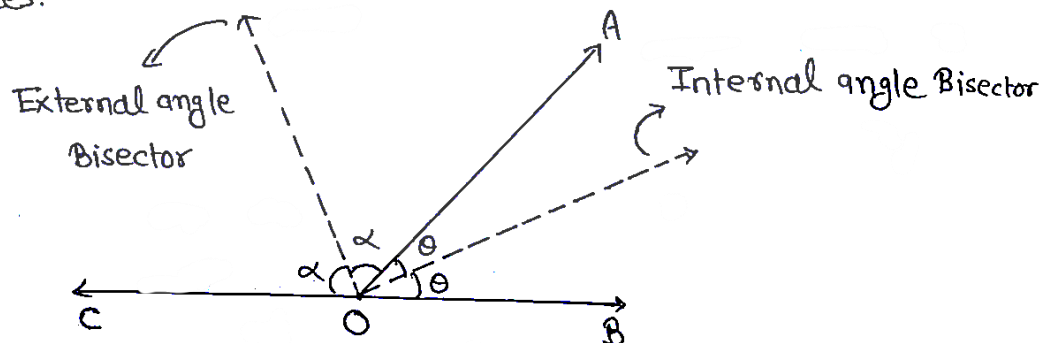


Vertically opposite Angle :-



Angle Bisector :-

A angle bisector divides the angle into two equal measures.

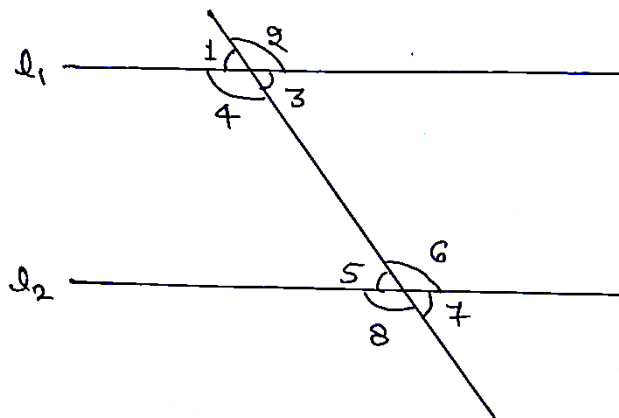


* Angle between External Angle bisector and Internal Angle Bisector is 90°

$$\alpha + \theta = 90^\circ$$

*

$l_1 \parallel l_2$



1. Corresponding Angle

$$\angle 1 = \angle 5$$

$$\angle 2 = \angle 6$$

$$\angle 3 = \angle 7$$

$$\angle 4 = \angle 8$$

Alternate Angle

$$\angle 3 = \angle 5$$

$$\angle 4 = \angle 6$$

Interior Angle

$$\angle 3 + \angle 6 = 180^\circ$$

$$\angle 4 + \angle 5 = 180^\circ$$

Vertically opposite Angle:-

$$\angle 3 = \angle 1$$

$$\angle 2 = \angle 4$$

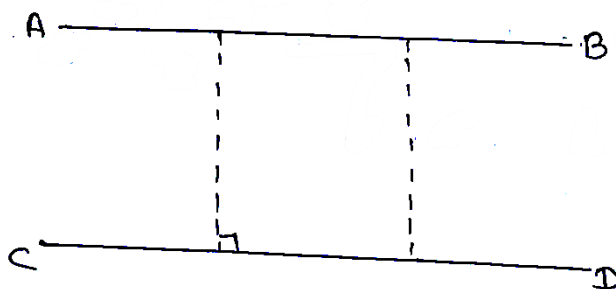
$$\angle 5 = \angle 7$$

$$\angle 6 = \angle 8$$

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* Parallel Lines:-

Parallel lines are two lines that are always the same distance apart and never touch. They do not intersect each other.



$AB \parallel CD$

Que.1. The measure of angle is 14° less than the measure of its complementary angle. then find the value of angle.

Soln:-

$$\theta = (90 - \theta) - 14^\circ$$

$$2\theta = 76$$

$$\theta = 38^\circ \text{ Ans}$$

Que.2. The measure of the supplementary of an angle is 20° more than three times of original angle. find the original value of angle.

Soln:-

$$180^\circ - \theta = 3\theta + 20^\circ$$

$$4\theta = 160^\circ$$

$$\theta = \boxed{40^\circ} \text{ Ans}$$

Que.3 The measure of supplement of LA is 40° larger than twice the measure of the complement of LA. What is the sum in degree of the measure of the supplement and complement of LA ?

Soln:-

$$180^\circ - A = 2(90 - A) + 40^\circ$$

$$180 - A = 180 - 2A + 40^\circ$$

$$A = 40^\circ$$

$$90 - A + 180 - A = 270 - 2A = 270 - 80 = \boxed{190^\circ} \text{ Ans}$$

Que.4:- X and y are two supplementary angles. If supplementary of angle x is equals to half the complementary of y. then find x:y.

$$\text{Soln:- } x + y = 180^\circ \quad \left| \quad y = \frac{1}{2}(90 - y) \right.$$

$$2y = 90 - y$$

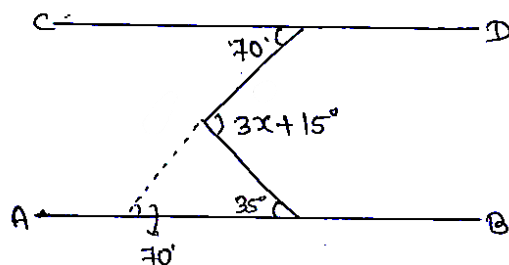
$$x + 30^\circ = 180^\circ \quad \leftarrow \quad y = 30^\circ$$

$$x = 150^\circ$$

$$x:y = 150:30 = \boxed{5:1} \text{ Ans}$$

Ques. 5 In the given figure below $AB \parallel CD$. Find the value of x ?

Soln.



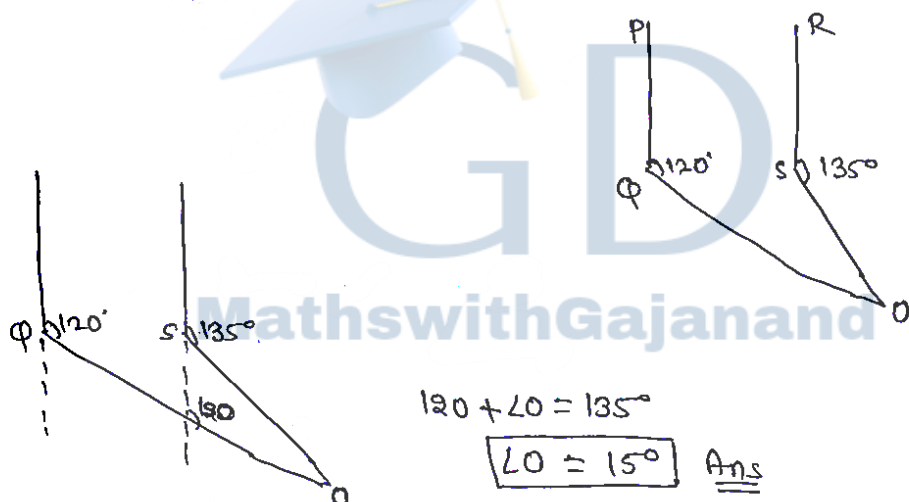
$$3x + 15^\circ = 70^\circ + 35^\circ$$

$$3x = 90$$

$$x = 30^\circ \text{ Ans}$$

Que. 6 :- In the given figure $PQ \parallel RS$, then find $\angle QOS$?

Soln.



$$120 + \angle O = 135^\circ$$

$$\angle O = 15^\circ \text{ Ans}$$

Que. 7 :- In the given figure, if $AC \parallel DE$ and $\angle CAB = 38^\circ$, then the value of $\angle ABC + 5\angle CBD$ is:

{ SSC CHSL-2020 }

Soln. $AC \parallel DE$

$$\text{so } \angle ABE = 2a = 38^\circ$$

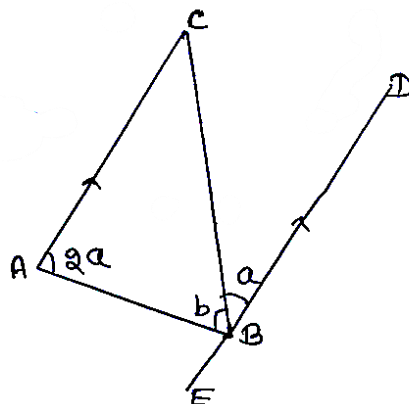
$$a = 19^\circ$$

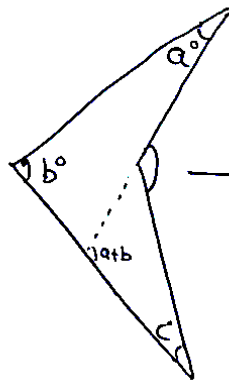
$$\text{2 } 2a + b + a = 180^\circ$$

$$38 + b + 19 = 180^\circ$$

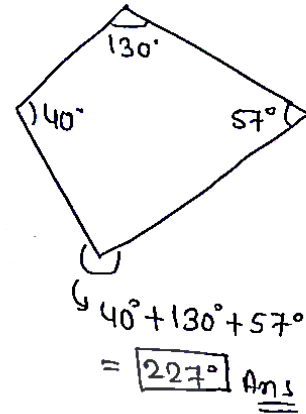
$$b = 123^\circ$$

$$\begin{aligned} \text{Now } \angle ABC + 5\angle CBD &= 123 + 5 \times 19^\circ \\ &= 218^\circ \text{ Ans} \end{aligned}$$





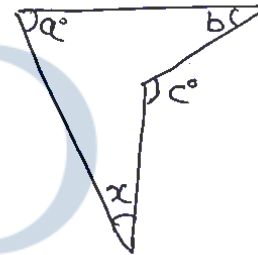
$$a^\circ + b^\circ + c^\circ$$



Que.8:- What is the value of x in the given figure.?

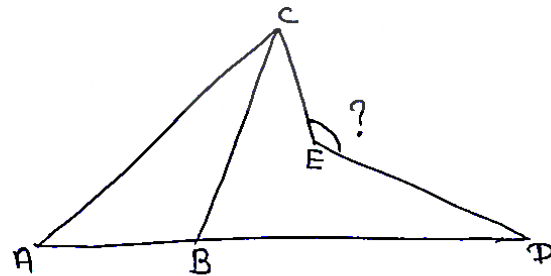
Soln:- $a^\circ + b^\circ + x^\circ = c^\circ$

$$x^\circ = c^\circ - a^\circ - b^\circ \quad \underline{\text{Ans}}$$

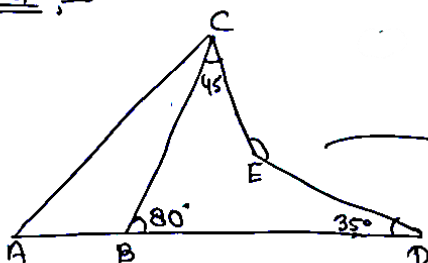


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Que.9:- If in the given figure $\angle ACB + \angle BAC = 80^\circ$, $\angle BDE = 35^\circ$, $\angle BCE = 45^\circ$, then the marked angle $\angle CED$ is.

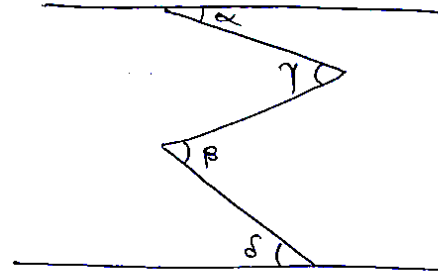
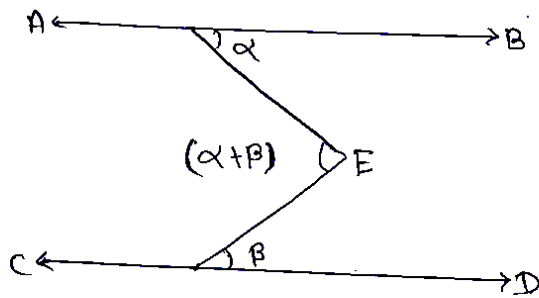


Soln:-



$$80^\circ + 45^\circ + 35^\circ = 160^\circ \quad \underline{\text{Ans}}$$

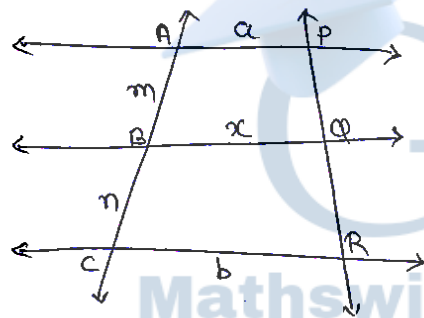
⇒ If AB and CD are parallel lines



$$* \boxed{\alpha + \beta = \gamma + \delta}$$

Three parallel lines and two transversal lines

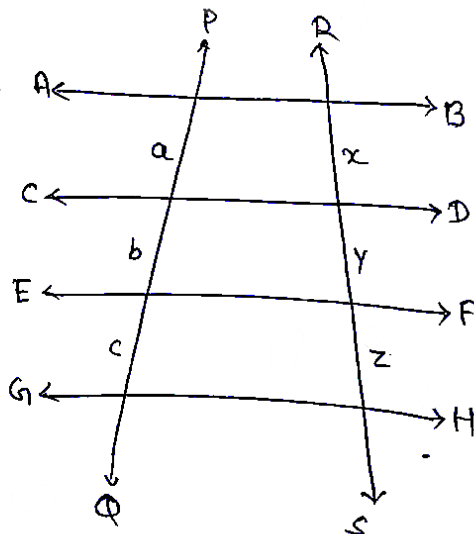
⇒



$$\boxed{\frac{AB}{BC} = \frac{PQ}{QR} = \frac{m}{n}}$$

$$\boxed{BQ = \frac{an + bm}{m+n}}$$

⇒ There are four parallel lines AB, CD, EF & GH and two transversal line PQ and RS.



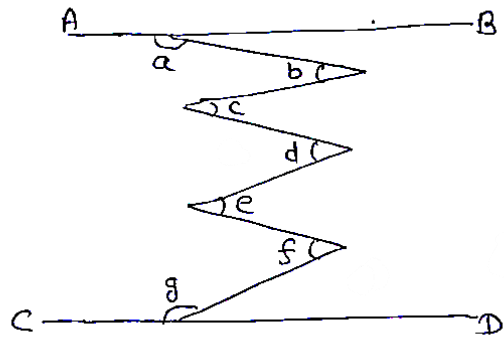
$$* \boxed{a:b:c = x:y:z}$$

$$* \boxed{\frac{a}{b+c} = \frac{x}{y+z}}$$

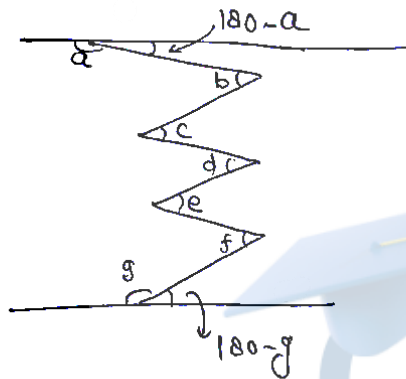
$$* \boxed{\frac{b}{a+b+c} = \frac{y}{x+y+z}}$$

Ques 10:- In the given figure $AB \parallel CD$ then choose correct one -

- (A) $360^\circ + c + e = b + d + f + a + g$
 (B) $a + c + e + g = b + d + f$
 (C) $180^\circ + b + d + f = a + c + e + g$
 (D) NONE



Soln:-



$$180 - a + c + e + 180 - g = b + d + f$$

$$360^\circ + c + e = b + d + f + a + g$$

Ques 11:- In the given fig. below $AB \parallel CD \parallel EF$ If $AB = 29\text{cm}$, $EF = 57\text{cm}$, $AC = \frac{3}{4}CE$, and BD is $x\text{cm}$ less than DF then find $CD = ?$

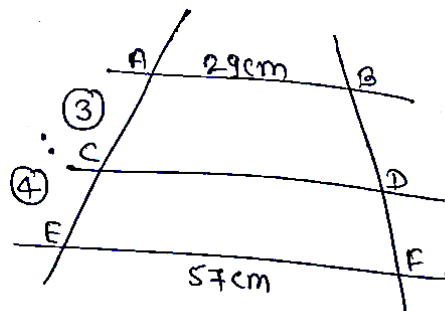
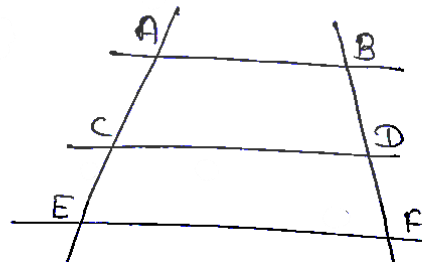
Soln:-

$$CD = \frac{29 \times 4 + 57 \times 3}{3 + 4}$$

$$= \frac{116 + 171}{7}$$

$$= \frac{287}{7}$$

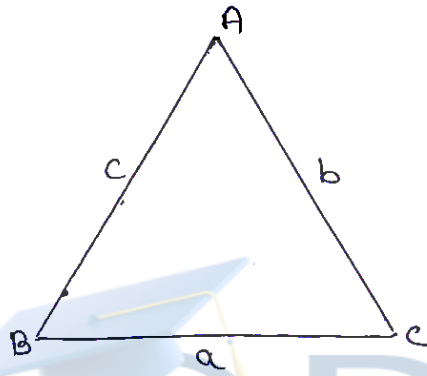
$$= \boxed{41\text{cm}} \text{ Ans}$$



Triangle

Triangle:-

A plane figure bounded with three sides is called "Triangle"

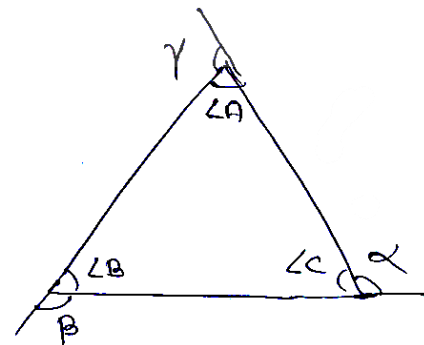


<i> Sum of all angles of a triangle is 180°

$$\angle A + \angle B + \angle C = 180^\circ$$

<ii> [Exterior angle = Sum of Interior opposite Angle]

$$\left\{ \begin{array}{l} \alpha = \angle A + \angle B \\ \beta = \angle A + \angle C \\ \gamma = \angle B + \angle C \end{array} \right\}$$



<iii> Sum of Exterior angle of a triangle is 360°

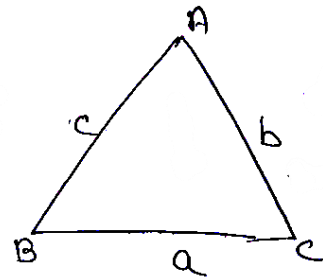
$$\alpha + \beta + \gamma = 360^\circ$$

<iv> Sum of any two sides of triangle is always greater than third side.

$$a+b > c$$

$$b+c > a$$

$$c+a > b$$



Similarly:-

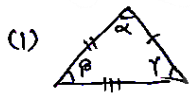
$$* \left\{ \begin{array}{l} |c-a| < b < c+a \\ |b-c| < a < b+c \\ |a-b| < c < a+b \end{array} \right\}$$

<v> Types of Triangle :-

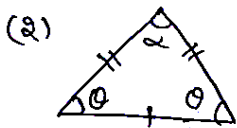


Based on Sides

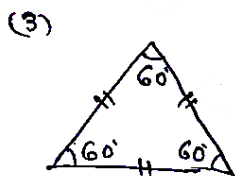
Based on Angle



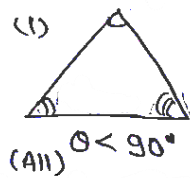
Scalene triangle



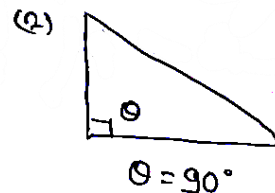
Isosceles triangle



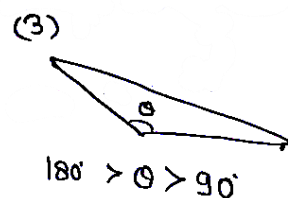
Equilateral triangle



Acute Angle triangle

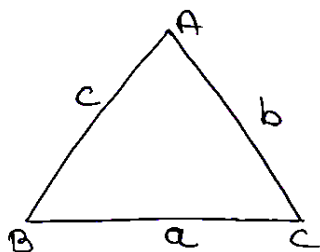


Right Angle triangle



Obtuse Angle triangle.

vi> If sides given And Asked type of triangle:-



(1) $a^2 + b^2 > c^2 \Rightarrow$ Acute Angle triangle

(2) $a^2 + b^2 = c^2 \Rightarrow$ Right Angle triangle

(3) $a^2 + b^2 < c^2 \Rightarrow$ Obtuse Angle triangle

Note:- (Here c is largest side)

$\Rightarrow$ largest Angle $\Rightarrow$ opposite side will be largest

Que.12 In a triangle ABC, if $\angle A = 55^\circ$ and $\angle C = 80^\circ$, then which one is true:-

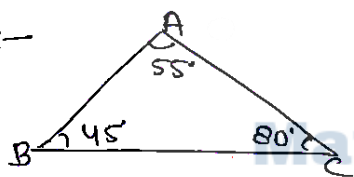
(A) $AB > AC > BC$

(B) $BC > AB > AC$

(C) $CA > AB > BC$

(D) $AB > BC > AC$

Soln:-



As $\angle C > \angle A > \angle B$

then AB > BC > AC Ans

Que.13 If $\triangle ABC$, $\angle A - \angle B = 33^\circ$, $\angle B - \angle C = 18^\circ$. What is the sum of the smallest and the largest angles of a triangle?

Soln:- $\angle A + \angle B + \angle C = 180^\circ$ — (1) (we know)

Given $\angle A - \angle B = 33^\circ \Rightarrow \angle A = 33^\circ + \angle B$ /

$\angle B - \angle C = 18^\circ \Rightarrow \angle B = 18^\circ + \angle C \Rightarrow \angle C = \angle B - 18^\circ$

from (1)

$33^\circ + \angle B + \angle B + \angle B - 18^\circ = 180^\circ$

$3\angle B + 15^\circ = 180^\circ$

$\angle B = 55^\circ$

$\angle A = 88^\circ$

$\angle C = 37^\circ$

largest + smallest

$= 88^\circ + 37^\circ$

$= \boxed{125^\circ}$ Ans

- Q.14** Two sides of a triangle are of length 3cm and 8cm. If the length of the third side is 'x' cm then,
 (a) $5 < x$ (b) $5 < x < 11$ (c) $0 < x < 11$ (d) $x > 11$

Soln:-

$$|a-b| < c < (a+b)$$

$$5 < x < 11 \quad \underline{\text{Ans}}$$

- Q.15** Three sides of a triangle measure 6cm, 10cm and x cm. The minimum integral value of x is:

Soln:-

$a+b > c$ (As we know in triangle sum of two sides is always greater than the 3rd side)

So

$$x = 5$$

check:

$$5+6 > 10 \quad \checkmark$$

$$5+10 > 6 \quad \checkmark$$

- Q.16** Which of the set of three sides can't form a triangle?

- (a) 5cm, 6cm, 7cm (b) 5cm, 8cm, 15cm
 (c) 8cm, 15cm, 18cm (d) 6cm, 7cm, 11cm

Soln:- Sum of two sides $>$ third side

- Q.17** The angles of a triangle are in the ratio 2:3:7. what is the measure of the largest angle of the triangle?

Soln

$$\angle A + \angle B + \angle C = 180^\circ$$

{ SSC CGL-2023 Pre }

$$2+3+7 \Rightarrow 12 \text{ Unit} \rightarrow 180^\circ$$

$$1 \text{ Unit} \rightarrow 15^\circ$$

$$\text{largest Angle} = 7 \text{ Unit} \Rightarrow 7 \times 15^\circ$$

$$\Rightarrow \boxed{105^\circ} \quad \underline{\text{Ans}}$$

Q.18 If the angles of a triangle are $(x-46)$ degrees, $(x+96)^\circ$ and $8x^\circ$, then what is the value of $2x$?

Soln:-

$$\angle A + \angle B + \angle C = 180^\circ$$

$$(x-46) + (x+96) + 8x = 180^\circ$$

$$10x + 50 = 180^\circ$$

$$\boxed{x = 13^\circ}$$

$$2x = 2 \times 13 = \boxed{26^\circ} \text{ Ans}$$

Q.19 Which of the following sets of lengths will give three sides of an obtuse angle triangle?

(a) 15, 62, 64 (b) 17, 64, 66 (c) 16, 63, 65 (d) 18, 65, 67

Soln:-

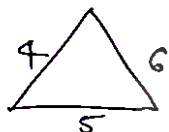
$a^2 + b^2 < c^2$ for obtuse angle triangle

check:- $15^2 + 62^2 < 64^2$

Q.20 The side of triangle are in the ratio 4:5:6. The triangle is:

(a) Acute angle Δ (b) Right Angle Δ (c) obtuse Δ (d) None

Soln:-



$$6^2 < 4^2 + 5^2$$

$$36 < 41$$

$\boxed{\text{Acute Angle } \Delta} \text{ Ans}$

Q.21 If three altitudes of a triangle is in ratio 3:4:5 then triangle is:

Soln:-

$$h_1 : h_2 : h_3 = 3 : 4 : 5$$

$$a : b : c = \frac{1}{3} : \frac{1}{4} : \frac{1}{5}$$

$$\Rightarrow 20 : 15 : 12$$

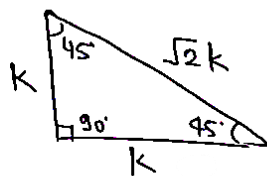
$$20^2 > 15^2 + 12^2$$

$$400 > 369$$

its $\boxed{\text{obtuse Angle } \Delta} \text{ Ans}$

Q.22 In $\triangle ABC$, $AB = BC = k$, $AC = \sqrt{2}k$, then $\triangle ABC$ is a:

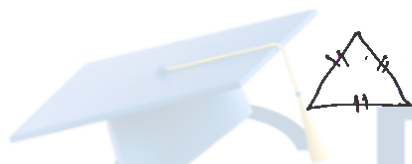
- ☒ (a) Right isosceles $\triangle$
- (b) Isosceles $\triangle$
- (c) Right-angled $\triangle$
- (d) Equilateral $\triangle$



Q.23 Length of the sides of a triangle are a , b and c respectively. If $a^2 + b^2 + c^2 = ab + bc + ca$ then the triangle is:

- (a) isosceles.
- ☒ (b) equilateral
- (c) Scalene
- (d) right-angled

It is possible when
 $\rightarrow a = b = c$



equilateral $\triangle$ Ans

Q.24 The three sides of an acute angled triangle are 15, 17 and x cm. if x is an integer. then how many such triangles are possible?

Soln:-

$$15, 17, x$$

$$2 < x < 32$$

Case I if 17 is largest side

$$17^2 < 15^2 + x^2$$

$$289 < 225 + x^2$$

$$64 < x^2$$

$$8 < x$$

$$x \geq 9 \quad x \leq 9$$

Case II When x is largest

$$15^2 + 17^2 > x^2$$

$$225 + 289 > x^2$$

$$514 > x^2$$

$$x \leq 22$$

total possible value :- $\{9, 10, 11, \dots, 22\}$

$$\Rightarrow (22 - 9) + 1$$

$$= \boxed{14} \text{ Ans}$$

Q.25 In an obtuse angle triangle the length of side opposite to obtuse angle is k cm and length of remaining two sides is 12 cm and 17 cm. How many such triangles are possible?

Soln:- $(17-12) < k < (17+12)$

$$5 < k < 29$$

$$12^2 + 17^2 < k^2$$

$$144 + 289 < k^2$$

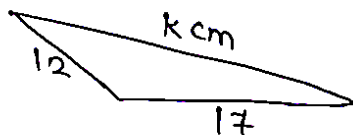
$$433 < k^2$$

$$k^2 > 441 = 49^2$$

$$k > 21$$

$$(21, 22, 23 \dots 28)$$

$\Rightarrow$ 8 possible triangle. Ans



Q.26 The measure of one of the exterior angles of a triangle is twice of the interior opposite angles and the measure of the other interior opposite angles is 60° . The triangle is a:

(a) right angle Δ

(b) scalene triangle

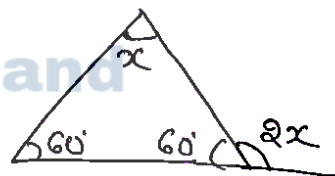
(c) Isosceles Δ

☒ (d) equilateral Δ

$$2x = x + 60^\circ$$

$$x = 60^\circ$$

equilateral Δ Ans



Q.27 The exterior angle of a triangle is 115° and the corresponding interior opposite angles are in the ratio 2:3. The measure of greatest angle of the triangle is:

Soln:- Exterior angle = sum of opposite angle

$$115^\circ = a + b$$

(a:b = 2:3) Given

$$\text{greatest angle} = \frac{115^\circ}{(2+3)} \times 3$$

$$\Rightarrow \boxed{69^\circ} \text{ Ans}$$

- Q 28** The side BC of a triangle ABC is extended to the point D. If $\angle ACD = 132^\circ$ and $\angle B = \frac{4}{7} \angle A$ then the measure of $\angle A$ is equal to?

Soln:-

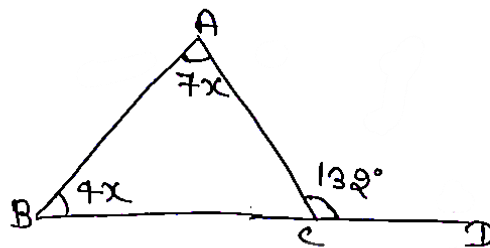
$$7x + 4x = 132^\circ$$

$$11x = 132^\circ$$

$$x = 12^\circ$$

$$\angle A = 7x = 7 \times 12$$

$$= \boxed{84^\circ} \text{ Ans}$$



- Q 29** If in a triangle ABC as drawn in the figure, $AB = AC$ and $\angle ACD = 130^\circ$, then $\angle BAC$ is equal to:

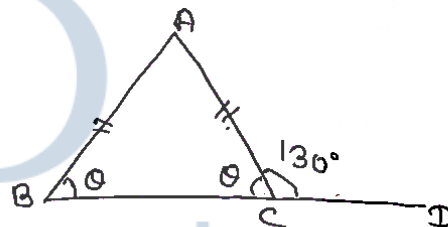
Soln:-

$$\angle A + \theta = 130^\circ$$

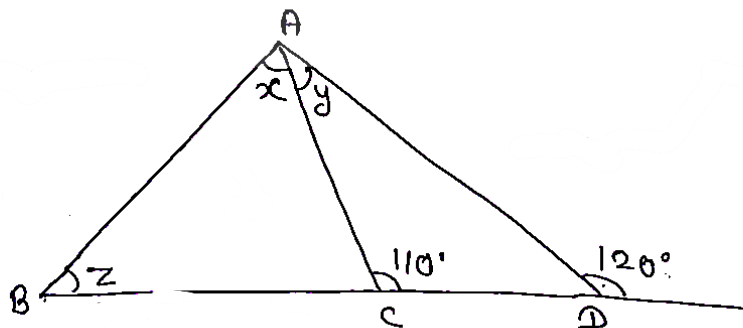
$$\& \quad \theta = 180 - 130^\circ$$

$$= 50^\circ$$

$$\angle A = 130 - 50 = \boxed{80^\circ} \text{ Ans}$$



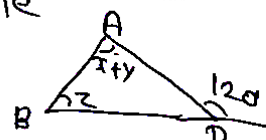
- Q 30** From the following figure find $x + y + z$.



In $\triangle ABD$

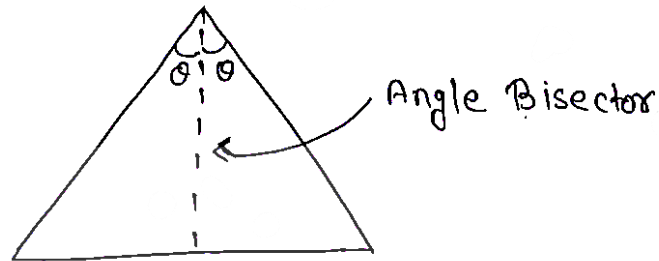
Sum of interior angle = Exterior angle

$$(x+y) + z = \boxed{120^\circ} \text{ Ans}$$



Angle Bisector

⇒ A Line which divides a angle in two equally part, called Angle Bisector.



⇒ Interior and Exterior Angle Bisector:-



SSC CHSL DEO
&
LDC-2014

→ [Angle Between Interior and Exterior A.B = 90°]

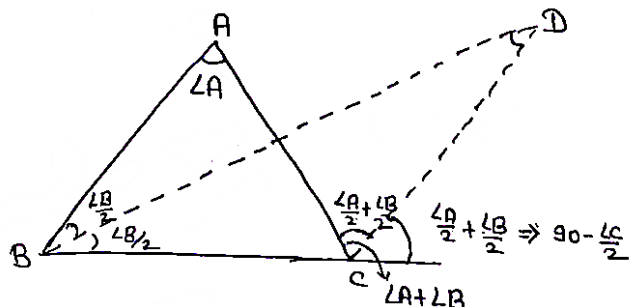
⇒ **Theorem**:- ABC is a Triangle and BD is internal angle bisector of $\angle B$, CD is External angle Bisector of $\angle C$ then

$$* \quad \angle D = \frac{\angle A}{2}$$

Proof:- $\angle A + \angle B + \angle C = 180^\circ$

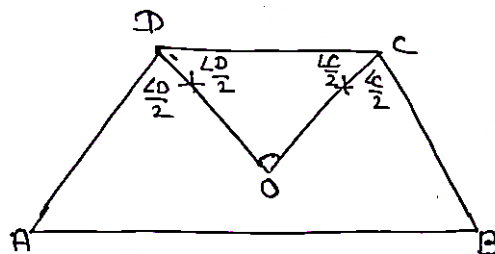
$$\frac{\angle A}{2} + \frac{\angle B}{2} + \frac{\angle C}{2} = 90^\circ$$

$$\angle D = \left(90 - \frac{\angle C}{2}\right) - \frac{\angle B}{2} = \frac{\angle A}{2} \quad \checkmark$$



⇒ Theorem :-

$$* \quad \angle COD = \frac{\angle A + \angle B}{2}$$

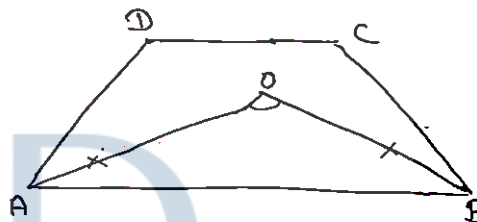


Proof:- $\angle A + \angle B + \angle C + \angle D = 360^\circ$

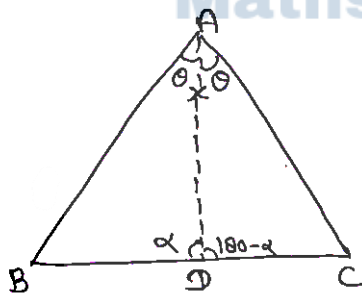
$$\begin{aligned} \angle COD &= 180 - \left(\frac{\angle C + \angle D}{2} \right) \\ &= 180 - \left(180 - \frac{\angle A + \angle B}{2} \right) = \frac{\angle A + \angle B}{2} \end{aligned}$$

Same As:-

$$\angle AOB = \frac{\angle C + \angle D}{2}$$



⇒ Interior Angle Bisector Theorem :- v.v. Imp.



$$* \quad \frac{AB}{AC} = \frac{BD}{DC}$$

Proof:- In $\triangle ABD$,

$$\frac{\sin \theta}{\sin \alpha} = \frac{BD}{AB} \quad (\text{by Sine Rule}) \quad \text{--- ①}$$

In $\triangle ADC$,

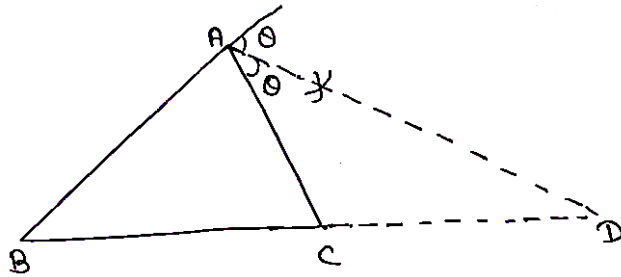
$$\frac{\sin \theta}{\sin \alpha} = \frac{DC}{AC} \quad \text{--- ②}$$

$\sin(180 - \alpha) = \sin \alpha$

by ① & ②

$$\frac{BD}{DC} = \frac{AB}{AC}$$

⇒ **Exterior Angle Bisector Theorem** :-

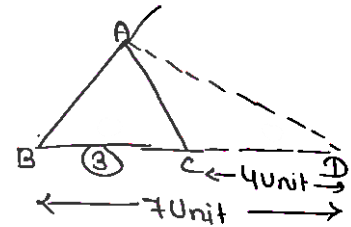


$$* \frac{AB}{AC} = \frac{BD}{CD}$$

Example:- find the value of CD ?

If $AB = 28 \text{ cm}$
 $AC = 16 \text{ cm}$
 $BC = 15 \text{ cm}$

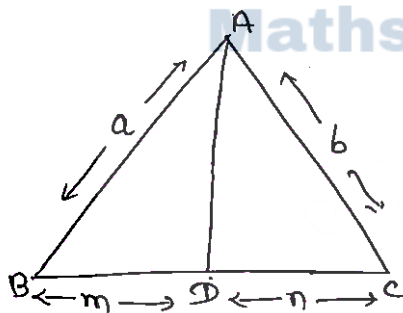
$$AB : AC \\ 28 : 16 \\ \boxed{7 : 4}$$



$$\begin{aligned} 3 \text{ Unit} &\rightarrow 15 \\ 1 \text{ Unit} &\rightarrow 5 \\ 4 \text{ Unit} &\rightarrow 20 \end{aligned}$$

$$\boxed{CD = 20 \text{ cm}} \quad \underline{\text{Ans}}$$

⇒ **Length of Angle Bisector** :-

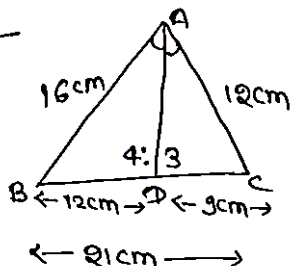


$$* \frac{a}{b} = \frac{m}{n}$$

$$* x^2 = ab - mn$$

Example:- In the given $\triangle ABC$, $AB = 16 \text{ cm}$, $AC = 12 \text{ cm}$ and $BC = 21 \text{ cm}$ then find the length of Angle bisector AD ?

Soln:-

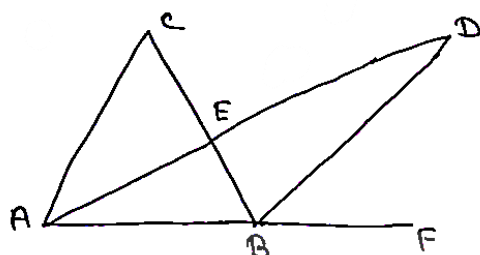


$$\begin{aligned} AD^2 &= ab - mn \\ &= 12 \times 16 - 12 \times 9 \\ &= 12 \times 7 \\ &= 84 \end{aligned}$$

$$AD = \boxed{\sqrt{84} \text{ cm}} \quad \underline{\text{Ans}}$$

Q.31:- In the given figure, AD is bisector of angle $\angle CAB$ and BD is bisector of angle $\angle CBF$. If the angle at C is 34° , the angle $\angle ADB$ is:

Soln:-



$$\begin{aligned}\angle ADB &= \frac{\angle C}{2} \\ &= \frac{34}{2} = \boxed{17^\circ} \text{ Ans.}\end{aligned}$$

Q.32:- In quadrilateral ABCD, $\angle C = 72^\circ$ and $\angle D = 28^\circ$. The bisectors of $\angle A$ and $\angle B$ meet in O. What is the measure of $\angle AOB$

Soln:-
$$\angle AOB = \frac{\angle C + \angle D}{2} = \frac{72 + 28}{2} \Rightarrow \frac{100}{2} = \boxed{50^\circ} \text{ Ans}$$

Q.33 In quadrilateral ABCD, the bisectors of $\angle A$ and $\angle B$ meet at O and $\angle AOB = 64^\circ$. $\angle C + \angle D$ is equal to:-

Soln

$$\begin{aligned}\angle AOB &= \frac{\angle C + \angle D}{2} \\ \angle C + \angle D &= 2\angle AOB = 2 \times 64 = \boxed{128^\circ} \text{ Ans}\end{aligned}$$

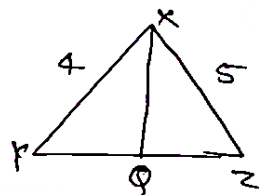
Q.34 In a quadrilateral ABCD, the bisectors of $\angle C$ and $\angle D$ meet at point E. If $\angle CED = 57^\circ$ and $\angle A = 47^\circ$, then the measure of $\angle B$ is:

Soln:-

$$\begin{aligned}\angle CED &= \frac{\angle A + \angle B}{2} \\ 57^\circ &= \frac{47^\circ + \angle B}{2} \\ 114^\circ &= 47^\circ + \angle B \\ \angle B &= \boxed{67^\circ} \text{ Ans}\end{aligned}$$

Q.35 If in ΔXYZ , $XY = 4$ and $XZ = 5$ cm, and Q is a point on YZ such that XQ bisects $\angle X$, then $YQ : QZ$ is:

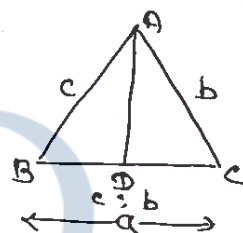
Soln:- $YQ : QZ = \boxed{4 : 5}$ Ans



Q.36 In ΔABC , AD is the bisector of $\angle A$, meet BC at D . If $BC = a$, $AC = b$ and $AB = c$, then what is the value of $BD - DC$?

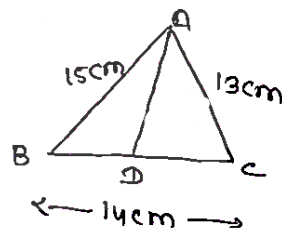
Soln:- (a) $\frac{ac}{b+c}$ (b) $\frac{a(b+c)}{c-b}$ (c) $\frac{a(c-b)}{c+b}$ (d) $\frac{ab}{b+c}$

$(c+b)$ Unit $\rightarrow a$
 1 Unit $\rightarrow \frac{a}{(c+b)}$
 $(c-b)$ Unit $\rightarrow \frac{a(c-b)}{(c+b)}$ Ans



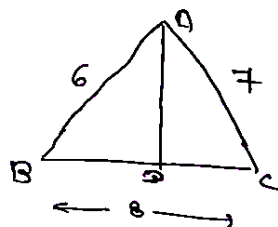
Q.37 The bisector of $\angle A$ in ΔABC meets BC in D . If $AB = 15$ cm, $AC = 13$ cm and $BC = 14$ cm, then $DC = ?$

Soln:- $BD : DC = 15 : 13$
 $DC = \frac{14}{(15+13)} \times 13$
 $= \frac{14}{28} \times 13 \Rightarrow \boxed{6.5 \text{ cm}}$ Ans



Q.38 In ΔABC with sides 6 cm, 7 cm and 8 cm, the angle bisector of the largest angle, divides the opposite side into two segments. What is the length of shorter segments?

Soln:- $BD : DC = 6 : 7$
 Shorter = $\frac{8}{(6+7)} \times 6 \Rightarrow \boxed{\frac{48}{13} \text{ cm}}$ Ans



Q.33 In the following figure, AD bisects angle BAC. Find the length of BD.

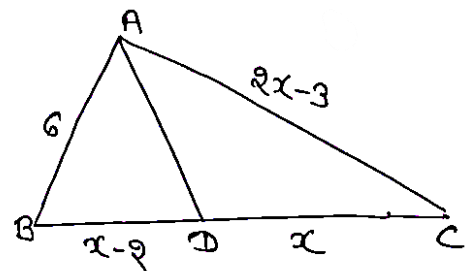
(a) 4

(b) 5

(c) 9

(d) 6

[CGL-2022]



Soln.:- $\frac{AB}{AC} = \frac{BD}{DC}$

$$\Rightarrow \frac{6}{2x-3} = \frac{x-2}{x}$$

$$\Rightarrow 6x = 2x^2 - 4x - 3x + 6$$

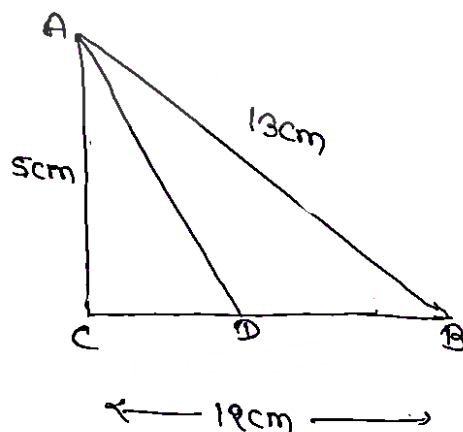
$$\Rightarrow 2x^2 - 13x + 6 = 0$$

Solve $x = 6, \frac{1}{2}$ (not valid)

$$BD = x - 2 \Rightarrow 6 - 2 = \boxed{4\text{cm}} \text{ Ans}$$

Q.40 In $\triangle ABC$, $\angle C = 90^\circ$ and D is a point on CB such that AD is the bisector of $\angle A$. If $AC = 5\text{cm}$ and $BC = 12\text{cm}$, then what is length of AD?

Soln.:-



$$CD : DB = 5 : 12$$

$$CD = \frac{12}{5+12} \times 5$$

$$= \frac{10}{3}\text{cm}$$

In $\triangle ACD$

$$AD = \sqrt{(5)^2 + \left(\frac{10}{3}\right)^2}$$

$$\Rightarrow 5\sqrt{1 + \frac{4}{9}}$$

$$\Rightarrow \boxed{\frac{5\sqrt{13}}{3}\text{cm}} \text{ Ans}$$

Centers of Triangle

- (i) Incenter
- (ii) Circum Center
- (iii) Orthocenter
- (iv) Centroid

(i) Incenter :-

The intersection-point of all three Angle-Bisectors of Triangle is called Incenter

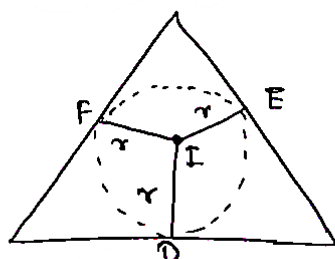


⇒ Incenter always lie inside the triangle.

⇒ Center of Incircle is also called Incenter.

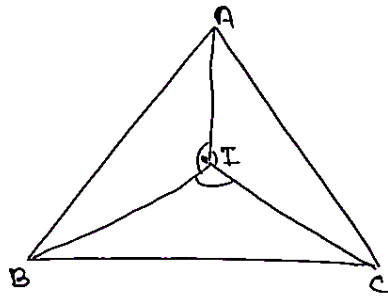


⇒ Incenter is equi-distant from all side of triangle.



$$ID = IE = IF$$

⇒ Angles on Incenter :-



$$* \angle BIC = 90^\circ + \frac{\angle A}{2}$$

$$* \angle AIB = 90^\circ + \frac{\angle C}{2}$$

$$* \angle AIC = 90^\circ + \frac{\angle B}{2}$$

Proof:- $\angle A + \angle B + \angle C = 180^\circ$

$$\frac{\angle A}{2} + \frac{\angle B}{2} + \frac{\angle C}{2} = 90^\circ$$

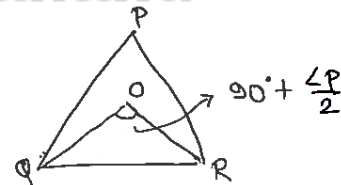
$$\text{In } \triangle BIC \Rightarrow \frac{\angle B}{2} + \frac{\angle C}{2} + \angle BIC = 180^\circ$$

$$\angle BIC = 180^\circ - \left(\frac{\angle B}{2} + \frac{\angle C}{2} \right)$$

$$= 180^\circ - \left(90^\circ - \frac{\angle A}{2} \right) = \boxed{90^\circ + \frac{\angle A}{2}}$$

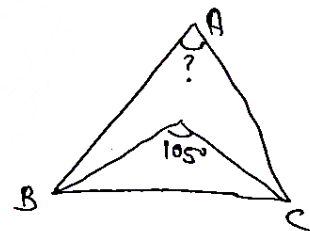
Q41 In $\triangle PQR$, O is the incenter and $\angle P = 42^\circ$, then what is measure of $\angle QOR$?

Soln:- $\angle QOR = 90^\circ + \frac{42}{2}$
 $= 90 + 21$
 $= \boxed{101^\circ} \text{ Ans}$



Q42 In a $\triangle ABC$, the internal bisectors of the angle B and the angle C intersect at an angle of 105° . What is the value of $\angle A$?

Soln:- $105^\circ = 90^\circ + \frac{\angle A}{2}$
 $\angle A = (105 - 90) \times 2$
 $= 15^\circ \times 2$
 $= \boxed{30^\circ} \text{ Ans}$



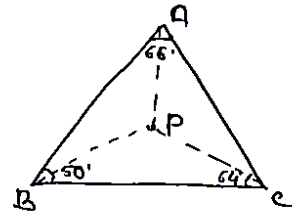
Q43 In $\triangle ABC$, $\angle A = 66^\circ$ and $\angle B = 50^\circ$. If the bisectors of $\angle B$ and $\angle C$ meet at P , then $\angle BPC - \angle PCA = ?$

Soln:- $\angle C = 180^\circ - (50 + 66^\circ)$
 $= 64^\circ$

$\angle PCA = \frac{64^\circ}{2} = \boxed{32^\circ}$

$\angle BPC = 90^\circ + \frac{66^\circ}{2} \Rightarrow \boxed{123^\circ}$

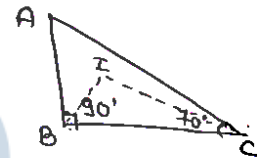
$\angle BPC - \angle PCA = 123^\circ - 32^\circ = \boxed{91^\circ}$ Ans



Q44 Incenter of $\triangle ABC$ is I : $\angle ABC = 90^\circ$ and $\angle ACB = 70^\circ$. $\angle BIC$ is?

Soln:- $\angle BIC = 90^\circ + \frac{\angle A}{2}$
 $\angle A = 90^\circ - 70^\circ = 20^\circ$

$\angle BIC = 90^\circ + 10^\circ = \boxed{100^\circ}$ Ans



Q45 In a triangle ABC , $\angle A = 70^\circ$, $\angle B = 80^\circ$ and D is the incenter of $\triangle ABC$, $\angle ACB = 2x^\circ$ and $\angle BDC = y^\circ$. The values of x and y , respectively are.

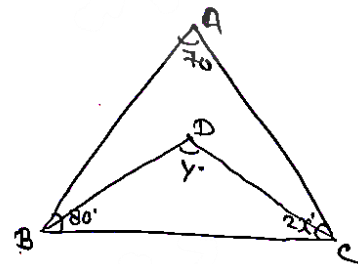
Soln:- $\angle ACB = 180^\circ - (70 + 80)$
 $= 180 - 150 = 30^\circ$

$2x = 30^\circ$

$x = 15^\circ$

$y = 90 + \frac{70}{2} = 125^\circ$

$x, y = 15^\circ, 125^\circ$ Ans



⇒ Inradius :-

$$r = \frac{\Delta}{S}$$

$$\Delta = r \cdot S$$

Here Δ = Area of Triangle

S = Semiperimeter

$$= \frac{a+b+c}{2}$$

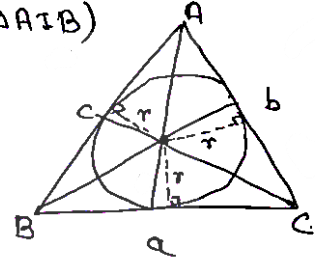
Proof:-

Area of $\Delta ABC = r(\Delta BIC) + r(\Delta AIC) + r(\Delta AIB)$

$$\Delta = \frac{1}{2} \cdot r \cdot a + \frac{1}{2} \cdot b \cdot r + \frac{1}{2} \cdot c \cdot r$$

$$= \left(\frac{a}{2} + \frac{b}{2} + \frac{c}{2} \right) \cdot r$$

$$\Delta = r \cdot S$$



Q.46 The area of triangle is 15 m^2 and radius of its incircle is 3 m . Find its perimeter?

Soln:-

$$\Delta = 15$$

$$r = 3$$

$$S = 5 \text{ m}$$

$$\text{Perimeter} = 2 \times 5$$

$$= \boxed{10 \text{ m}} \text{ Ans}$$

Q.47 What is the radius of incircle of the triangle whose sides are 51 cm , 37 cm , and 20 cm ?

Soln:-

$$S = \frac{51 + 37 + 20}{2} = 54$$

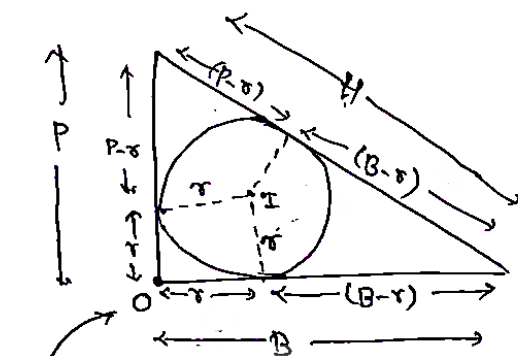
$$\Delta = \sqrt{54 \times 3 \times 17 \times 34}$$

$$= 17 \times 18$$

$$\left[\because \sqrt{s(s-a)(s-b)(s-c)} = \Delta \right]$$

$$r = \frac{\Delta}{S} = \frac{17 \times 18}{54} = \boxed{\frac{17}{3}} \text{ Ans}$$

⇒ Inradius of Right Angle Triangle :-

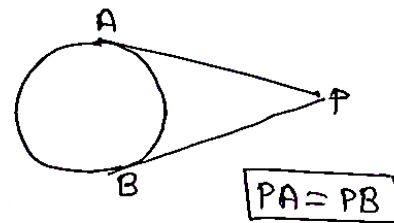


Orthocenter

$$P-r + B-r = H$$

$$P + B - H = 2r$$

$$r = \frac{P+B-H}{2}$$



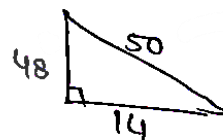
* distance between orthocenter & incentre

$$r\sqrt{2}$$

Q48 ABC is a right angled triangle, right angled at A. A circle is inscribed in it. The lengths of two sides containing the right angles are 48cm and 14cm. The radius of the inscribed circle is:

Soln:- Triplet:- 14, 48, 50

$$\begin{aligned} \text{radius} &= \frac{14+48-50}{2} \\ &= \boxed{6\text{cm}} \text{ Ans} \end{aligned}$$



Q49 find inradius of a triangle whose sides are 27cm, 36cm and 45cm.

$$\begin{aligned} \text{Soln:- } r &= \frac{27+36-45}{2} \\ &= \boxed{9\text{cm}} \text{ Ans} \end{aligned}$$

OR

3	:	4	:	5	⇒	$r = 1$
×9	↓	×9	↓	×9		↓×9
27		36		45		$\boxed{9\text{cm}}$
						<u>Ans</u>

always

⇒ Concept:-

$$\Delta = \frac{1}{2} a \cdot h_1 = \frac{1}{2} b \cdot h_2 = \frac{1}{2} c \cdot h_3$$

$$\frac{\Delta}{h_1} = \frac{a}{2}$$

$$\frac{\Delta}{h_2} = \frac{b}{2}$$

$$\frac{\Delta}{h_3} = \frac{c}{2}$$

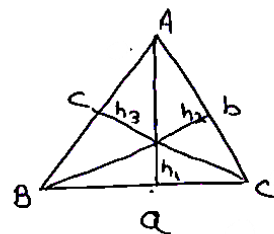
$$\Delta \left(\frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3} \right) = \left(\frac{a}{2} + \frac{b}{2} + \frac{c}{2} \right)$$

$$\Delta \left(\frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3} \right) = s$$

$$\frac{s}{\Delta} = \left(\frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3} \right)$$

$$\left[r = \frac{\Delta}{s} \right]$$

$$* \quad \frac{1}{r} = \frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3}$$



Q 50 If the three altitudes of a triangle are 18cm, 18cm and 20cm then find the inradius of a triangle?

Soln:- $\frac{1}{r} = \frac{1}{12} + \frac{1}{18} + \frac{1}{20} \Rightarrow \frac{15+10+9}{180}$

$$\frac{1}{r} = \frac{17}{90}$$

$$r = \frac{90}{17}$$

Ans

⇒ Inradius of equilateral Triangle :-

$$\text{height} = \frac{\sqrt{3}a}{2}$$

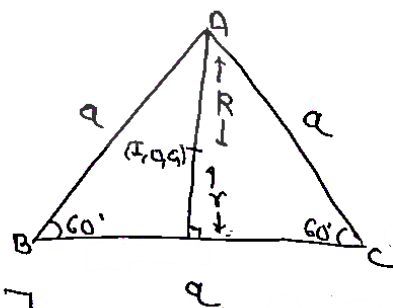
$$\text{Inradius} = \frac{\frac{\sqrt{3}a}{2}}{3} \times 1$$

In Radius

$$r = \frac{a}{2\sqrt{3}}$$

Circum Radius

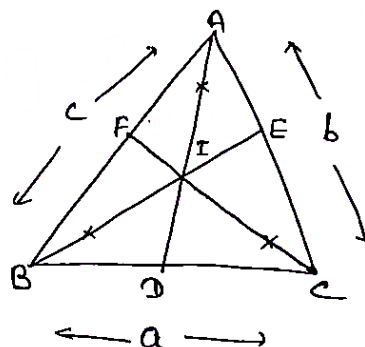
$$R = \frac{a}{\sqrt{3}}$$



$$[\because R:r=2:1]$$

⇒ Concepts -

$$\left\{ \begin{array}{l} \frac{AI}{ID} = \frac{b+c}{a} \\ \frac{BI}{IE} = \frac{c+a}{b} \\ \frac{CI}{IF} = \frac{a+b}{c} \end{array} \right\}$$



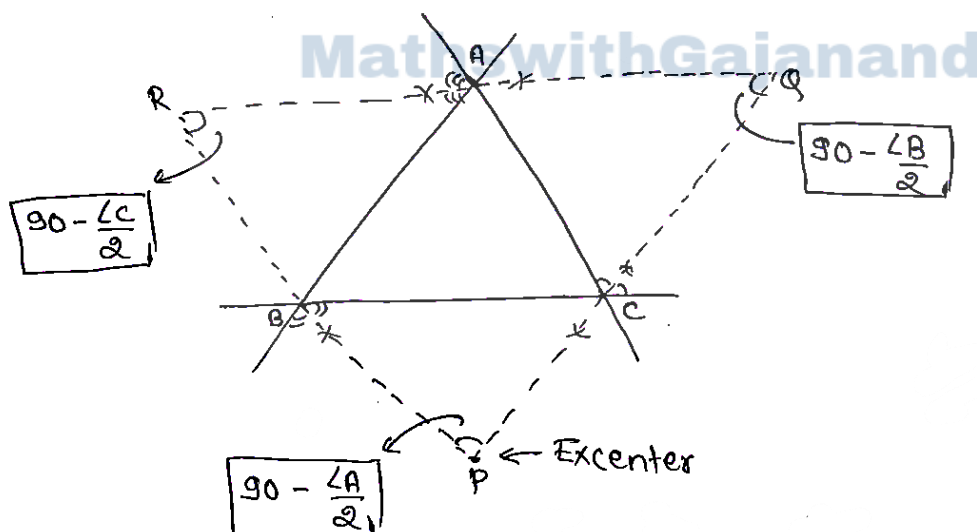
Q51 The perimeter of $\triangle ABC$ is 24cm and its side $BC = 9$ cm, AD is the bisector of $\angle BAC$, while I is the incentre $AI:ID$ is equal to:-

Soln:- $BC = 9$ cm, $AB + AC = 15$ cm

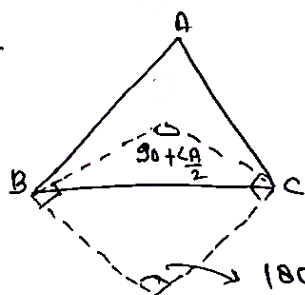
$$\frac{AI}{ID} = \frac{15}{9} = \frac{5}{3}$$

5:3 Ans.

⇒ Angle on Excenter:-



Proof:-

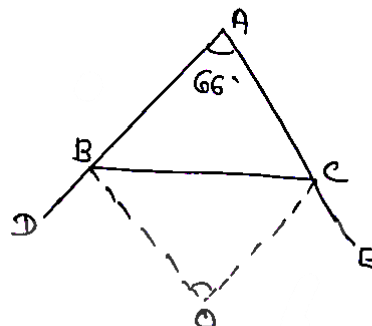


$$180 - \left(90 + \frac{\angle A}{2}\right) \Rightarrow 90 - \frac{\angle A}{2}$$

Q.52 In $\triangle ABC$, $\angle A = 66^\circ$, AB and AC are produced to points D and E , respectively. If the bisectors of $\angle CBD$ and $\angle BCE$ meet at the point O , then $\angle BOC$ is equal to:

Soln:

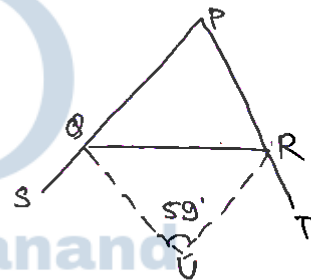
$$\begin{aligned}\angle BOC &= 90 - \frac{\angle A}{2} \\ &= 90 - 33 \\ &= \boxed{57^\circ} \text{ Ans}\end{aligned}$$



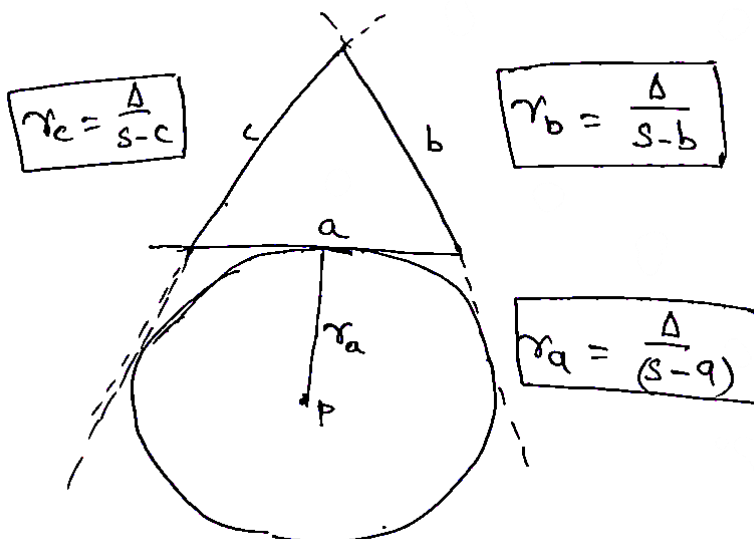
Q.53 The side PQ and PR and $\triangle PQR$ are produced to point S and T respectively. The bisectors of $\angle SQR$ and $\angle TRQ$ meet at U . If $\angle QUR = 59^\circ$, then the measure of $\angle P$ is:

Soln:-

$$\begin{aligned}\angle QUR &= 90 - \frac{\angle P}{2} \\ 59^\circ &= 90 - \frac{\angle P}{2} \\ \frac{\angle P}{2} &= 90 - 59 \\ \angle P &= 31 \times 2 = \boxed{62^\circ} \text{ Ans}\end{aligned}$$



Concept 2 Radius of Excircle :-

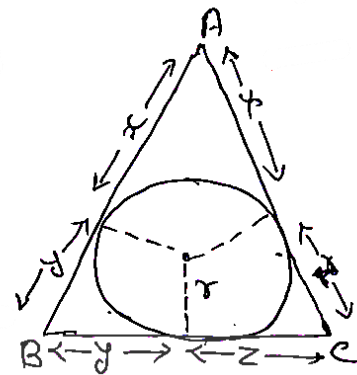


Concept:-

$$\text{Semi perimeter } (s) = x + y + z$$

$$\begin{aligned} \text{area} &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{(x+y+z)xyz} \end{aligned}$$

$$* \quad r = \frac{\sqrt{xyz}}{\sqrt{x+y+z}}$$



Q.54 A circle is inscribed in $\triangle PQR$ touching the sides QR , PR and PQ at the points S , U and T respectively. $PQ = (QR + 5)$ cm, $PQ = (PR + 2)$ cm. If the perimeter of $\triangle PQR$ is 32 cm then PR is equal to:

Soln:-

$$PR = x$$

$$PQ = (PR + 2) = x + 2$$

$$PQ = (QR + 5)$$

$$x + 2 = QR + 5$$

$$QR = x - 3$$

$$\text{Perimeter} = 32$$

$$PQ + QR + PR = 32$$

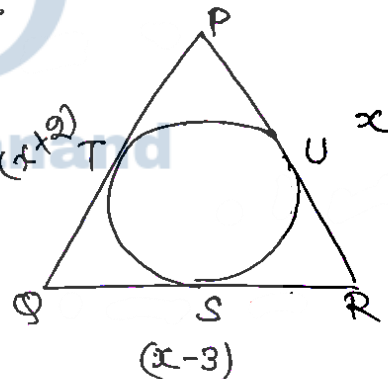
$$(x + 2) + (x - 3) + (x + 2) = 32$$

$$3x - 1 = 32$$

$$3x = 33$$

$$x = 11$$

$$\boxed{PR = 11 \text{ cm}} \quad \text{Ans}$$

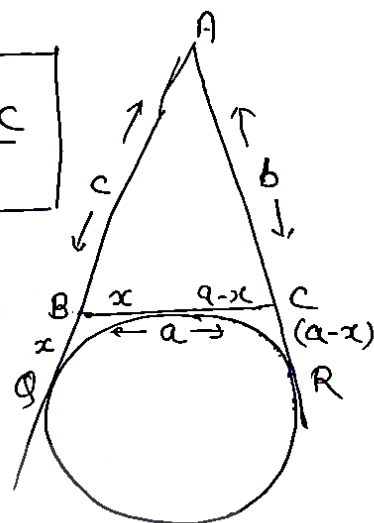


Concept :-

$$* AQ = AR = \frac{\text{Perimeter of } \triangle ABC}{2}$$

Proof :-

$$\begin{aligned} AQ &= AR \\ c+x &= b+a-x \\ 2x &= b+a-c \\ x &= \frac{b+a-c}{2} \end{aligned}$$



$$AQ = c+x$$

$$AQ = c + \frac{b+a-c}{2}$$

$$AQ = \frac{2c+b+a-c}{2}$$

$$AQ = \frac{a+b+c}{2} = \frac{\text{Perimeter}}{2}$$

MathswithGajanand

Example:- In $\triangle ABC$, $AB = 11\text{cm}$, $BC = 12\text{cm}$, $AC = 13\text{cm}$

then $AQ = ?$

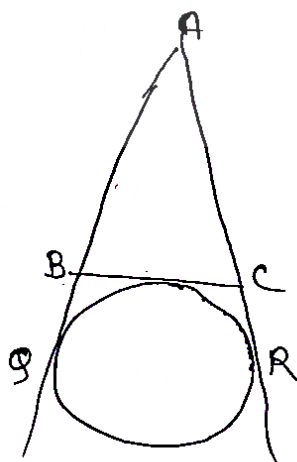
Soln

$$AQ = \frac{AB+BC+CA}{2}$$

$$= \frac{11+12+13}{2}$$

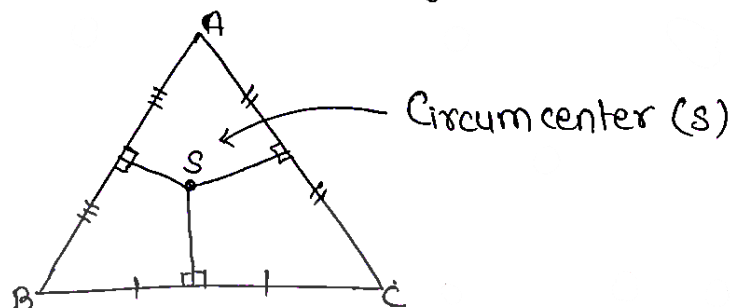
$$= \frac{36}{2}$$

$$AQ = \boxed{18\text{cm}} \text{ Ans}$$



CIRCUMCENTER

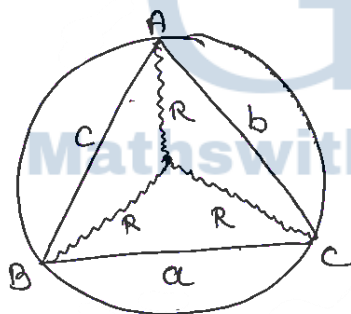
⇒ The intersection point of all three perpendicular bisector of sides of a triangle called 'Circumcenter'



⇒ The center of Circumcircle of a triangle is circumcenter.

⇒ Circumcenter may lie inside, outside or on the triangle.

⇒ Circumcenter is equidistance from all three vertices of Δ



(Circum-radius)

$$* R = \frac{abc}{4\Delta}$$

Δ = Area of Δ

Proof:- $\Delta = \frac{1}{2} \times b \times c \times \sin A$ $\left| \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R \right.$
(Sine Rule)

$$\Delta = \frac{1}{2} bc \times \frac{a}{2R}$$

$$\frac{a}{\sin A} = 2R$$

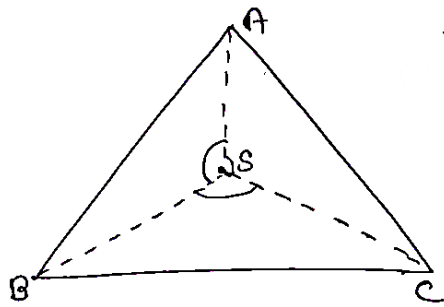
$$\Delta = \frac{abc}{4R}$$

$$\frac{a}{2R} = \sin A$$

$$R = \frac{abc}{4\Delta}$$

R is Circum Radius

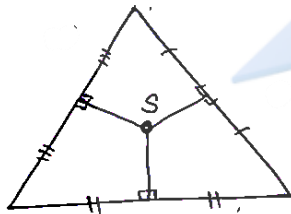
* Angles on Circumcenter :-



$$* \left\{ \begin{array}{l} \angle BSC = 2\angle A \\ \angle ASC = 2\angle B \\ \angle ASB = 2\angle C \end{array} \right\} *$$

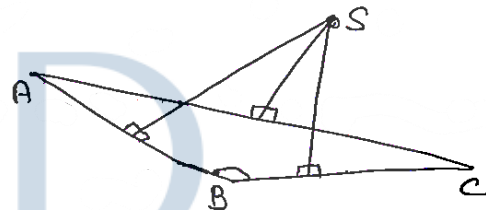
* Position of Circumcenter :-

(i) Acute Angle Δ



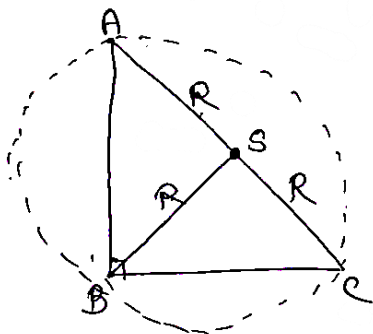
(Inside)

(ii) Obtuse angle Δ



(Outside)

(iii) Right Angle triangle :- Circumcenter is mid-point of Hypotenuse of Right Angle triangle

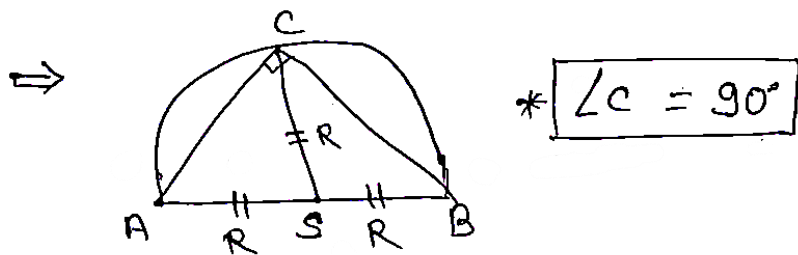


$$AS = SR = BS$$

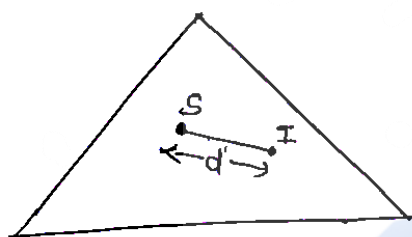
$$* R = \frac{\text{Hypotenuse}}{2}$$

$\Rightarrow$ BS is the median of the triangle (ΔABC)

$$[BS = R = \frac{H}{2}]$$



⇒ Distance between Incenter and circumcenter:-



I = Incenter
O = Circumcenter

* $d = \sqrt{R^2 - 2Rr}$

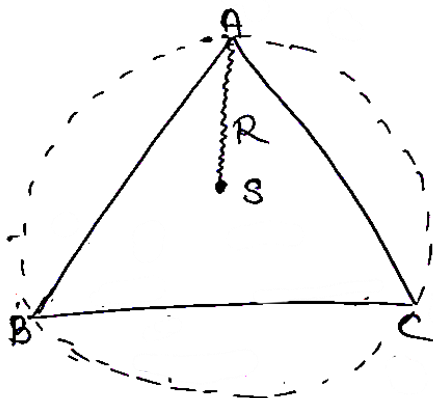
$d \geq 0$

$R^2 - 2Rr \geq 0$

$R \geq 2r, \frac{R}{r} \geq 2$

$\left(\frac{R}{r}\right)_{\min} = 2$ in equilateral triangle

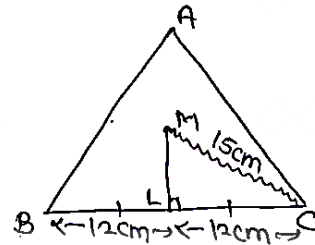
⇒ Equilateral triangle:-



* $R = \frac{\text{Side}}{\sqrt{3}}$

Ques 55 M is the circumcenter of $\triangle ABC$ with circumradius 15cm. Let $BC = 24\text{cm}$ and ML is perpendicular to BC . Then the length of ML is:

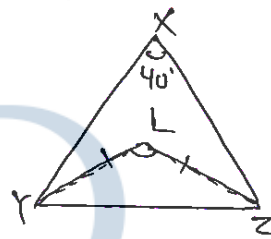
Soln:- $ML = 9\text{cm}$
(Triplet $\rightarrow 9, 12, 15$)



Q.56 If L is the circumcenter of $\triangle XYZ$ and angle X is 40° , then the value of $\angle YZL$ is:

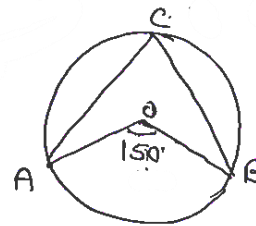
Soln:- $\angle YLZ = 2\angle X$
 $= 2 \times 40^\circ = 80^\circ$

$$\angle YZL = \frac{180 - 80}{2} = \frac{100}{2} = 50^\circ \text{ Ans}$$



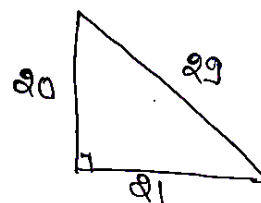
Q.57 In the figure 'G' is the centre of the circle. If $\angle AGB = 150^\circ$, Find the angle ACB

Soln:- $\angle C = \frac{\angle AOB}{2}$
 $= \frac{150^\circ}{2} = 75^\circ \text{ Ans}$



Q.58 The length of the two sides forming the right angle of a right-angled triangle are 21cm and 20cm. What is the radius of the circle circumscribing the triangle?

Soln:- $R = \frac{H}{2}$
 $= \frac{29}{2}$
 $= 14.5\text{cm} \text{ Ans}$



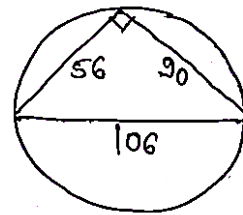
Q.59 The sides of a triangle are 56cm, 90cm. and 106cm.
The circumference of its circumcircle is.

Soln:- $R = \frac{abc}{4\Delta}$

$$R = \frac{56 \times 90 \times 106}{4 \times \frac{1}{2} \times 56 \times 90}$$

$$R = 53$$

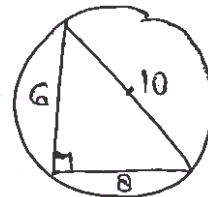
$$\text{Circumference} = 2\pi R = 2 \times \pi \times 53 = \boxed{106\pi} \text{ Ans}$$



Q.60 What is the area (in cm²) of the circumcircle of a triangle whose sides are 6cm, 8cm and 10cm respectively?

Soln:- $R = \frac{H}{2} = \frac{10}{2} = 5\text{cm}$

$$\begin{aligned} \text{Area} &= \pi r^2 = \frac{22}{7} \times 5 \times 5 \\ &= \boxed{\frac{550}{7} \text{ cm}^2} \text{ Ans} \end{aligned}$$



Q.61 O is the circumcenter of the isosceles $\triangle ABC$. Given that $AB = AC = 17\text{cm}$ and $BC = 6\text{cm}$. The radius of the circle is:

Soln:- $a = b = 17\text{cm}$, $c = 6\text{cm}$

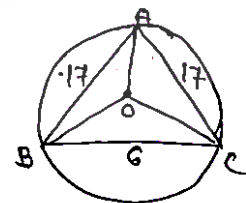
$$R = \frac{abc}{4\Delta} = \frac{17 \times 17 \times 6}{4 \times \sqrt{s(s-a)(s-b)(s-c)}}$$

$$s = \frac{17+17+6}{2} = 20\text{cm}$$

$$R = \frac{17 \times 17 \times 6}{4 \sqrt{20(3)(3)(14)}}$$

$$R = \frac{17 \times 17 \times 6}{4 \times 3 \times 2 \sqrt{70}}$$

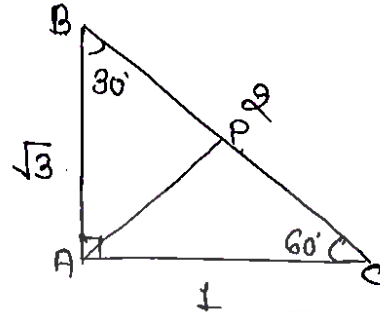
$$R = \boxed{8.636 \text{ cm}} \text{ Ans}$$



Q62 ABC is a right angled triangle. $\angle BAC = 90^\circ$ and $\angle ACB = 60^\circ$. What is the ratio of the circum radius of the triangle to the side AB?

Soln: $R = \frac{2}{2} = 1$

$R : AB = \boxed{1 : \sqrt{3}}$ Ans



Q63 O is the circumcenter of a triangle PQR whose $\angle P = 56^\circ$. If the bisector of $\angle QOR$ and $\angle ORQ$ intersect at M then what is the measure of $\angle OMR$?

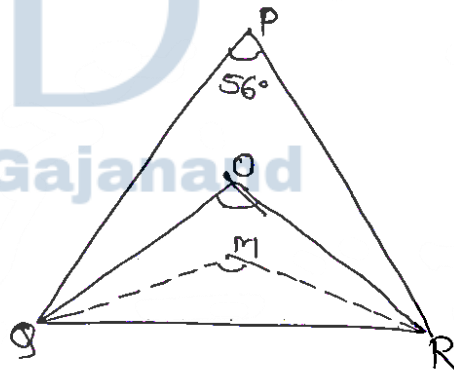
Soln:-

$$\begin{aligned}\angle QOR &= 2 \times \angle P \\ &= 2 \times 56^\circ \\ &= 112^\circ\end{aligned}$$

In $\triangle QOR$

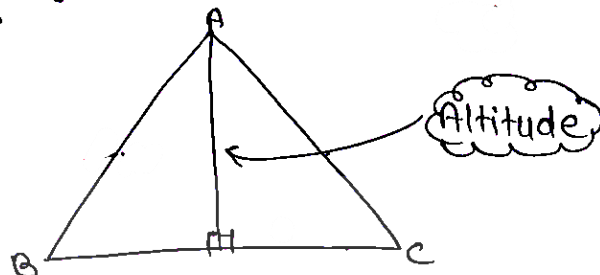
M is Incenter

$$\begin{aligned}\angle OMR &= 90^\circ + \frac{\angle Q}{2} \\ &= 90^\circ + \frac{112^\circ}{2} \\ &= 90^\circ + 56^\circ \\ &= \boxed{146^\circ} \text{ Ans}\end{aligned}$$

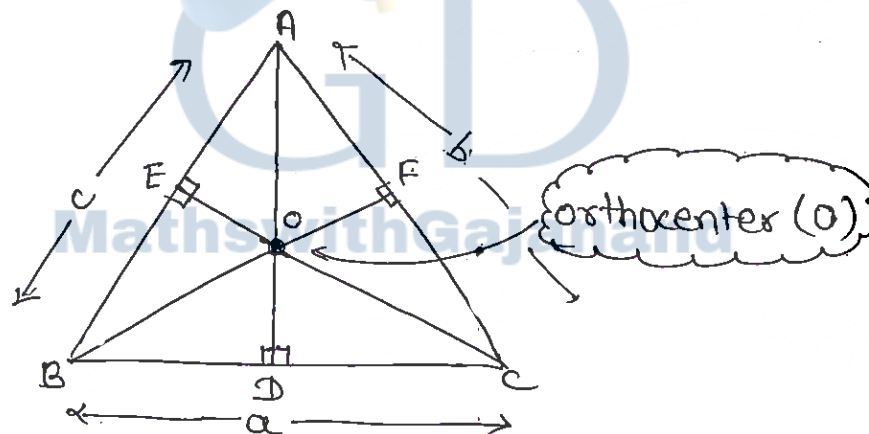


ORTHOCENTER

Altitude of a triangle :- An altitude of a triangle is the perpendicular segment from a vertex of a triangle to the opposite side.



Orthocenter :- orthocenter is intersection of all three altitudes of a triangle.



⇒ Orthocenter may lie outside or on the Δ

$$* \quad \Delta = \frac{1}{2}ah_1 = \frac{1}{2}bh_2 = \frac{1}{2}ch_3$$

$$ah_1 = bh_2 = ch_3 = \text{Constant}$$

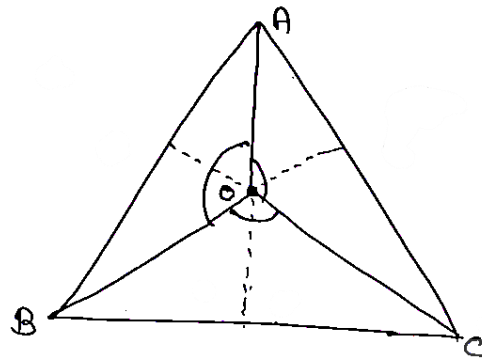
Imp. *

$$h_1 : h_2 : h_3 = \frac{1}{a} : \frac{1}{b} : \frac{1}{c}$$

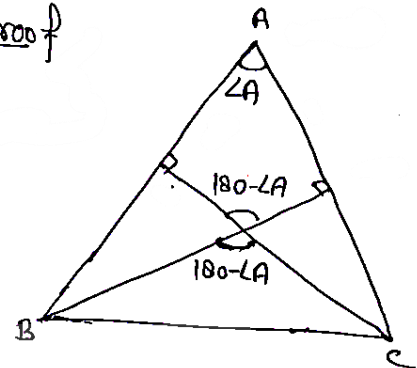
⇒

$$(AB + BC + CA) > (AD + BF + CE)$$

* Angles on orthocenter :-



Proof



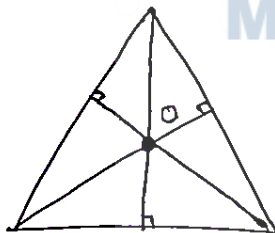
$$\angle BOC = 180^\circ - \angle A$$

$$\angle AOC = 180^\circ - \angle B$$

$$\angle AOB = 180^\circ - \angle C$$

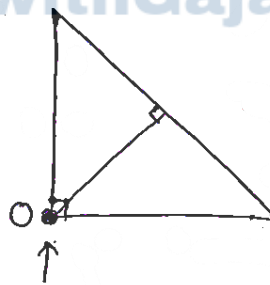
* Position of Orthocenter :-

(i) Acute angle Δ

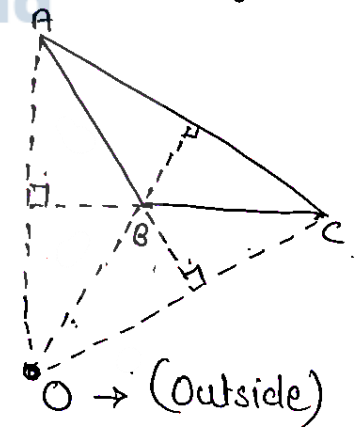


(Inside)

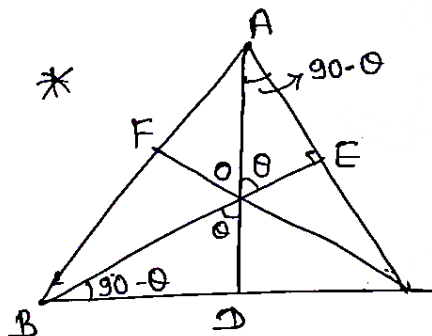
(ii) Right Angle Δ



(iii) Obtuse angle Δ



O $\rightarrow$ (Outside)



(In ΔAOE & ΔBOD)

$$\frac{AO}{BO} = \frac{OE}{OD}$$

$$AO \times OD = BO \times OE$$

$\Rightarrow$

$$AO \times OD = BO \times OE = CO \times OF$$

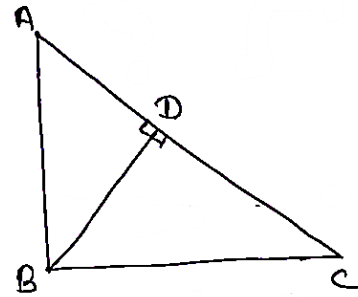
⇒ length of Altitude :-

$$* \quad BD = \frac{AB \times BC}{AC}$$

$$* \quad AD = \frac{AB^2}{AC}$$

$$* \quad CD = \frac{BC^2}{AC}$$

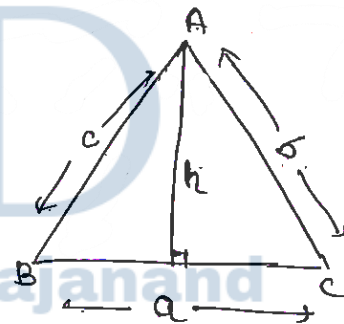
$$* \quad BD^2 = AD \times CD$$



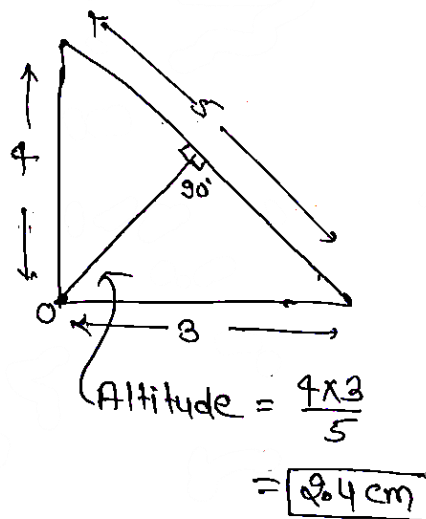
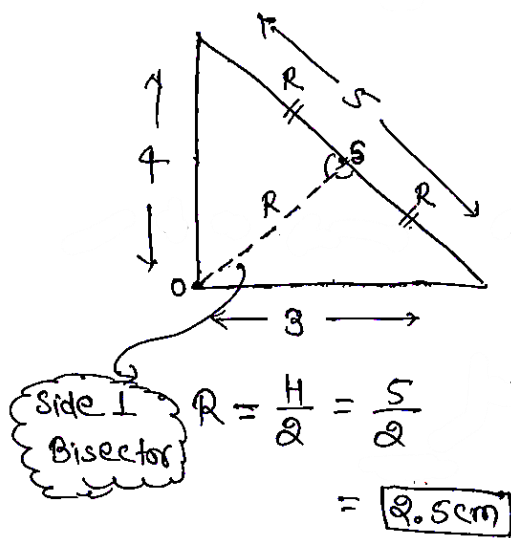
⇒ In Scalene Triangle :-

$$\frac{1}{2} \times a \times h = \sqrt{s(s-a)(s-b)(s-c)}$$

find h (it's altitude)



⇒ Difference between Altitude and side perpendicular Bisector

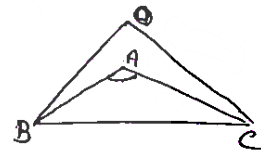


- Q.64** In any triangle if orthocenter lie on its one side then
- orthocenter lies on vertex (✓)
 - circumcenter lies outside the triangle
 - circumcenter also lies on that side.
 - centroid and orthocentre are on same point.

Q.65 In an obtuse angle triangle ABC, $\angle A$ is obtuse and O is orthocentre. If $\angle BOC = 54^\circ$ then find the value of $\angle BAC$?

Soln:-

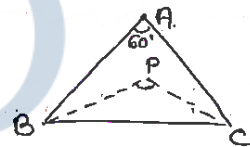
$$\begin{aligned}\angle BAC &= 180^\circ - 54^\circ \\ &= \boxed{126^\circ} \text{ Ans}\end{aligned}$$



Q.66 ABC is an equilateral triangle and P is the orthocentre of the triangle, then what is the value of $\angle BPC$?

Soln:-

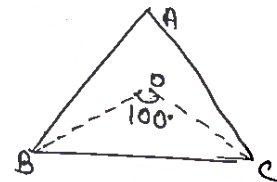
$$\begin{aligned}\angle BPC &= 180^\circ - 60^\circ \\ &= \boxed{120^\circ}\end{aligned}$$



Q.67 If O is the orthocentre of triangle ABC and $\angle BOC = 100^\circ$, the measure of $\angle BAC$ is:

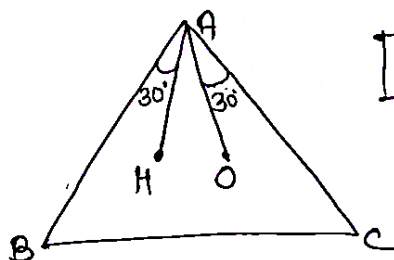
Soln

$$\begin{aligned}\angle BOC &= 180^\circ - \angle A \\ \angle A &= 180^\circ - \angle BOC \\ &= 180^\circ - 100^\circ \\ &= \boxed{80^\circ} \text{ Ans}\end{aligned}$$



Q.68 In $\triangle ABC$, H is orthocentre, O is circumcentre if $\angle BAH = 30^\circ$ then, find $\angle OAC = ?$

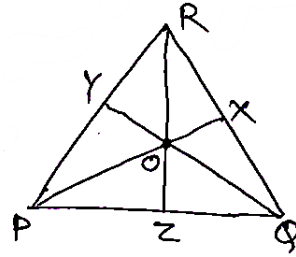
Soln



$$\boxed{\angle BAH = \angle OAC}$$

$$\therefore \angle OAC = \boxed{30^\circ} \text{ Ans}$$

Q.69 In a triangle PQR , PX , QY and RZ be altitudes intersecting at O . If $PO = 6\text{cm}$, $PX = 8\text{cm}$ and $QO = 4\text{cm}$, then what is the value of OY ?



Soln:-

$$PO \times OX = QO \times OY = RO \times OZ$$

$$6 \times 2 = 4 \times OY$$

$$OY = 3$$

$$\text{then } QY = QO + OY = 4 + 3 = \boxed{7\text{cm}}$$

Q.70 Given that the ratio of altitudes of two triangles is $4:5$, ratio of their areas is $3:2$. The ratio of their corresponding bases is:

Soln:- $\frac{\Delta_1}{\Delta_2} = \frac{\frac{1}{2} \times b_1 \times 4}{\frac{1}{2} \times b_2 \times 5} = \frac{3}{2}$

$$\frac{b_1}{b_2} = \frac{15}{8} \quad \boxed{15:8} \quad \underline{\underline{\text{Ans}}}$$

Q.71 Let ABC be an equilateral triangle and AD perpendicular to BC , Then $AB^2 + BC^2 + CA^2 = ?$

Soln $AB^2 = AD^2 + BD^2 \dots (i)$

$+ AC^2 = AD^2 + CD^2 \dots (ii)$

$$AB^2 + AC^2 = 2AD^2 + BD^2 + CD^2$$

$$AB^2 + BC^2 + CA^2 = ?$$

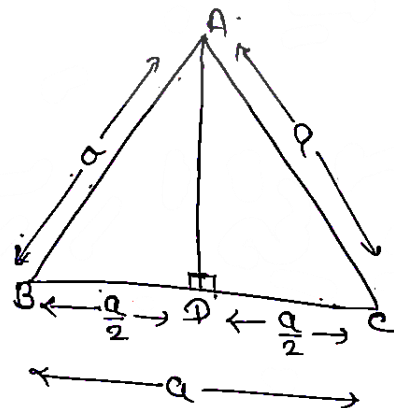
$$\Rightarrow 2AD^2 + BC^2 + BD^2 + CD^2$$

$$\Rightarrow 2AD^2 + a^2 + \frac{a^2}{4} + \frac{a^2}{4}$$

$$\Rightarrow 2AD^2 + a^2 + \frac{a^2}{2}$$

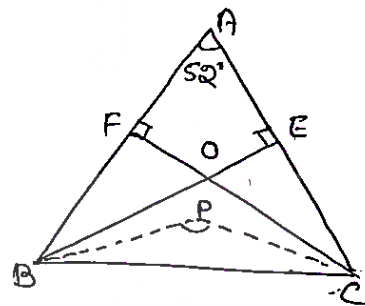
$$\Rightarrow \boxed{4AD^2} \quad \underline{\underline{\text{Ans}}}$$

$$\left\{ a^2 - \frac{a^2}{4} = AD^2 \right\} = \frac{3a^2}{4}$$



Q.72 In $\triangle ABC$, $\angle A = 52^\circ$ and O is the orthocentre of triangle (BO and CO meet AC and AB at E and F respectively when produced). If the bisectors of $\angle OBC$ and $\angle OCB$ meet at P , then the measure of $\angle BPC$ is?

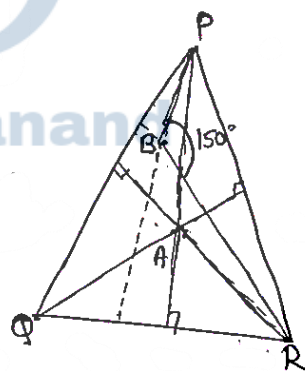
Soln:-
 $\angle BOC = 180^\circ - 52^\circ = 128^\circ$
 $\angle BPC = 90^\circ + \frac{\angle BOC}{2}$
 $= 90^\circ + \frac{128^\circ}{2}$
 $= 90^\circ + 64^\circ = \boxed{154^\circ}$ Ans



Q.73 In triangle PQR , A is the point of intersection of all the altitudes and B is the point of intersection of all the angle bisectors of the triangle. If $\angle PBR = 150^\circ$ then what is the value of $\angle PAR$?

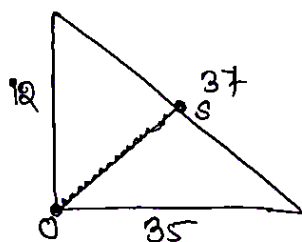
Soln:- $90^\circ + \frac{\angle Q}{2} = 150^\circ$
 $\angle Q = \frac{150^\circ - 90^\circ}{2} = 120^\circ$

A is orthocentre
 $\angle PAR = 180^\circ - \angle Q$
 $= 180^\circ - 120^\circ = \boxed{60^\circ}$ Ans

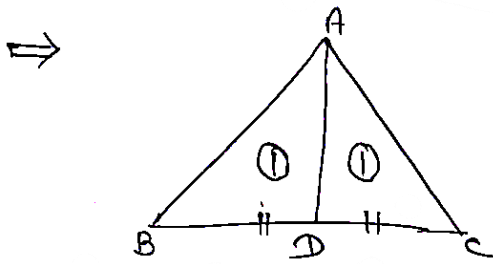


Q.74 What is the distance between the orthocenter and circumcenter of a triangle whose side measure 12cm, 35cm, and 37cm?

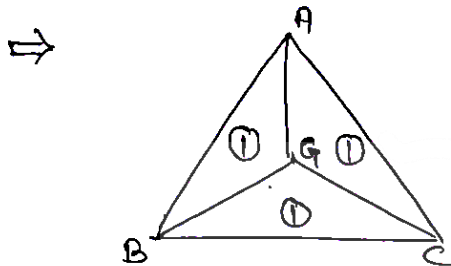
Soln:-



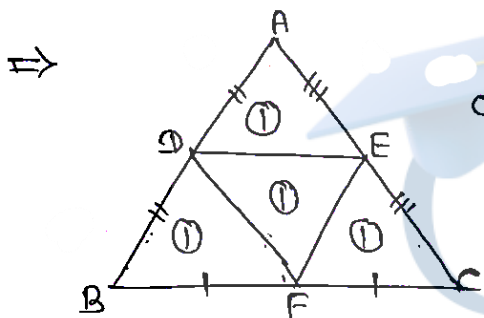
Distance $= \frac{37}{2}$
 $= \boxed{18.5\text{cm}}$ Ans



$$\text{Ar}(\triangle ABD) = \text{Ar}(\triangle ADC) = \frac{1}{2} \text{Ar}(\triangle ABC)$$

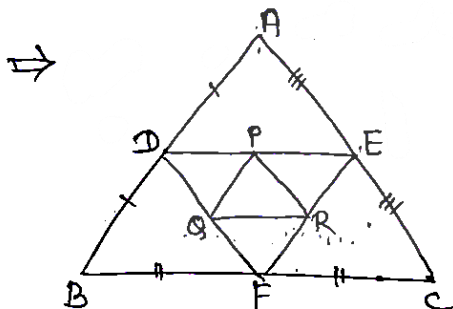


$$\begin{aligned} \text{Ar}(\triangle AGB) &= \text{Ar}(\triangle BGC) = \text{Ar}(\triangle AGC) \\ &= \frac{1}{3} \text{rd of Ar}(\triangle ABC) \end{aligned}$$



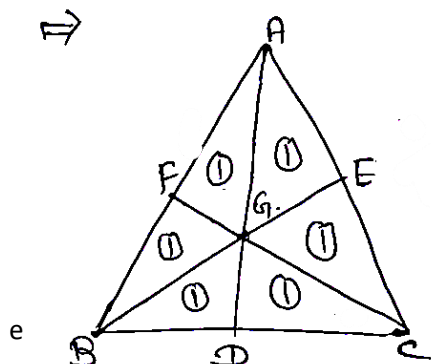
$$\begin{aligned} \text{Ar}(\triangle ADE) &= \text{Ar}(\triangle BDF) = \text{Ar}(\triangle EFC) = \text{Ar}(\triangle DEF) \\ &= \frac{1}{4} \text{ area of } \triangle ABC \end{aligned}$$

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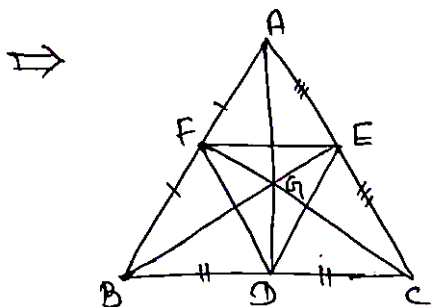


$$* \left[\text{Ar}(\triangle PQR) = \frac{1}{4} \text{ of area } \triangle DEF \right]$$

$$* \left[\text{Ar}(\triangle PQR) = \frac{1}{16} \text{ of area } \triangle ABC \right]$$

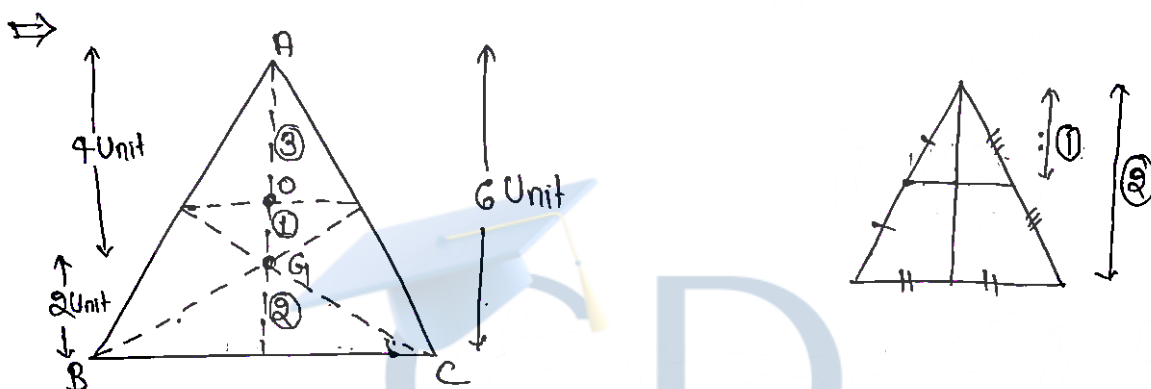


$$\begin{aligned} \text{Ar}(\triangle AFG) &= \text{Ar}(\triangle AEG) = \text{Ar}(\triangle EGC) \\ &= \text{Ar}(\triangle CGD) = \text{Ar}(\triangle BGD) = \text{Ar}(\triangle BGF) \\ &= \frac{1}{6} \text{th of area } \triangle ABC \end{aligned}$$

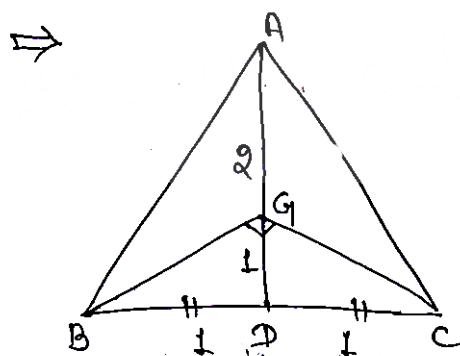


$$\frac{\text{area of } \triangle EGF}{\text{area of } \triangle ABC} = \frac{1}{12}$$

$$\Rightarrow \text{area } \triangle EGF = \frac{1}{12} \text{ of area } \triangle ABC$$



$$* AG : GD = 2 : 1$$



$$\text{If } AG = BG \text{ OR } AD = 1.5 BC \text{ then } \angle BGC = 90^\circ$$

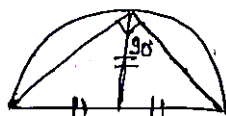
Proof

$$AG = \frac{2}{3} AD$$

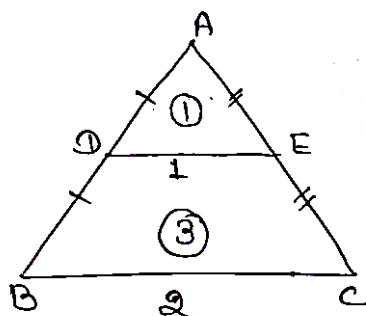
$$\frac{2}{3} AD = \frac{2}{3} BC$$

$$AD = \frac{3}{2} BC$$

$$AD = 1.5 BC$$



⇒



$$* DE = \frac{BC}{2}$$

OR

$$* BC = 2DE$$

$$* DE : BC = 1 : 2$$

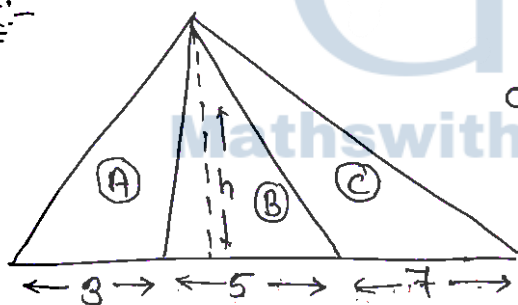
$$* \frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle ABC)} = \frac{1}{4}$$

$$* \frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle DEC)} = \frac{1}{3}$$

⇒

Ratio of Area = Ratio of Base/side

Ex:-

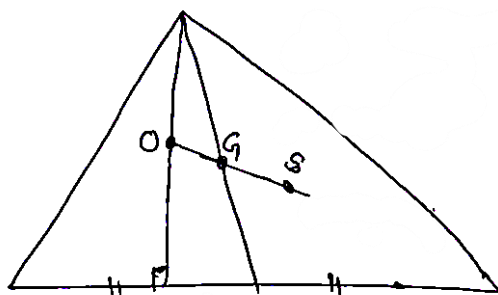


$$\text{area : } (A) : (B) : (C) \\ 3 : 5 : 7$$

$$\text{Proof } \frac{1}{2} \times 3 \times h = \frac{1}{2} \times 5 \times h = \frac{1}{2} \times 7 \times h$$

$$3 : 5 : 7$$

⇒



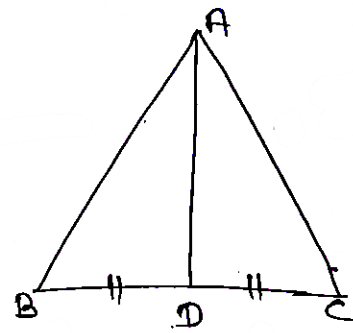
O = orthocentre
G = Centroid
S = Circumcentre

(O, G, S will be linear)

$$* OG : GS = 2 : 1$$

Apollonius theorem :-
(For length of median)

$$\begin{aligned} AB^2 + AC^2 &= 2(AD^2 + BD^2) \\ AB^2 + AC^2 &= 2\left(AD^2 + \frac{BC^2}{4}\right) \end{aligned}$$



Proof :- By Cosine Rule :-

$$\begin{aligned} AB^2 &= AD^2 + BD^2 - 2AD \cdot BD \cos D \\ AC^2 &= AD^2 + CD^2 - 2AD \cdot CD \cos D \end{aligned} \quad \left\{ \because BD = CD \right\}$$

Add both equation

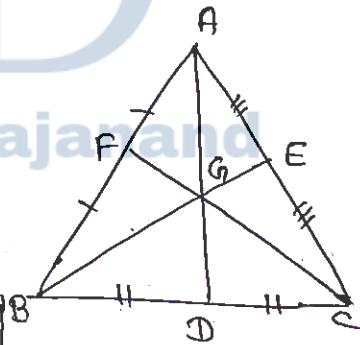
$$[AB^2 + AC^2 = 2(AD^2 + BD^2)]$$

$\Rightarrow$ Important concept :-

$$* \frac{AB^2 + BC^2 + CA^2}{AD^2 + BE^2 + CF^2} = \frac{4}{3}$$

OR

$$* 3(AB^2 + BC^2 + CA^2) = 4(AD^2 + BE^2 + CF^2)$$



Proof :- by Apollonius theorem :-

$$AB^2 + AC^2 = 2\left(AD^2 + \frac{BC^2}{4}\right)$$

$$AB^2 + BC^2 = 2\left(BE^2 + \frac{AC^2}{4}\right)$$

$$\oplus \quad AC^2 + CA^2 = 2\left(CF^2 + \frac{AB^2}{4}\right)$$

$$2(AB^2 + BC^2 + CA^2) = 2(AD^2 + BE^2 + CF^2) + \frac{1}{2}(AB^2 + BC^2 + CA^2)$$

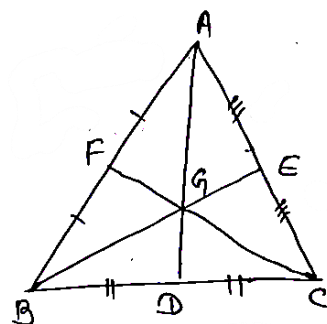
$$\Rightarrow [3(AB^2 + BC^2 + CA^2) = 4(AD^2 + BE^2 + CF^2)]$$

⇒ Important Concept:-

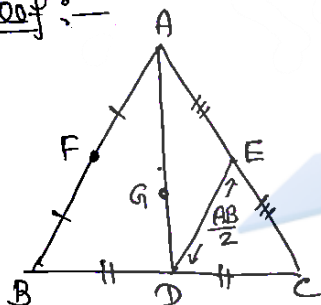
$$\bullet \quad \boxed{AB + BC + CA > AD + BE + CF}$$

$$\bullet \quad \boxed{3(AB + BC + CA) < 4(AD + BE + CF)}$$

$$\bullet \quad \boxed{1 < \frac{AB + BC + CA}{AD + BE + CF} < \frac{4}{3}}$$



Proof:-



Sum of two sides of a triangle is always greater than third side.

In $\triangle ADE$

$$\frac{AE}{2} + \frac{AD}{2} > DE$$

$$AE + AD > 2DE$$

Similarly $AB + BC > 2BE$

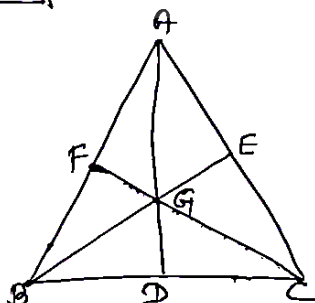
$$AC + BC > 2CF$$

$$\left[\frac{AB + BE + CA}{AD + BE + CF} > 1 \right]$$

$$2(AB + BC + AC) > 2(AD + BE + CF)$$

$$\text{OR } \boxed{AB + BC + AC > AD + BE + CF} \quad \text{Proved}$$

Proof:-



$$BG + GC > BC$$

$$AG + GE > AC$$

$$AG + GB > AB$$

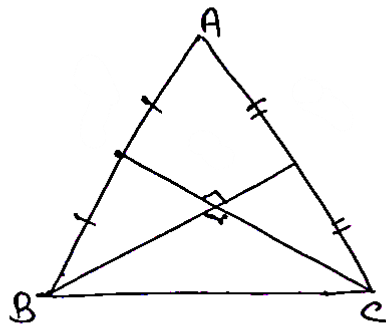
$$2(AG + BG + CG) > (AB + BC + CA)$$

$$\frac{4}{3}(AD + BE + CF) > (AB + BC + CA)$$

$$\left. \begin{aligned} AG &= \frac{2}{3} AD \\ BG &= \frac{2}{3} BE \\ CG &= \frac{2}{3} CF \end{aligned} \right\}$$

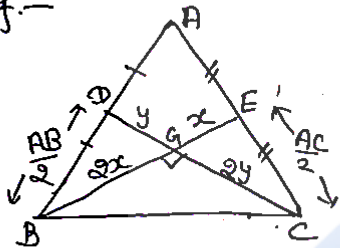
$$\left[\frac{4}{3} > \frac{AB + BC + CA}{AD + BE + CF} \right] \quad \text{Proved}$$

⇒ When two median of a triangle intersect each other at go



$$* \boxed{AB^2 + AC^2 = 5BC^2}$$

Proof:-



In $\triangle BDG$ & $\triangle CGE$

$$4x^2 + y^2 = \frac{AB^2}{4}$$

$$x^2 + 4y^2 = \frac{AC^2}{4}$$

$$+ \quad \frac{5(x^2 + y^2)}{4} = \frac{AB^2 + AC^2}{4}$$

$$5(x^2 + y^2) = AB^2 + AC^2$$

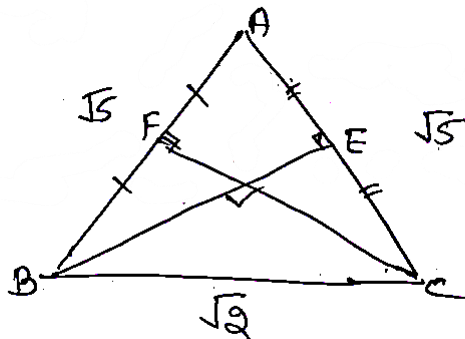
$$\boxed{5BC^2 = AB^2 + AC^2}$$

⇒ Special Case:-

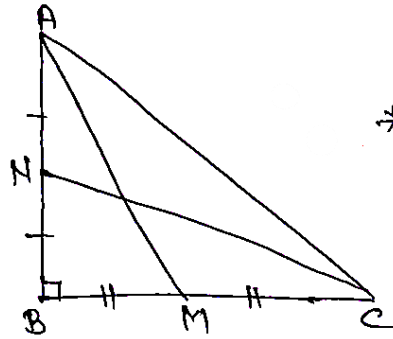
For isosceles Triangle $(AB = AC)$

$$* \boxed{\frac{AB}{BC} = \frac{\sqrt{5}}{\sqrt{2}}}$$

$$* \boxed{\frac{AB^2}{BC^2} = \frac{5}{2}}$$



⇒ Point :-

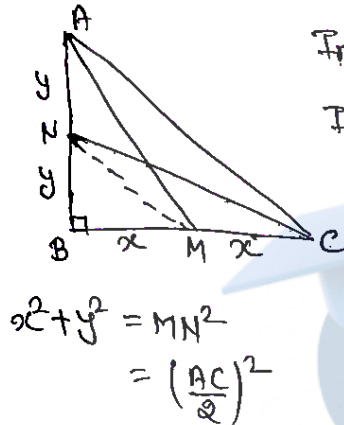


$$* AM^2 + CN^2 = \frac{5}{4} AC^2$$

OR

$$\frac{AM^2 + CN^2}{AC^2} = \frac{5}{4} \quad \left\{ \begin{array}{l} * \text{SSC CHSL-2020} \\ * \text{SSC CGT Mains 2020} \end{array} \right\}$$

Proof :-



$$\text{In } \triangle ABM \Rightarrow x^2 + 4y^2 = AM^2$$

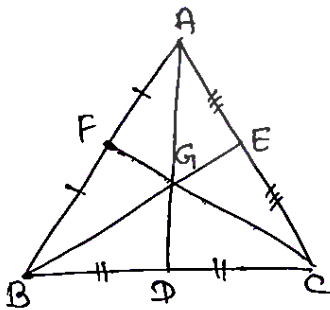
$$\text{In } \triangle ABC \Rightarrow 4x^2 + y^2 = CN^2$$

$$5(x^2 + y^2) = AM^2 + CN^2$$

$$\left[5\left(\frac{AC^2}{4}\right) = AM^2 + CN^2 \right]$$

⇒ Concept :- Area of Triangle :-

(When length of all medians is given)



$$\left. \begin{array}{l} AD = m_1 \\ BE = m_2 \\ CF = m_3 \end{array} \right\}$$

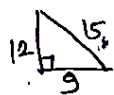
$$\text{Area of } \triangle ABC = \frac{4}{3} \sqrt{s(s-m_1)(s-m_2)(s-m_3)}$$

$$\text{where } s = \frac{m_1 + m_2 + m_3}{2}$$

If Right Angle Δ is given

$$\text{Area} = \frac{4}{3} \left[\frac{1}{2} \times \text{Base} \times \text{height} \right]$$

Ex:- $m_1 = 9\text{cm}, m_2 = 12\text{cm}, m_3 = 15\text{cm}$

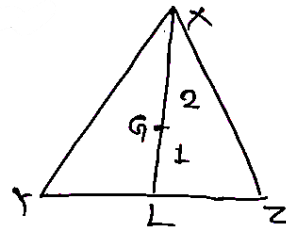


$$\text{Area} = \frac{4}{3} \left[\frac{1}{2} \times 9 \times 12 \right] \text{ Ans}$$

Q.75 In $\triangle XYZ$, if G is the centroid and XL is the median with length 18cm, then the length of XG is:

Soln:- $XG : GL = 2 : 1$

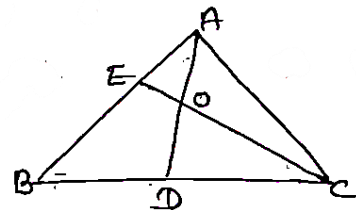
$$XG = \frac{18}{3} \times 2 = \boxed{12\text{cm}} \text{ Ans}$$



Q.76 AD and CE are two medians $\triangle ABC$. If $EO = 7\text{cm}$, then the length of CE is:

Soln:- $CO : OE = 2 : 1$

$$CE = \frac{7}{1} \times 3 = \boxed{21\text{cm}} \text{ Ans}$$



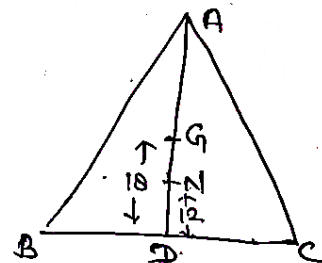
Q.77 In $\triangle ABC$, D is the mid-point of BC . Length AD is 27cm. N is a point in AD such that the length of DN is 12cm. The distance of N from the centroid of $\triangle ABC$ is equal to:

Soln:- $AD = 27\text{cm}$

$$GD = \frac{27}{3} \times 2 = 18\text{cm}$$

$$DN = 12\text{cm}$$

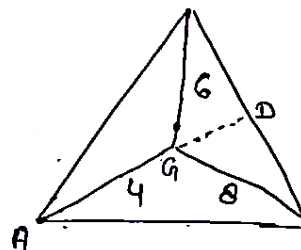
$$GN = 18 - 12 = \boxed{6\text{cm}} \text{ Ans}$$



Q.78 In a triangle, the distance of the centroid and three vertices is 4cm, 6cm, and 8cm respectively. Then the length of the smallest median is:

Soln:- $AG : GD = 2 : 1$

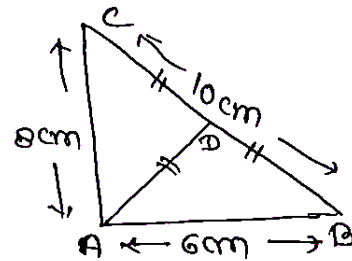
$$AD = \frac{4\text{cm} \times 3}{2} = \boxed{6\text{cm}} \text{ Ans}$$



Q.79 In $\triangle ABC$, D is the median from A to BC . $AB = 6\text{cm}$, $AC = 8\text{cm}$, and $BC = 10\text{cm}$. The length of median AD is:

Soln:-

$$AD = \frac{10\text{cm}}{2} = \boxed{5\text{cm}} \text{ Ans}$$



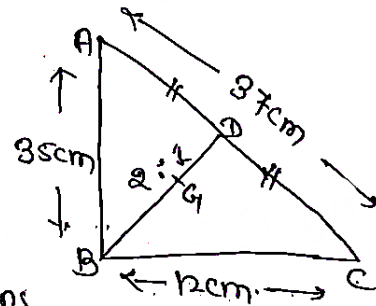
Q.80 G is the centroid of a triangle ABC , whose sides $AB = 35\text{cm}$, $BC = 12\text{cm}$ and $AC = 37\text{cm}$. The length of BG is (Correct to one decimal place):

Soln:-

$$BD = \frac{37\text{cm}}{2}$$

$$BG : GD = 2 : 1$$

$$BG = \frac{37}{2} \times \frac{2}{3} = \boxed{12.3\text{cm}} \text{ Ans}$$



Q.81 In $\triangle ABC$, $AB = 6\text{cm}$, $AC = 8\text{cm}$, and 9cm . The length of median AD is:

Soln:-

$$AB^2 + AC^2 = 2[AD^2 + BD^2]$$

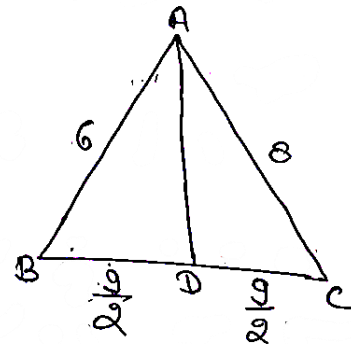
$$6^2 + 8^2 = 2[AD^2 + (\frac{9}{2})^2]$$

$$36 + 64 = 2[AD^2 + \frac{81}{4}]$$

$$100 = 2[AD^2 + \frac{81}{4}]$$

$$AD^2 = 50 - \frac{81}{4}$$

$$AD = \sqrt{\frac{119}{4}} \text{ Ans}$$



Q.82 In $\triangle ABD$, C is the midpoint of BD . If $AB = 10\text{cm}$, $AD = 12\text{cm}$ and $AC = 9\text{cm}$, then $BD = ?$

Soln :- $AB^2 + AD^2 = 2(AC^2 + BC^2)$

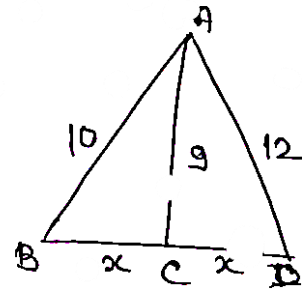
$$10^2 + 144 = 2(81 + x^2)$$

$$\frac{244}{2} = 81 + x^2$$

$$x^2 = 122 - 81$$

$$x = \sqrt{41}$$

$$BD = \boxed{2\sqrt{41}} \text{ Ans}$$



Q.83 In triangle PQR , C is the centroid. $PQ = 30\text{cm}$, $QR = 36\text{cm}$ and $PR = 50\text{cm}$. If D is the midpoint of QR , then what is the length of CD ?

Soln :- $(PQ)^2 + (PR)^2 = 2(PD^2 + QD^2)$

$$(30)^2 + (50)^2 = 2[PD^2 + (18)^2]$$

$$900 + 2500 = 2[PD^2 + 324]$$

$$\frac{3400}{2} = PD^2 + 324$$

$$PD^2 = 1700 - 324$$

$$PD^2 = 1376$$

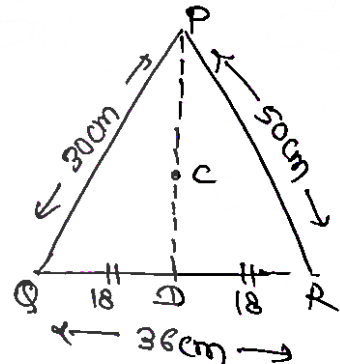
$$PD = 4\sqrt{86}$$

$$\boxed{PC : CD = 2 : 1} \text{ so } CD = \frac{1}{3} PD$$

so

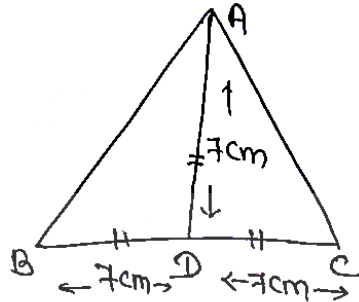
$$\boxed{CD = \frac{4\sqrt{86}}{3}}$$

Ans



Q.84 In $\triangle ABC$, the median AD is 7cm and CB is 14cm, measure of $\angle CAB$ is?

Soln:-



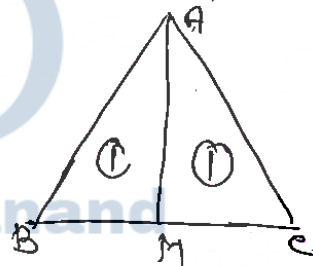
$\angle A = 90^\circ$ Ans

Q.85 If M is the mid-point of the side BC of $\triangle ABC$, and the area of $\triangle ABM$ is 18cm^2 , then the area of $\triangle ABC$ is?

Soln:-

$$\begin{aligned} \text{ar}(\triangle ABC) &= 2 \text{ar}(\triangle ABM) \\ &= 2 \times 18 \\ &= \boxed{36\text{cm}^2} \end{aligned}$$

Ans

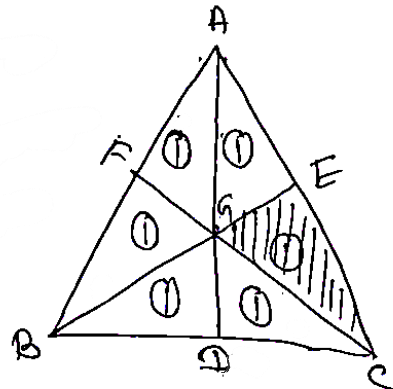


Q.86 Three medians AD , BE and CF of $\triangle ABC$ intersect at G ; area of $\triangle ABC$ is 36cm^2 . Then the area of $\triangle CGE$ is.

Soln

$$\begin{aligned} \text{ar}(\triangle CGE) &= \frac{1}{6} \text{ar}(\triangle ABC) \\ &= \frac{1}{6} \times 36 \\ &= \boxed{6\text{cm}^2} \end{aligned}$$

Ans

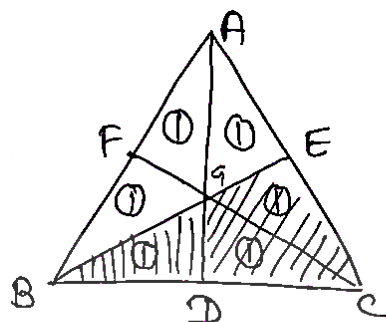


Q87 In $\triangle ABC$, the medians AD and BE meet at G .

The ratio of the areas of $\triangle BDG$ and the quadrilateral $GDCE$ is:

Soln :- $ar(\triangle BDG) : ar(GDCE)$

$$\boxed{1 : 2} \text{ Ans}$$



Q88 If two medians BE and CF of a triangle ABC , intersect each other at G and if $BG = CG$, $\angle BGC = 60^\circ$, $BC = 8\text{cm}$, then area of the triangle ABC is:

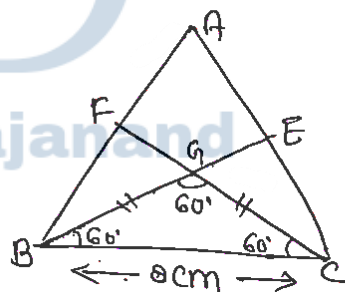
Soln :- $\triangle BGC$ is equilateral $\triangle$

$$\begin{aligned} \text{So Area}(\triangle BGC) &= \frac{\sqrt{3}}{4} \times (8)^2 \\ &= 16\sqrt{3} \end{aligned}$$

$$ar(\triangle ABC) = 3 [ar(\triangle BGC)]$$



$$= 3 \times 16\sqrt{3} \Rightarrow \boxed{48\sqrt{3} \text{ cm}^2} \text{ Ans}$$



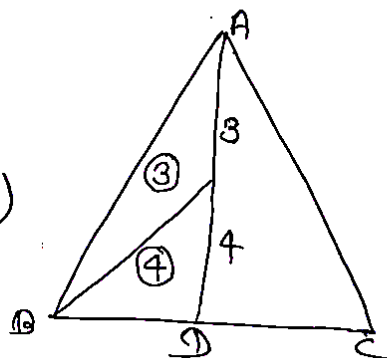
Q89 In $\triangle ABC$, AD is a median and P is a point on AC such that $AP : PD = 3 : 4$, then $ar(\triangle APB) : ar(\triangle ABC)$ is:

$$\text{Soln :- } ar(\triangle APB) = 3$$

$$ar(\triangle ABC) = 7 + 7 = 14$$

(median divide into two equal part)

$$\triangle APB : \triangle ABC = \boxed{3 : 14} \text{ Ans}$$

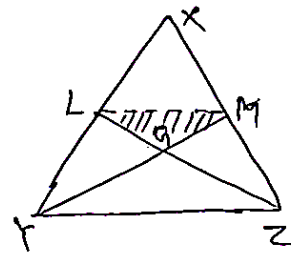


Q.90 XYZ is a triangle. If the medians ZL and YM intersect each other at G, then (Area of ΔGLM : Area of ΔXYZ)

Soln:-

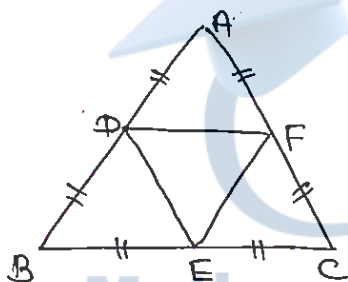
$$\Delta GLM = \frac{1}{12} \text{ of } (\Delta XYZ)$$

1 : 12 Ans



Q.91 ABC is an equilateral triangle. Points D, E and F are taken as the mid-point on sides AB, BC, CA respectively, so that AD = BE = CF. Then AE, BF, CD enclosed a triangle which is:

Soln:-



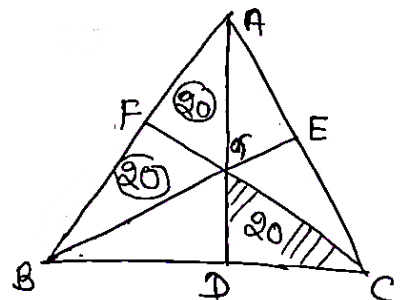
- ☒ (a) equilateral
- (b) isosceles Δ
- (c) right angle Δ
- (d) None of these

Given $AD = BE = CF$ so $AB = BC = CA$ (equilateral Δ)
 $\Rightarrow (DB = AF = CE)$

Q.92 In ΔABC , AD, BE and CF are medians and G is the centroid of the triangle. If the area of the ΔDGC is 20cm^2 , then the area of ΔAGF + the Area of ΔBGF is equal to:

Soln:-

$$\begin{aligned} \Delta BGF + \Delta AGF &= 20 + 20 \\ &= \boxed{40\text{cm}^2} \text{ Ans } \end{aligned}$$



Q.93 In an equilateral $\triangle ABC$, the medians AD , BE and CF intersect to each other at point G . If the area of quadrilateral $BDGF$ is $12\sqrt{3} \text{ cm}^2$, then the side of $\triangle ABC$ is:

Soln:-

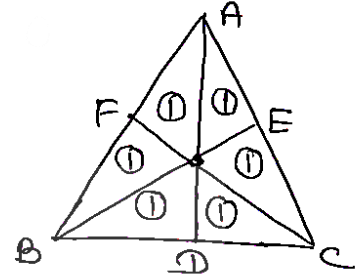
$$2 \text{ Unit} \rightarrow 12\sqrt{3}$$

$$6 \text{ Unit} \rightarrow 12\sqrt{3} \times 3 = 36\sqrt{3}$$

$$\text{ar}(\triangle ABC) = 36\sqrt{3}$$

$$\frac{\sqrt{3}}{4}(\text{side})^2 = 36\sqrt{3}$$

$$\text{Side} = \boxed{12\text{cm}} \text{ Ans}$$



Q.94 If D , E and F are the mid points of BC , CA and AB respectively of the $\triangle ABC$. The ratio of area of the parallelogram $DEFB$ and area of the trapezium $CAFD$ is:

Soln:-

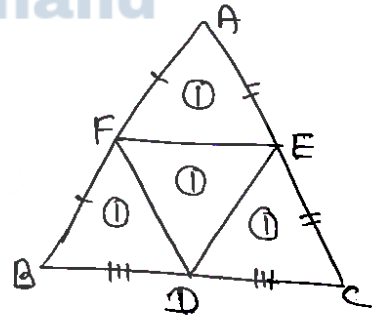
$$\text{area of } (DEFB) = ① + ①$$

$$= 2$$

$$\text{area of } (CAFD) = ① + ① + ①$$

$$= 3$$

Required ratio $\Rightarrow \boxed{2:3} \text{ Ans}$



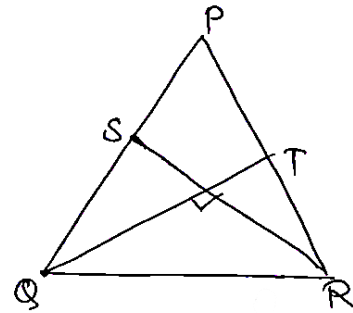
Q.95 PQR is a triangle such that $PQ = PR$. RS and QT are the median to the sides PQ and PR respectively. If the medians RS and QT intersect at right angle, then what is the value of $(PQ/QR)^2 = ?$

Soln:- $PQ = PR$

$$PQ^2 + PR^2 = 5 QR^2$$

$$2PQ^2 = 5QR^2$$

$$\left(\frac{PQ}{QR}\right)^2 = \frac{5}{2} \quad \underline{\underline{\text{Ans}}}$$



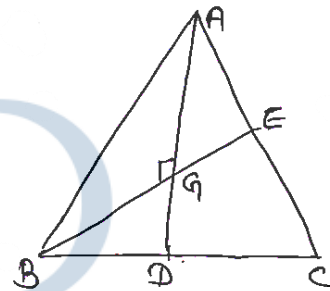
[Q.96] G is the centroid of $\triangle ABC$. The median AD and BE intersect at right angles. If the lengths of AD and BE are 9cm and 12cm respectively then the length of AB is ?

Soln:- $AG = \frac{2}{3} AD$
 $= \frac{2}{3} \times 9 = 6\text{cm}$

$$BG = \frac{2}{3} BE$$

$$= \frac{2}{3} \times 12 = 8\text{cm}$$

$$AB = \sqrt{(6)^2 + (8)^2} = \underline{\underline{10\text{cm}}} \quad \underline{\underline{\text{Ans}}}$$



[Q.97] In a right-angled triangle the length of the medians from the vertices of acute angles are 7cm and $4\sqrt{6}\text{cm}$. what is the length of the hypotenuse of the triangle?

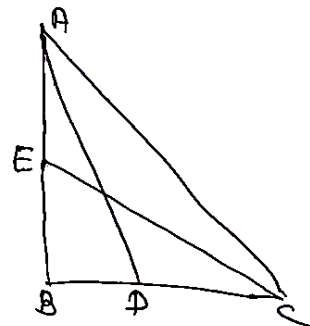
Soln:- $AD^2 + CE^2 = \frac{5}{4} AC^2$ {SSC CHL-Pre-2022}

$$49 + 16 \times 6 = \frac{5}{4} AC^2$$

$$AC^2 = \frac{4}{5} \times 145$$

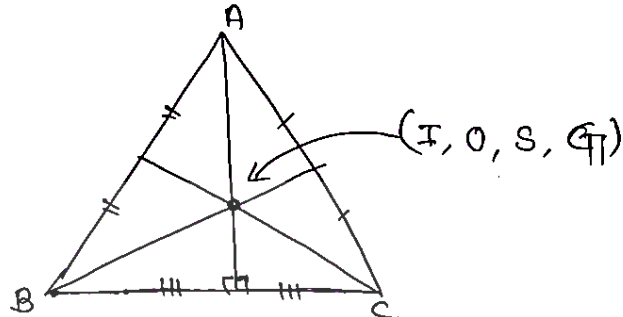
$$AC^2 = 4 \times 29$$

$$AC = \underline{\underline{2\sqrt{29}\text{cm}}} \quad \underline{\underline{\text{Ans}}}$$



⇒ Equilateral Triangle :-

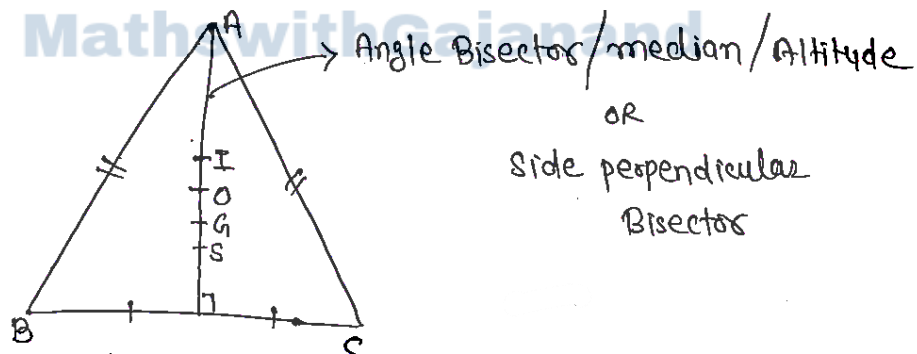
In equilateral Δ all centers (Circumcenter, incenter, orthocentre, centroid) will be at same point.



* If length of all three median/Altitude are same then it will be equilateral Δ .

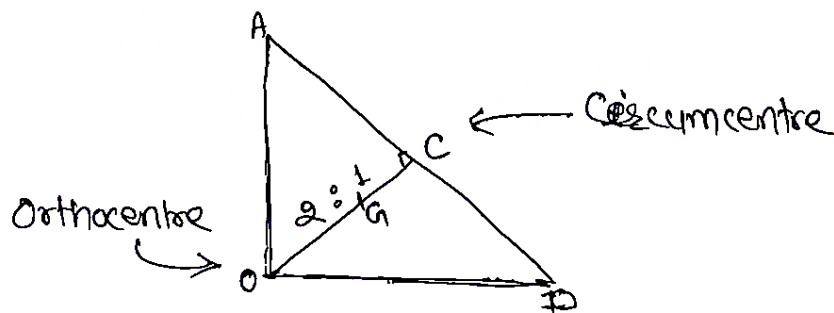
⇒ Isosceles Triangle :-

In isosceles Triangle all centres (Circumcentre, incentre, orthocentre, centroid) will be Colinear (at same line)



* If any two median/Altitude length are same then it will be isosceles Δ .

⇒ Right Angle Triangle :-



Q.98 which of the following is a FALSE statement?

- (a) The circumcentre is the centre of a circle which circumscribes the triangle
- ✓ (b) The angle formed by any side at the incentre is always 90° more than the opposite vertex.
- (c) Incentre is the point of concurrency of angle bisectors of a triangle
- (d) Centroid of the triangle divides each median in the ratio of 2:1.

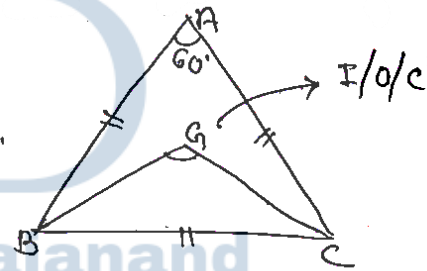
Q.99 G is the centroid of $\triangle ABC$. If $AB = BC = AC$, then measure of $\angle BGC$ is.

Soln:-

$$\begin{aligned}\angle BGC &= 90^\circ + \frac{\angle A}{2} \\ &= 90^\circ + \frac{60^\circ}{2} \\ &= 120^\circ \text{ Ans}\end{aligned}$$

OR

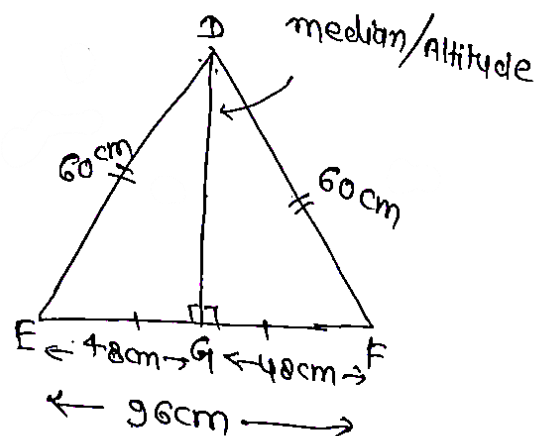
$$\begin{aligned}\angle BGC &= 2\angle A \\ &= 2 \times 60^\circ \\ &= 120^\circ\end{aligned}$$



Q.100 DEF is an isosceles triangle with such that $DE = DF = 60\text{cm}$ and $EF = 96\text{cm}$. DG is a median to base EF. What is the length of DG?

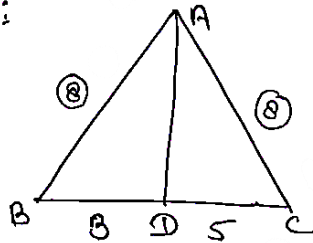
Soln:-

$$\begin{aligned}DG &= \sqrt{(60)^2 - (48)^2} \\ &= \sqrt{3600 - 2304} \\ &= \sqrt{1296} \\ &= 36\text{cm} \text{ Ans}\end{aligned}$$



Q.101 $\triangle ABC$ is an equilateral triangle in which $BD:DC = 3:5$ then $AD:DC = ?$

Soln:



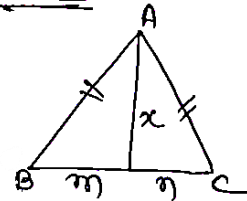
$$AD^2 = 8^2 - 3 \times 5 = 49$$

$$AD = 7$$

$$AD:AC = 7:8 \text{ Ans}$$

Stewart theorem

$$x^2 = a^2 - mn$$



Q.102 In a $\triangle PQR$, $\angle Q = 55^\circ$ and $\angle R = 35^\circ$. Find the ratio of angles subtended by side QR on circumcentre, incentre and orthocentre of the triangle.

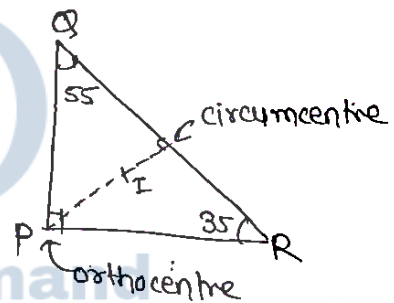
Soln:- circumcentre = 180°

$$\text{Incentre} = 90^\circ + \frac{\angle P}{2} = 135^\circ$$

$$\text{orthocentre} = 90^\circ$$

$$\text{Ratio :- } 180:135:90$$

$$= 4:3:2 \text{ Ans}$$



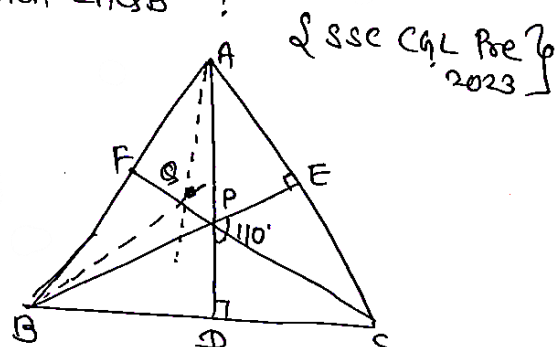
Q.103 In $\triangle ABC$, the perpendiculars drawn from A, B and C meet the opposite sides at points D, E and F respectively. AD, BE and CF intersect at point P. If $\angle EPD = 110^\circ$ and the bisectors of $\angle A$ and $\angle B$ meet at point Q, then $\angle AQB = ?$

Soln:- $\angle C = 180^\circ - 110^\circ = 70^\circ$

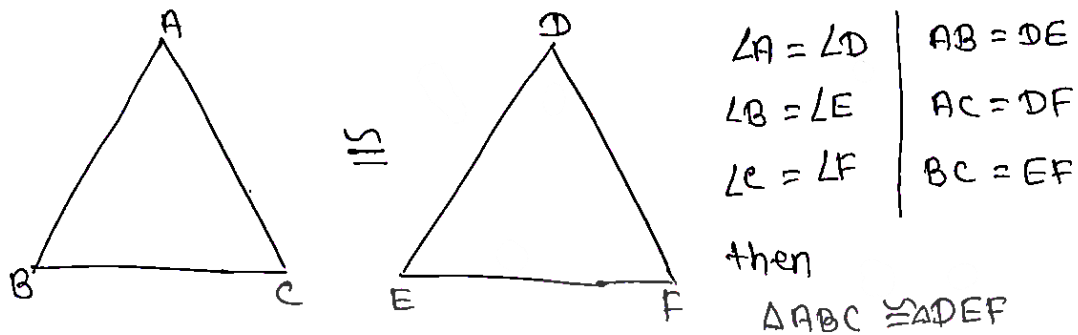
$$\angle AQB = 90^\circ + \frac{\angle C}{2}$$

$$= 90^\circ + \frac{70^\circ}{2}$$

$$125^\circ$$

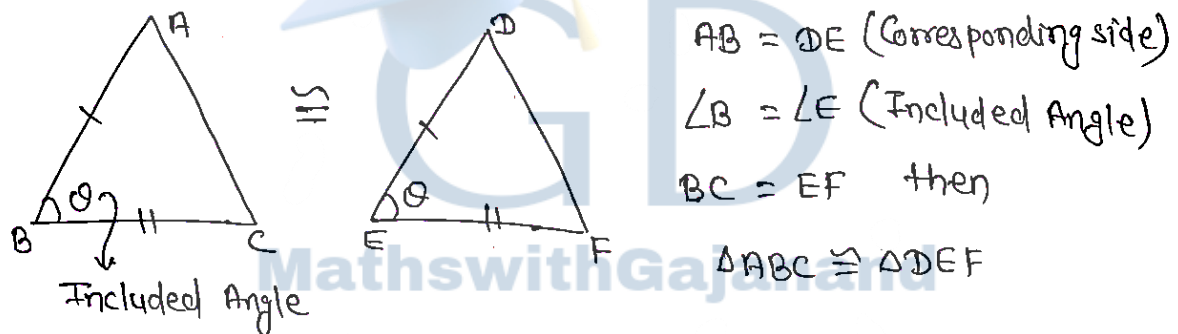


CONGRUENCY OF TRIANGLE

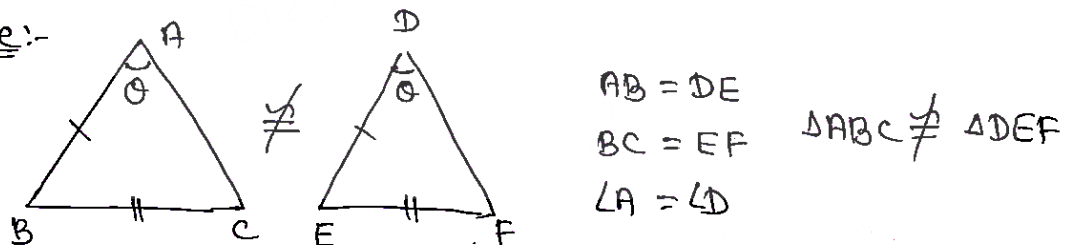


Condition for Congruency :-

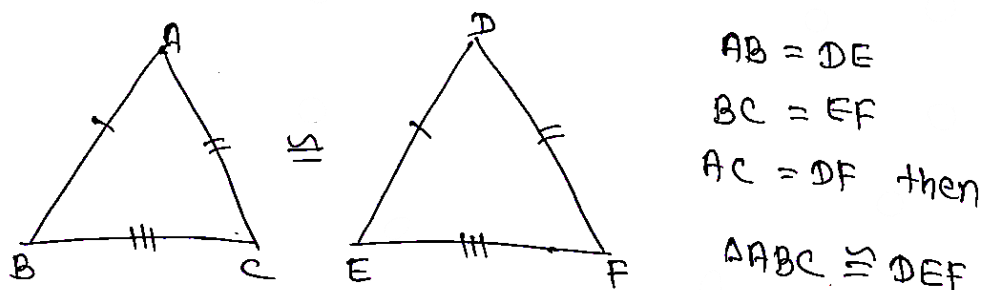
(i) S-A-S (side-angle-side) :-



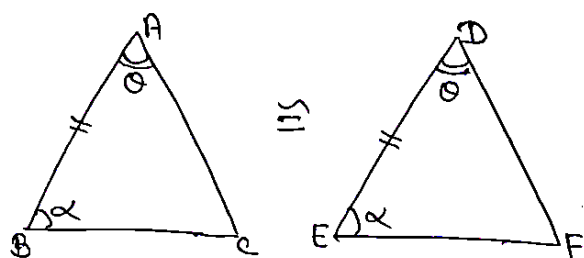
Note:-



(ii) S-S-S (side-side-side) :-

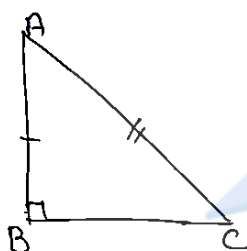


<iii> A-A-S (Angle-Angle-side) :-

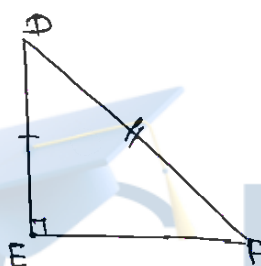


$$\begin{aligned}\angle A &= \angle D \\ \angle B &= \angle E \\ AB &= DE \text{ then} \\ \Delta ABC &\cong \Delta DEF\end{aligned}$$

<iv> R-H-S (Right Angle-Hypotenuse-side) :-



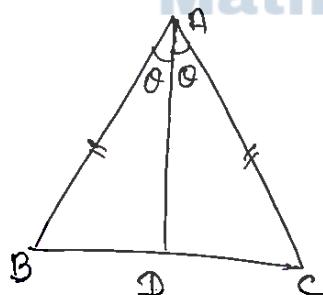
$\cong$



$$\angle B = \angle E = 90^\circ$$

$$\begin{aligned}AC &= DF \\ AB &= DE \text{ then} \\ \Delta ABC &\cong \Delta DEF\end{aligned}$$

$\Rightarrow$ Special Case of Isosceles triangle:-



AD $\rightarrow$ perpendicular / height /
median / angle bisector /
side perpendicular bisector

$$\Delta ADB \cong \Delta ADC$$

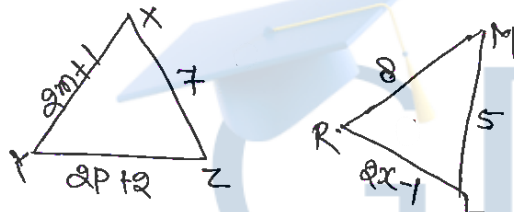
$\Rightarrow$ If two Triangles are congruent then:-

- <i> one of them can be superimposed (अतिरोपित) of the other Δ .
- <ii> The perimeter of both the triangle will be equal.
- <iii> The ratio of the angles in both the triangles will be same.
- <iv> Their Areas will be equal.

Q.104 Which of the following statement is FALSE?

- (a) Two triangles are congruent if the size and shape of the triangle may or may not be equal.
- (b) SAS and SSS are both conditions of congruency of triangles.
- (c) If two angles and the included side of one triangle is equal to two angles and the included side of other triangle, then the triangle are congruent.
- (d) If two triangles are congruent, then one of them can be superimposed of the other triangle.

Q.105 If $\triangle XYZ \cong \triangle LMR$, then $m + x + p = ?$

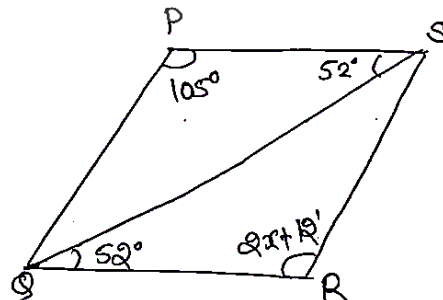


Soln:- $XY = LM$ $2m+1 = 8 \Rightarrow m = 2$
 $YZ = MR \Rightarrow 2p+2 = 5 \Rightarrow p = 3$
 $XZ = LR$ $2x-1 = 7 \Rightarrow x = 4$

$\therefore m + x + p = 2 + 4 + 3 = 9$ Ans

Q.106 In the given figure, $\triangle QPS \cong \triangle SRQ$. Find $\angle PSR = ?$

Soln:- $\triangle QPS \cong \triangle SRQ$
 $\angle P = \angle R = 105^\circ$
 $\angle QSR = 180 - (52^\circ + 105^\circ)$
 $= 23^\circ$
 $\angle PSQ = 52^\circ$
 $\angle PSR = 23^\circ + 52^\circ$
 $= 75^\circ$ Ans



Q.107 Let $\triangle ABC, \triangle PQR$ be two congruent triangles such that $\angle A = \angle P = 90^\circ$. If $BC = 13\text{cm}$, $PR = 5\text{cm}$, find AB .

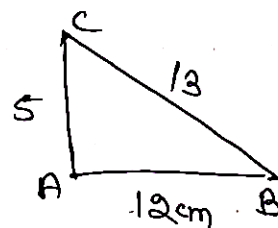
Soln:-

$$\triangle ABC \cong \triangle PQR$$

$$BC = QR = 13\text{cm}$$

$$AC = PR = 5\text{cm}$$

$$AB = PQ \Rightarrow \sqrt{(13)^2 - (5)^2} = \boxed{12\text{cm}}$$



Q.108 It is given that $\triangle ABC \cong \triangle PQR$, $AB = 5\text{cm}$, $\angle B = 40^\circ$ & $\angle A = 80^\circ$. Which of the following options is true?

(a) $PQ = 5\text{cm}$, $\angle R = 60^\circ$

(b) $QR = 5\text{cm}$, $\angle R = 60^\circ$

(c) $QR = 5\text{cm}$, $\angle Q = 60^\circ$

(d) $PQ = 5\text{cm}$, $\angle P = 60^\circ$

$$\triangle ABC \cong \triangle PQR$$

$$\angle B = \angle Q = 40^\circ$$

$$\angle A = \angle P = 80^\circ$$

$$AB = PQ = 5\text{cm}$$

$$\angle R = 180 - (40 + 80) = 60^\circ$$

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Q.109 In $\triangle ABC$, M is midpoint of the side AB . N is a point in the interior of $\triangle ABC$ such that CN is the bisector of $\angle C$ and $CN \perp NB$, what is the length of MN , if $BC = 10\text{cm}$ and $AC = 15\text{cm}$?

Soln:-

$$\triangle CNB \cong \triangle CND$$

$$NB = ND$$

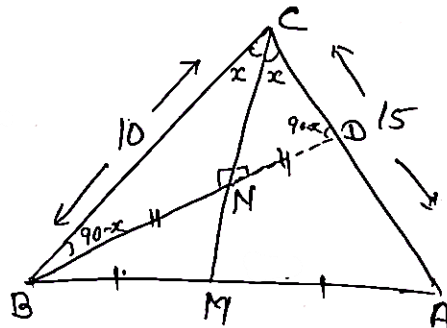
$$CB = CD = 10\text{cm}$$

$$AD = 15 - 10 = 5\text{cm}$$

In $\triangle ABD$

$$MN = \frac{AD}{2} = \frac{5}{2}$$

$$= \boxed{2.5\text{cm}} \text{ Ans}$$

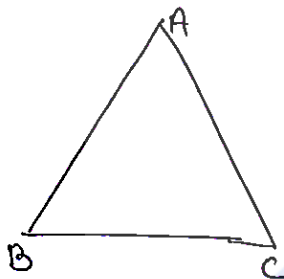


Similarity of Triangle

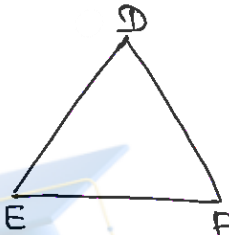
Condition for Similarity :-

(i) A-A-A (Angle-Angle-Angle) :-

If two corresponding angles of the two triangle are equal, the triangle must be similar.



~



$$\angle A = \angle D$$

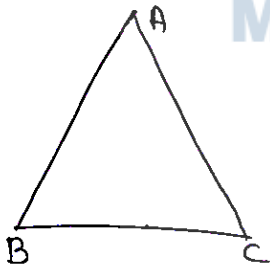
$$\angle B = \angle E$$

$$\angle C = \angle F \text{ then}$$

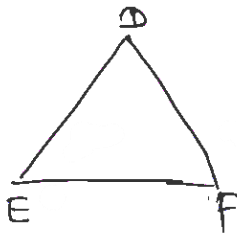
$$\triangle ABC \sim \triangle DEF$$

(ii) S-S-S (side-side-side) :-

If the corresponding sides of the two triangles are proportional the triangle must be similar.



~



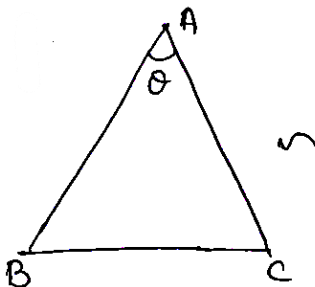
$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$$

then

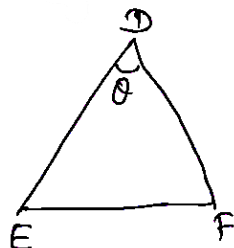
$$\triangle ABC \sim \triangle DEF$$

(iii) S-A-S (side-Angle-side) :-

If two sides of two triangles are proportional and one included angle is same then the two triangles are similar.



~

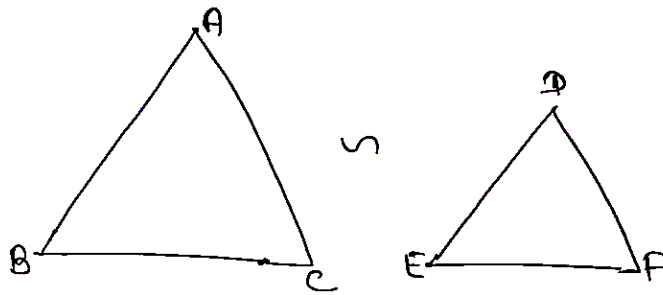


$$\frac{AB}{DE} = \frac{AC}{DF}$$

$$\angle A = \angle D \text{ then}$$

$$\triangle ABC \sim \triangle DEF$$

Most Important Concept:-



$$\frac{\text{Perimeter of } \triangle ABC}{\text{Perimeter of } \triangle DEF} = \frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} = \frac{A \cdot B_1}{A \cdot B_2} = \frac{A \cdot L_1}{A \cdot L_2} = \frac{m_1}{m_2} = \frac{r_1}{r_2} = \frac{R_1}{R_2}$$

$A \cdot B \rightarrow$ Angle Bisector

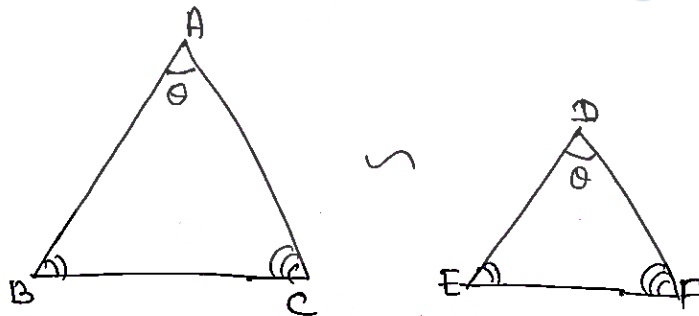
$A \cdot L \rightarrow$ Altitude (height)

$m \rightarrow$ median

$r \rightarrow$ In radius

$R \rightarrow$ Circumradius

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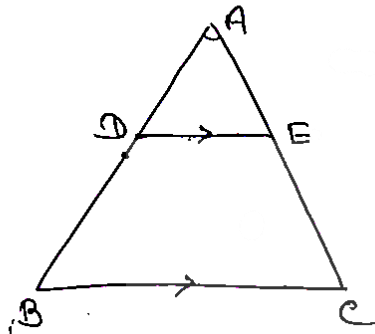
2.

$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle DEF} = \frac{\frac{1}{2} AB \cdot AC \cdot \sin \theta}{\frac{1}{2} DE \cdot DF \cdot \sin \theta} = \frac{AB \cdot AC}{DE \cdot DF} = \frac{AB^2}{DE^2} = \frac{AC^2}{DF^2}$$

$$= \frac{BC^2}{EF^2} = \frac{A \cdot L_1^2}{A \cdot L_2^2} = \frac{m_1^2}{m_2^2} = \frac{r_1^2}{r_2^2} = \frac{R_1^2}{R_2^2}$$

Basic Proportionality Theorem (BPT) :-

If a line is parallel to a side of a triangle which intersects other two sides in distinct points, then the line divides other two sides in proportion.



$$\triangle ADE \sim \triangle ABC$$

$$\left[\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC} \right]$$

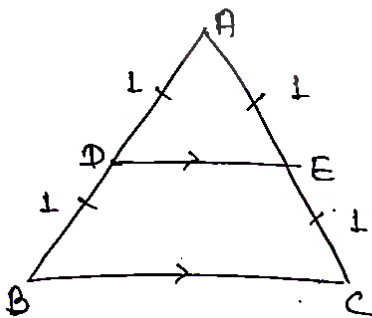
$$\left[\frac{AD}{BD} = \frac{AE}{EC} \right] \text{ (Thales theorem)}$$

$$\left[\frac{\text{Area of } \triangle ADE}{\text{Area of } \triangle ABC} = \left(\frac{AD}{AB} \right)^2 = \left(\frac{AE}{AC} \right)^2 = \left(\frac{DE}{BC} \right)^2 \right]$$

Convergence of BPT :-

$$\text{If } \frac{AD}{BD} = \frac{AE}{EC} \Rightarrow \text{then } DE \parallel BC$$

MID-Point Theorem :-



D and E are the midpoint of AB and AC

$$DE \parallel BC$$

$$2 \left[DE = \frac{BC}{2} \right]$$

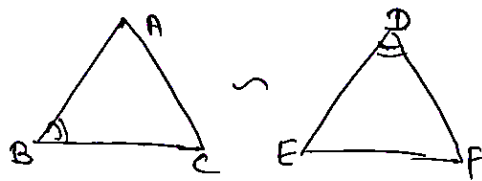
$$[DE : BC = 1 : 2]$$

$$\frac{\text{Area of } \triangle ADE}{\text{Area of } \triangle ABC} = \left(\frac{1}{2} \right)^2 = \frac{1}{4}$$

Q.110 If in $\triangle ABC$ and $\triangle DEF$, $\frac{AB}{DE} = \frac{BC}{FD}$, then they will be similar when?

- (a) $\angle B = \angle D$ (b) $\angle A = \angle D$ (c) $\angle A = \angle F$ (d) $\angle B = \angle E$

Soln:-

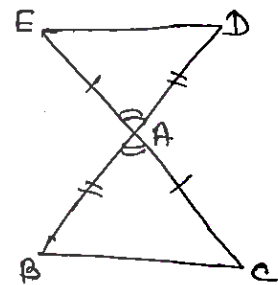


by SAS
 $\angle B = \angle D$

Q.111 In the figure $AB = AD = 9\text{cm}$ and $AC = AE = 13\text{cm}$ and $BC = 15\text{cm}$. Find ED ?

Soln SAS:- $\triangle AED \cong \triangle ACB$

$$ED = BC = \boxed{15\text{cm}} \text{ Ans}$$



Q.112 The perimeters of two similar triangles ABC and PQR are 78cm and 46.8cm , respectively. If $PQ = 11.7$, then the length of AB is?

Soln:- $\frac{78}{46.8} = \frac{AB}{11.7}$

$$AB = \boxed{19.5\text{cm}} \text{ Ans}$$

Q.113 $\triangle ABC$ is similar to $\triangle PQR$. If ratio of perimeter of $\triangle ABC$: perimeter of $\triangle PQR$ is $5:9$ and if $PQ = 45\text{cm}$, then the length of AB is?

Soln:-

$$\frac{5}{9} = \frac{AB}{45}$$

$$AB = \boxed{25\text{cm}} \text{ Ans}$$

Q.114 If $\triangle ABC \sim \triangle PQR$, $AB = 4\text{cm}$, $PQ = 6\text{cm}$, $QR = 9\text{cm}$ and $RP = 12\text{cm}$, then find the perimeter of $\triangle ABC$

Soln:- Perimeter of $\triangle PQR = PQ + QR + RP = 6 + 9 + 12 = 27\text{cm}$

$$\frac{\text{Pr}(\triangle ABC)}{27} = \frac{AB}{PQ}$$

$$\text{Pr}(\triangle ABC) = 27 \times \frac{4}{6} = \boxed{18\text{cm}} \text{ Ans}$$

Q.115 The ratio of the areas of two similar triangles is $196 : 625$, then what will be the ratio of corresponding sides?

Soln:-

$$\frac{196}{625} = \left(\frac{\text{side}_1}{\text{side}_2}\right)^2$$

$$\frac{\text{side}_1}{\text{side}_2} = \sqrt{\frac{196}{625}} = \boxed{14 : 25} \text{ Ans}$$

Q.116 Given that $\triangle DEF \sim \triangle ABC$, if the area of $\triangle ABC$ is 9cm^2 and that of $\triangle DEF = 12\text{cm}^2$ and $BC = 2.1\text{cm}$, then the length EF is:

Soln:-

$$\frac{\text{ar}(\triangle DEF)}{\text{ar}(\triangle ABC)} = \frac{(EF)^2}{(BC)^2}$$

$$\Rightarrow \frac{12}{9} = \frac{EF^2}{(2.1)^2}$$

$$\Rightarrow EF = \sqrt{\frac{12}{9} \times 4.41} = \frac{2.1 \times 2}{\sqrt{3}}$$

$$\Rightarrow EF = \frac{2.1 \times 2 \times \sqrt{3}}{10 \times 3}$$

$$\Rightarrow EF = \boxed{\frac{7\sqrt{3}}{5}} \text{ Ans}$$

[117] Let $\triangle ABC \sim \triangle RPQ$ and $\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PQR)} = \frac{4}{9}$. If $AB = 3\text{cm}$,

$BC = 4\text{cm}$ and $AC = 5\text{cm}$, then RP is equal to:

Soln:-

$$\frac{AB}{RP} = \sqrt{\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PQR)}}$$

$$\frac{3}{RP} = \sqrt{\frac{4}{9}} \Rightarrow \frac{2}{3}$$

$$\boxed{RP = 4.5\text{cm}} \text{ Ans}$$

[118] The areas of two similar triangles are 324cm^2 and 225cm^2 . If the altitude of the smaller triangle is 10cm , then the altitude of the bigger triangle is,

Soln:-

$$\frac{x}{10} = \sqrt{\frac{324}{225}}$$

$$x = \frac{18}{15} \times 10 = \boxed{12\text{cm}} \text{ Ans}$$

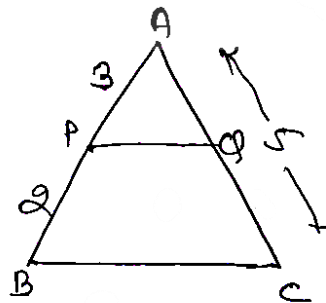
[119] A line PQ intersect the sides AB, AC of the triangle ABC , at P, Q respectively in such a way that $AP:PB = 3:2$ then $\text{ar}(\triangle APQ) : \text{ar}(\triangle ABC)$ is

Soln:-

$$\frac{\text{ar}(\triangle APQ)}{\text{ar}(\triangle ABC)} = \frac{AP^2}{AB^2}$$

$$= \left(\frac{3}{5}\right)^2$$

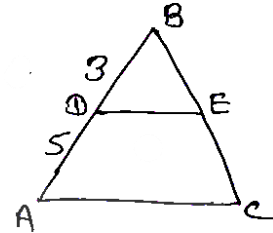
$$= \boxed{\frac{9}{25}} \text{ Ans}$$



Q.120 $\triangle ABC$, D and E are the points on sides AB and BC respectively such that $DE \parallel AC$. If $AD:DB = 5:3$, then what is the ratio of the area of $\triangle BDE$ to that of the trapezium ACED?

Soln:- $\triangle BDE \sim \triangle BAC$

$$\frac{\text{Area of } \triangle BDE}{\text{Area of } \triangle BAC} = \left(\frac{3}{8}\right)^2 = \frac{9}{64}$$



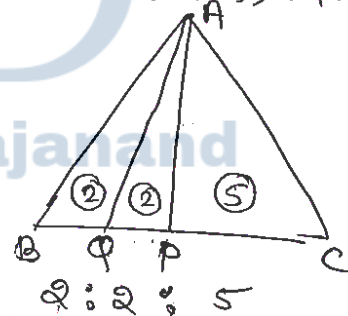
$$\frac{\square ACED}{\triangle BDE} = \frac{64-9}{9} = \frac{55}{9}$$

$$\triangle BDE : \square ACED = \boxed{9 : 55} \text{ Ans.}$$

Q.121 In $\triangle ABC$, P is a point on BC such that $BP:PC = 4:5$ and Q is the mid point of BP. then $\text{ar}(\triangle ABQ) : \text{ar}(\triangle ABC)$ is?

Soln:- $\text{ar}(\triangle ABQ) : \text{ar}(\triangle ABC)$

$$\boxed{2 : 9} \text{ Ans}$$

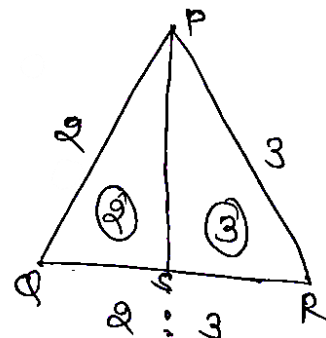


Q.122 PQR is a triangle, whose area is 100cm^2 . S is a point on side QR, such that PS is the angle bisector of $\angle QPR$. If $PQ:PR = 2:3$, then what is the area of triangle PSR?

Soln:- $\text{ar}(\triangle PSR) : \text{ar}(\triangle PQR) = 3 : 5$

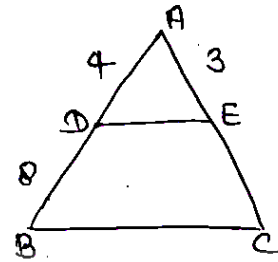
$$\frac{\text{ar}(\triangle PSR)}{100} = \frac{3}{5}$$

$$\text{ar}(\triangle PSR) = \boxed{108\text{cm}^2} \text{ Ans}$$



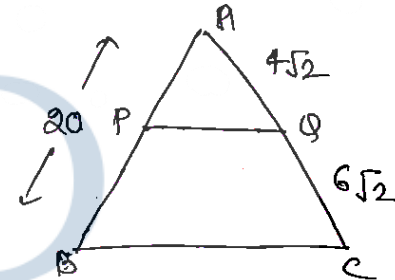
Q.123 In $\triangle ABC$, DE is parallel to BC when D and E are the points on AB and AC , respectively and $AD = 4\text{cm}$, $DB = 8\text{cm}$, $AE = 3\text{cm}$. Length of EC is:

Soln:- $\frac{AD}{DB} = \frac{AE}{EC}$
 $\frac{4}{8} = \frac{3}{EC}$
 $EC = \boxed{6\text{cm}}$ Ans.



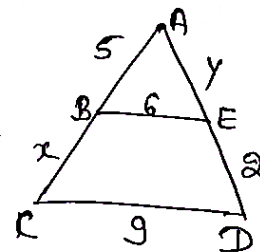
Q.124 In the given figure, $AQ = 4\sqrt{2}\text{cm}$, $QC = 6\sqrt{2}\text{cm}$ and $AB = 20\text{cm}$. If PQ is parallel to BC , then what is the value of PB ?

Soln:- $\frac{AP}{AB} = \frac{AQ}{AC}$
 $\frac{AP}{20} = \frac{4\sqrt{2}}{10\sqrt{2}}$
 $AP = 8\text{cm}$
 $PB = AB - AP = 20 - 8 = \boxed{12\text{cm}}$ Ans.



Q.125 In $\triangle ACD$, B and E are two points on side AC and AD respectively, such that BE is parallel to CD . $CD = 9\text{cm}$, $BE = 6\text{cm}$, $AB = 5\text{cm}$ and $ED = 2\text{cm}$. What are the measures of the length of AE and BC ?

Soln $\frac{AB}{AC} = \frac{BE}{CD}$
 $\Rightarrow \frac{5}{5+x} = \frac{6}{9}$
 $\Rightarrow x = \boxed{2.5\text{cm}} = BC$



$\frac{AB}{BC} = \frac{AE}{ED} \Rightarrow \frac{5}{2.5} = \frac{y}{2}$
 $\Rightarrow y = \boxed{4\text{cm}}$ Ans.

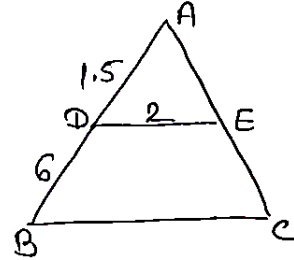
Q.127 If in $\triangle ABC$, $DE \parallel BC$, $AB = 7.5\text{cm}$, $BD = 6\text{cm}$ and $DE = 2\text{cm}$ then the length of BC in cm is,

Soln:- $\triangle ADE \sim \triangle ABC$

$$\frac{AD}{AB} = \frac{DE}{BC}$$

$$\frac{1.5}{7.5} = \frac{2}{BC}$$

$$BC = \boxed{10\text{cm}} \text{ Ans.}$$



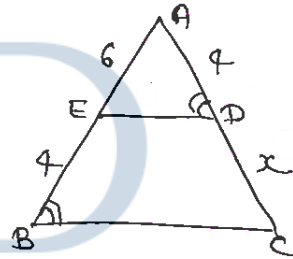
Q.128 In $\triangle ABC$, E and D are points on sides AB and AC , respectively, such that $\angle ABC = \angle ADE$. If $AE = 6\text{cm}$, $AD = 4\text{cm}$ and $EB = 4\text{cm}$, then the length of DC is:

Soln:- $\triangle AED \sim \triangle ACB$

$$\frac{AD}{AB} = \frac{AE}{AC}$$

$$\frac{4}{10} = \frac{6}{4+x}$$

$$x = \boxed{11\text{cm}} \text{ Ans.}$$



Q.129 In $\triangle ABC$, D and E are points on the sides BC and AB , respectively, such that $\angle ACB = \angle DEB$. If $AB = 12\text{cm}$, $BE = 5\text{cm}$ and $BD : CD = 1 : 2$, then BC is equal to:

Soln:- $\triangle BED \sim \triangle BCA$

$$\frac{BE}{BC} = \frac{BD}{AB}$$

$$\frac{5}{3x} = \frac{x}{12}$$

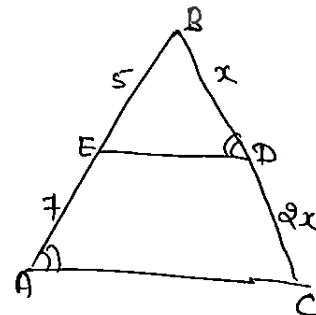
$$3x^2 = 60$$

$$x^2 = 20$$

$$x = 2\sqrt{5}$$

$$BC = 3x = 3 \times 2\sqrt{5}$$

$$= \boxed{6\sqrt{5}} \text{ Ans}$$



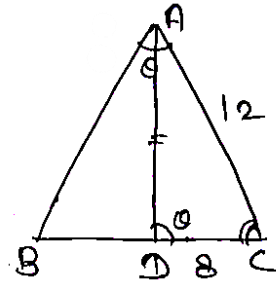
Q.130 In $\triangle ABC$, D is a point on side BC such that $\angle ADC = \angle BAC$. If $CA = 12$ cm, $CD = 8$ cm, then $CB = ?$

Solⁿ:- $\triangle ABC \sim \triangle DAC$ (by A-A-A)

$$\frac{BC}{AC} = \frac{AC}{DC}$$

$$\Rightarrow \frac{BC}{12} = \frac{12}{8}$$

$$\Rightarrow BC = \frac{144}{8} = \boxed{18 \text{ cm}} \text{ Ans}$$



Q.131 $\triangle ABC$, $AB = 20$ cm, $BC = 7$ cm and $CA = 15$ cm. Side BC is produced to D such that $\triangle DAB \sim \triangle DCA$, DC is equal to:

Solⁿ:- $\triangle DAB \sim \triangle DCA$

$$\frac{DA}{DC} = \frac{AB}{AC} = \frac{BD}{AD}$$

$$\frac{DA}{DC} = \frac{20}{15} \quad \left| \quad \frac{20}{15} = \frac{BD}{AD} \right.$$

$$AD = \frac{4}{3} \times DC \text{ --- (1)} \quad AD = \frac{3}{4} \times BD \text{ --- (2)}$$

$$\frac{4}{3} \times DC = \frac{3}{4} \times BD$$

$$\frac{DC}{BD} = \frac{9}{16}$$

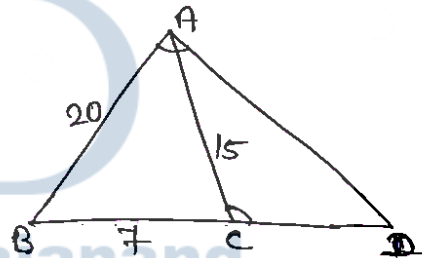
$$\frac{DC}{BC + DC} = \frac{9}{16}$$

$$\frac{DC}{7 + DC} = \frac{9}{16}$$

$$16DC = 63 + 9DC$$

$$7DC = 63$$

$$DC = \boxed{9 \text{ cm}} \text{ Ans}$$



Q.132 In $\triangle ADC$, E and B are the points on the sides AD and AC respectively such that $\angle ABE = \angle ADC$. If $AE = 6\text{cm}$, $BC = 2\text{cm}$, $BE = 3\text{cm}$ and $CD = 5\text{cm}$, then $(AB + DE)$ is equal to:

Solⁿ:- $\triangle AEB \sim \triangle ACD$

$$\frac{AE}{AC} = \frac{AB}{AD} = \frac{EB}{CD}$$

$$\frac{AE}{AC} = \frac{AB}{AD} = \frac{3}{5}$$

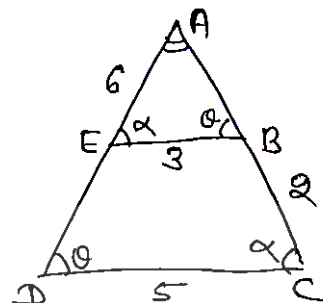
$$\frac{6}{2+AB} = \frac{3}{5} \quad \left| \quad \frac{AB}{6+DE} = \frac{3}{5} \right.$$

$$AB = 8$$

$$DE = \frac{22}{3}$$

$\Rightarrow$

$$AB + DE = 8 + \frac{22}{3} = \boxed{\frac{46}{3}} \text{ Ans}$$



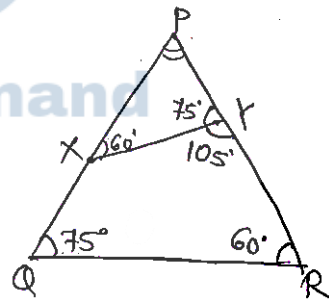
Q.133 In the given figure, if $\frac{QR}{XY} = \frac{14}{9}$ and $PY = 18\text{cm}$, then what is the value of PQ ?

Solⁿ:- $\triangle PXY \sim \triangle PRQ$ (A-A-A Rule)

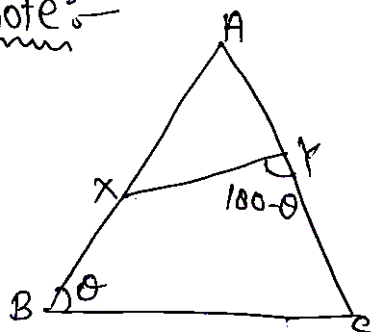
$$\frac{PY}{PQ} = \frac{XR}{QR}$$

$$\frac{18}{PQ} = \frac{9}{14}$$

$$PQ = \frac{18 \times 14}{9} = \boxed{28\text{cm}} \text{ Ans}$$



Note:-



If $\angle B + \angle Y = 180^\circ$ / $\angle X + \angle C = 180^\circ$
then

$$AX \cdot AB = AY \cdot AC$$

Q.134 S and T are points on the side PQ and PR, respectively, of ΔPQR such that $PS \times PR = PQ \times PT$. If $\angle Q = 96^\circ$ and $\angle PST = \angle PRQ + 34^\circ$, then $\angle QPR = ?$

Soln:- $PS \times PR = PQ \times PT$

$$\frac{PS}{PQ} = \frac{PT}{PR}$$

its mean $\Delta PQR \sim \Delta PST$

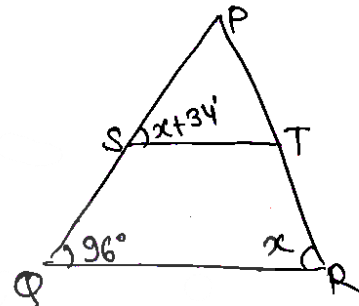
so $\angle S = \angle Q$, $\angle T = \angle R$

$$x + 34^\circ = 96^\circ$$

$$x = 62^\circ$$

$$\angle P = 180 - \angle Q - \angle R$$

$$= 180 - 96 - 62^\circ = \boxed{22^\circ} \text{ Ans.}$$



Q.135 In ΔABC , D and E are the points on AB and AC respectively such that $AD \times AC = AB \times AE$. If $\angle ADE = \angle ACB + 30^\circ$ and $\angle ABC = 78^\circ$ then $\angle A = ?$

Soln:- $AD \times AC = AB \times AE$

$$\frac{AD}{AB} = \frac{AE}{AC}$$

$\Delta ADE \sim \Delta ABC$

$$\angle B = \angle D$$

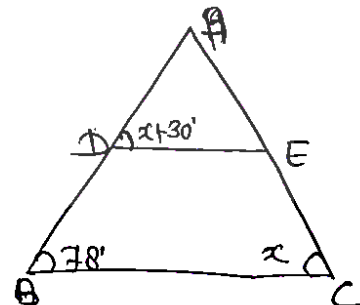
$$x + 30^\circ = 78^\circ$$

$$x = 48^\circ$$

$$\angle A = 180 - \angle B - \angle C$$

$$= 180^\circ - 78^\circ - 48^\circ$$

$$= \boxed{54^\circ} \text{ Ans}$$



Q.136 In a triangle ABC, P and Q are point on AB and AC, respectively, such that AP = 1cm, PB = 3cm, AQ = 1.5cm and CQ = 4.5cm. If the area of ΔAPQ is 12cm^2 , then find the area of BPQC.

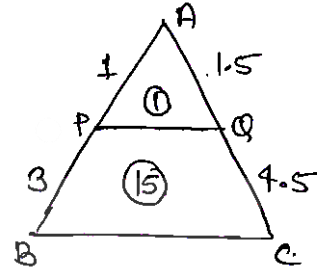
Soln:-

$$\frac{AP}{AB} = \frac{1}{4}$$

$$\frac{\text{ar}(\Delta APQ)}{\text{ar}(\Delta ABC)} = \left(\frac{1}{4}\right)^2 = \frac{1}{16} \xrightarrow{\times 1} 12\text{cm}^2$$

$$\text{ar}(BPQC) = 16 - 1 = 15\text{Unit}$$

$$\Rightarrow \xrightarrow{\times 12} \boxed{180\text{cm}^2} \text{ Ans}$$



Q.137 The mid points of AB and AC of a ΔABC are X and Y, respectively. If $BC + XY = 18$ units, then the value of $BC - XY$ is:

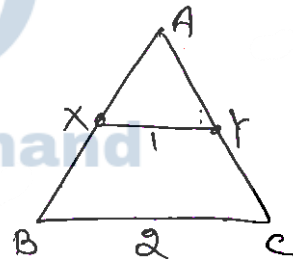
Soln:-

$$BC : XY = 2 : 1$$

$$\begin{aligned} +3\text{Unit} &\rightarrow 18 \\ 1\text{Unit} &\rightarrow 6 \end{aligned}$$

$$BC = 12$$

$$XY = 6 \quad BC - XY = \boxed{6\text{cm}} \text{ Ans}$$



Q.138 In ΔXYZ ; L and M are the mid points of the sides XY and XZ respectively. R is a point on the segment LM, such that $LR : RM = 1 : 2$. If $LR = 3\text{cm}$, then YZ is equal to:

Soln:-

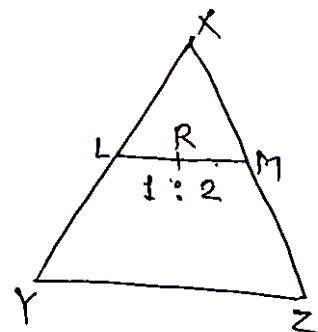
$$LR : RM = 1 : 2$$

$$1\text{Unit} \rightarrow 3\text{cm}$$

$$LM = 3\text{Unit} = 3 \times 3 = 9\text{cm}$$

$$YZ = 2LM$$

$$= 2 \times 9 = \boxed{18\text{cm}} \text{ Ans}$$



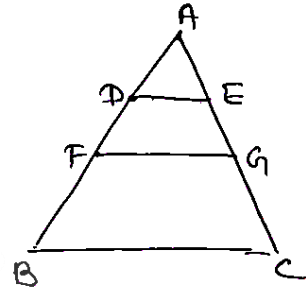
Q.139 In the given triangle, D and E are the mid-points of AF and AG, F and G are the mid-points of AB and AC. If $DE = 2.4 \text{ cm}$ then value of BC is:-

Soln:-

$$FG = 2 \times DE = 2 \times 2.4 = 4.8 \text{ cm}$$

$$BC = 2 \times FG = 2 \times 4.8 = \boxed{9.6 \text{ cm}}$$

Ans



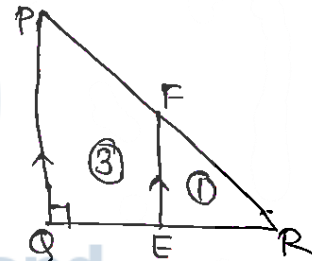
Q.140 A ΔPQR is right angled at Q. E and F are mid points of QR and PR, respectively. What will be the ratio of the area of the quadrilateral PQEF to the area of ΔPQR

Soln:- $EF \parallel QP$

$$\text{so } \frac{EF}{PQ} = \frac{1}{2}$$

$$\frac{\text{ar}(\Delta EFR)}{\text{ar}(\Delta PQR)} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$\frac{\text{ar}(PQEF)}{\text{ar}(\Delta PQR)} = \boxed{\frac{3}{4}} \text{ Ans.}$$

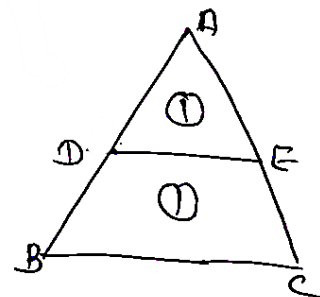


Q.141 In ΔABC , $DE \parallel BC$, where D is a point on AB and E is a point on AC. If DE divides the area of ΔABC into two equal parts, then $DB : AB$ is equal to:

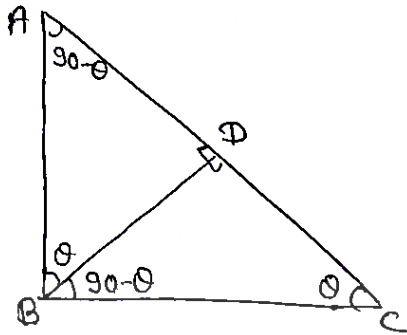
Soln:- $\frac{\text{ar}(\Delta ADE)}{\text{ar}(\Delta ABC)} = \frac{1}{2}$

side $\frac{AD}{AB} = \frac{1}{\sqrt{2}}$

$$\frac{DB}{AB} = \boxed{\frac{\sqrt{2}-1}{\sqrt{2}}} \text{ Ans}$$



Similarity in Right Angle Triangle



i) $\triangle ABD \sim \triangle ABC$

$$\frac{AB}{BC} = \frac{BD}{AB} = \frac{AD}{CD}$$

* $BD^2 = AD \times CD$

* $BD = \frac{AB \times BC}{AC}$

ii) $\triangle ABD \sim \triangle ACB$

$$\frac{AB}{AC} = \frac{BD}{CB} = \frac{AD}{AB}$$

* $AD = \frac{AB^2}{AC}$

iii) $\triangle ADC \sim \triangle ABC$

$$\frac{BD}{AB} = \frac{DC}{BC} = \frac{BC}{AC}$$

* $CD = \frac{BC^2}{AC}$

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Q.142 In a $\triangle ABC$, $\angle BAC = 90^\circ$ and AD is perpendicular to BC where D is a point on BC, If $BD = 4\text{cm}$ and $CD = 5\text{cm}$, then the length of AD is equal to:

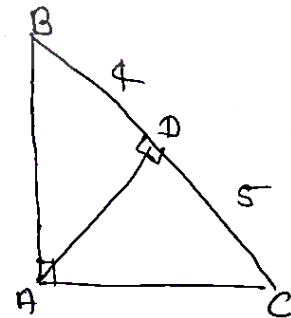
Soln:-

$$AD^2 = BD \times CD$$

$$AD^2 = 4 \times 5$$

$$AD = \sqrt{20}$$

$$= \boxed{2\sqrt{5}} \text{ Ans}$$



Q.143 In a right triangle ABC, right angled at B, altitude BD is drawn to the hypotenuse AC of the triangle. If AD = 6cm, CD = 5cm, then find the value of $AB^2 + BD^2$?

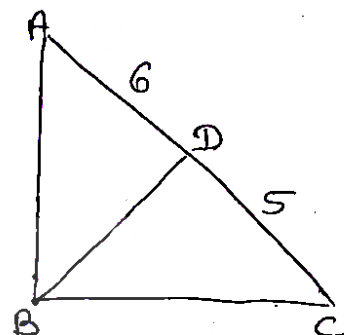
Soln:-

$$AD = \frac{AB^2}{AC}$$

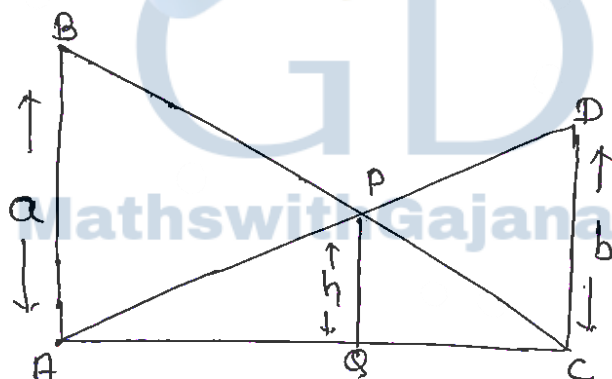
$$\begin{aligned} AB^2 &= AD \times AC \\ &= 6 \times 11 = 66\text{cm} \end{aligned}$$

$$\begin{aligned} \text{and } BD^2 &= AD \times DC \\ &= 6 \times 5 = 30\text{cm} \end{aligned}$$

$$AB^2 + BD^2 = 66 + 30 = \boxed{96\text{cm}} \text{ Ans}$$



Concept:-



$$* \quad \boxed{h = \frac{ab}{a+b}}$$

Proove:- By similarity

In $\triangle ABC$

$$\frac{h}{a} = \frac{CQ}{CA} \quad \text{--- (1)}$$

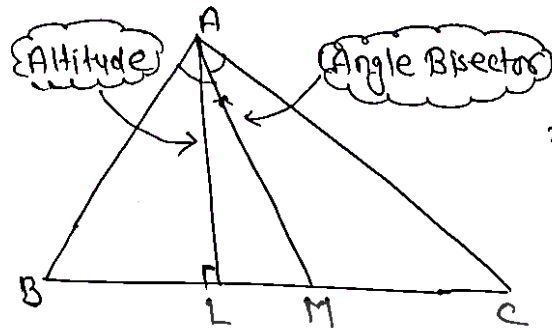
In $\triangle ADC$

$$\frac{h}{b} = \frac{AQ}{AC} \quad \text{--- (2)}$$

$$\text{(1) + (2)} \quad \frac{h}{a} + \frac{h}{b} = \frac{CQ}{CA} + \frac{AQ}{AC} = \frac{CA}{CA}$$

$$\boxed{\frac{1}{a} + \frac{1}{b} = \frac{1}{h}} \Rightarrow h = \frac{ab}{a+b}$$

Concept:- Angle Between Altitude and Angle Bisector:-



$$* \angle MAL = \frac{1}{2} |\angle B - \angle C| *$$

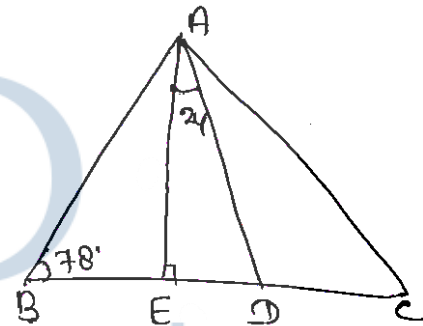
Q.144 $\triangle ABC$, $\angle B = 78^\circ$, AD is bisector of $\angle A$ meeting BC at D, and $AE \perp BC$ at E. If $\angle DAE = 24^\circ$, then the measure of $\angle ACB$ is:

Soln:-

$$\angle DAE = \frac{1}{2} (\angle B - \angle C)$$

$$24^\circ = \frac{1}{2} (78^\circ - \angle C)$$

$$\begin{aligned} \angle C &= 78^\circ - 48^\circ \\ &= \boxed{30^\circ} \text{ Ans} \end{aligned}$$



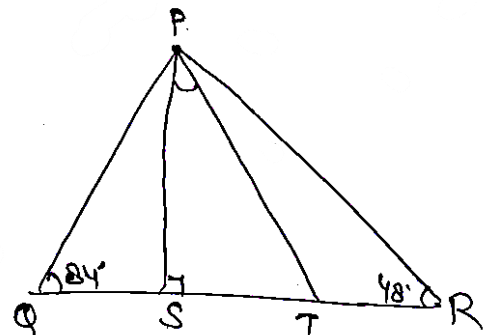
Q.145 In $\triangle PQR$, $\angle Q = 84^\circ$ and $\angle R = 48^\circ$, $PS \perp QR$ at S and the bisector of $\angle P$ meets QR at T. What is the measure of $\angle SPT$?

Soln:-

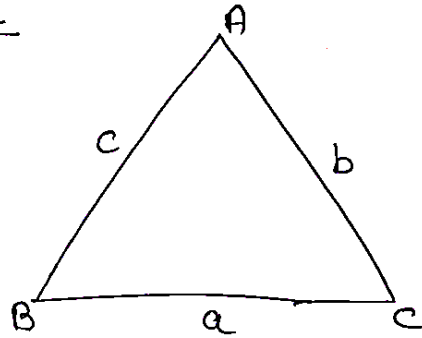
$$\angle SPT = \frac{\angle Q - \angle R}{2}$$

$$= \frac{84^\circ - 48^\circ}{2}$$

$$= \boxed{18^\circ} \text{ Ans}$$



Sine Rule:-



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k(\text{Constant}) = 2R$$

$R = \text{Circumradius}$

$$a = k \sin A, b = k \sin B, c = k \sin C$$

$$a : b : c = \sin A : \sin B : \sin C$$

Proof:- In $\triangle ADB$

$$\sin B = \frac{h}{c}$$

$$\Rightarrow h = c \sin B \quad \text{--- (1)}$$

In $\triangle ADC$

$$\sin C = \frac{h}{b}$$

$$\Rightarrow h = b \sin C \quad \text{--- (2)}$$

from (1) & (2)

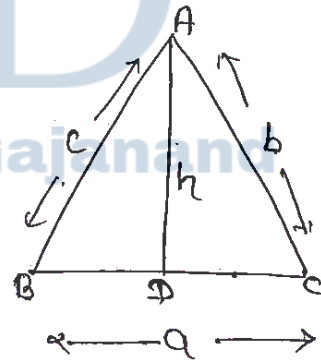
$$c \sin B = b \sin C$$

$$\left[\frac{\sin B}{b} = \frac{\sin C}{c} \right] \text{ Similarly } \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Area of $\triangle$:-

$$\frac{1}{2} \times a \times h$$

$$\Rightarrow \frac{1}{2} a c \sin B = \frac{1}{2} a b \sin C$$



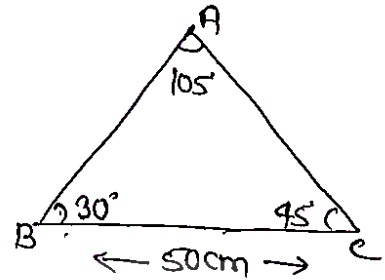
Q.146 In Triangle ABC, $\angle B = 30^\circ$ and $\angle C = 45^\circ$ if $BC = 50\text{cm}$ then find the length of AB?

Soln:- $\sin 105^\circ = \sin(180 - 75)$
 $= \sin 75^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}}$

Apply sine Rule:-

$$\frac{AB}{\sin 45^\circ} = \frac{50}{\sin 105^\circ}$$

$$AB = \frac{50 \times \frac{1}{\sqrt{2}}}{\frac{\sqrt{3}+1}{2\sqrt{2}}} = \boxed{50(\sqrt{3}-1)} \text{ Ans}$$



Q.147 O is a point in the interior of $\triangle ABC$ such that $OA = 12\text{cm}$, $OC = 9\text{cm}$, $\angle AOB = \angle BOC = \angle COA$ and $\angle ABC = 60^\circ$. What is the length of OB?

Soln:- In $\triangle AOB$

$$\frac{\sin \theta}{OB} = \frac{\sin(60-\theta)}{AO}$$

$$\Rightarrow \frac{\sin \theta}{\sin(60-\theta)} = \frac{OB}{AO} \quad \text{--- (1)}$$

In $\triangle BOC$

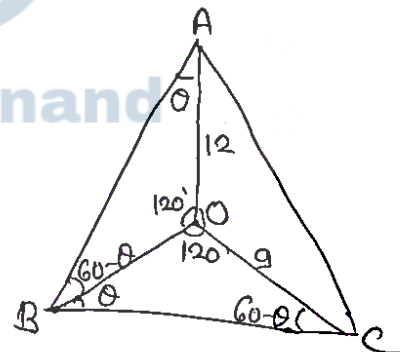
$$\frac{\sin \theta}{\sin(60-\theta)} = \frac{9}{OB} \quad \text{--- (2)}$$

from (1) & (2)

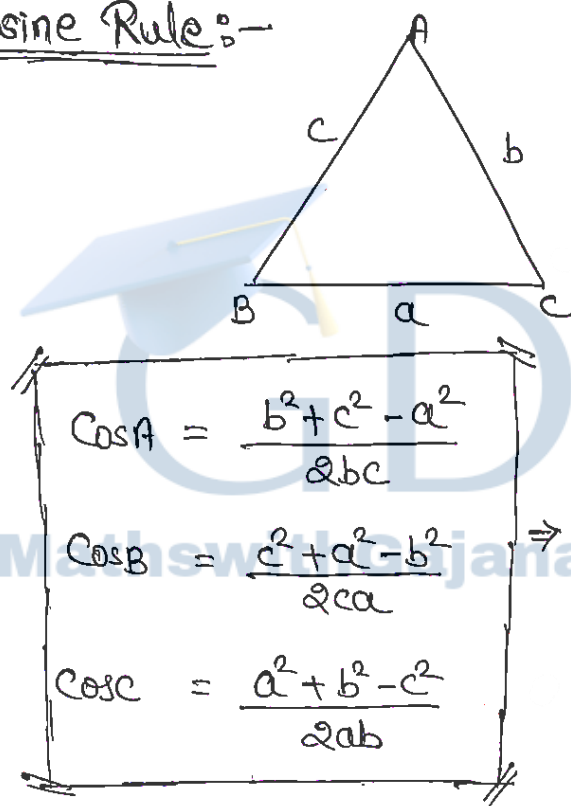
$$\frac{OB}{12} = \frac{9}{OB}$$

$$OB^2 = 108$$

$$OB = \boxed{6\sqrt{3}} \text{ Ans}$$



Cosine Rule:-

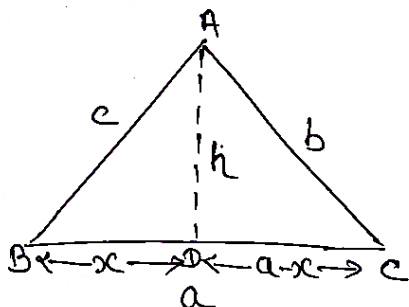


$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Rightarrow b^2 = c^2 + a^2 - 2ca \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Proof:-



$$\text{In } \triangle ABD \quad h^2 = c^2 - x^2 \quad \text{--- (1)}$$

$$\text{In } \triangle ADC \quad h^2 = b^2 - (a-x)^2 \quad \text{--- (2)}$$

from (1) & (2)

$$c^2 - x^2 = b^2 - (a-x)^2$$

$$c^2 - x^2 = b^2 - a^2 - x^2 + 2ax$$

$$c^2 + a^2 - b^2 = 2ax \quad \left[\frac{x}{c} = \cos B \right]$$

$$\left[\cos B = \frac{c^2 + a^2 - b^2}{2ac} \right]$$

Q.148 $\triangle LMN$, $LM = 5\sqrt{2}$ cm and $\angle LMN = 135^\circ$, what is the length of MN ?

Soln

$$\cos 135^\circ = \frac{(5\sqrt{2})^2 + x^2 - (13)^2}{2 \times 5\sqrt{2} \times x}$$

$$\downarrow$$

$$\cos(90+45) = -\sin 45^\circ$$

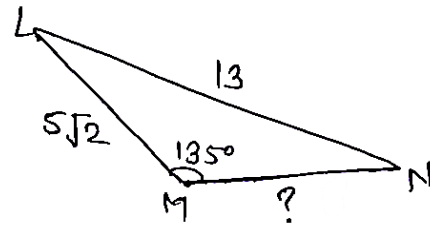
$$\downarrow$$

$$-\frac{1}{\sqrt{2}} = \frac{50 + x^2 - 169}{10\sqrt{2} \times x}$$

$$-10x = x^2 - 119$$

$$x^2 + 10x - 119 = 0$$

Solve $x = -17, 7$ $MN = 7\text{cm}$ Ans



Q.149 In $\triangle ABC$, $AB = AC$ and D is a point on BC . If $BD = 5$ cm, $AB = 12$ cm and $AD = 8$ cm, then the length of CD is:

Soln:-

$$\cos \theta = \frac{12^2 + 5^2 - 8^2}{2 \times 12 \times 5} = \frac{12^2 + x^2 - 8^2}{2 \times 12 \times x}$$

$$= \frac{144 + 25 - 64}{5} = \frac{144 + x^2 - 64}{x}$$

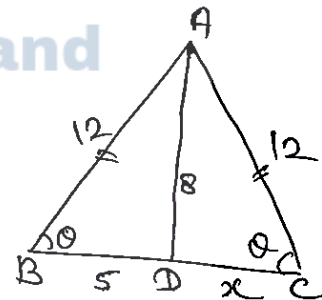
$$= \frac{105}{5} = \frac{80 + x^2}{x}$$

$$\Rightarrow x^2 - 21x + 80 = 0$$

Solve $x = 16, x = 5$
 $\times$ not possible

because at $x=5$, $AD \perp BC$ but there is no right angle triangle.

So. $x = 16\text{cm}$ Ans



Q.150 In $\triangle ABC$, Perimeter of $\triangle ABD =$ Perimeter of $\triangle BCD$ and $AB = 60\text{cm}$, $BC = 80\text{cm}$ and $AC = 100\text{cm}$. Then find BD .

Soln:-

Perimeter of $\triangle ABD =$ Perimeter of $\triangle BCD$

$$60 + x + y = 80 + y + (100 - x)$$

$$x = 60$$

$$CD = 100 - 60 = 40$$

Let $\angle ACB = \theta$

$$\text{In } \triangle ABC \rightarrow \cos \theta = \frac{80}{100} = \frac{4}{5}$$

Apply Cosine Rule in $\triangle CBD$:-

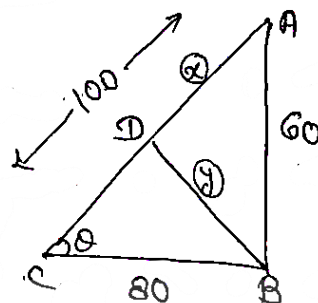
$$\cos \theta = \frac{80^2 + 40^2 - y^2}{2 \times 40 \times 80}$$

$$\frac{4}{5} \times 2 \times 40 \times 80 = 6400 + 1600 - y^2$$

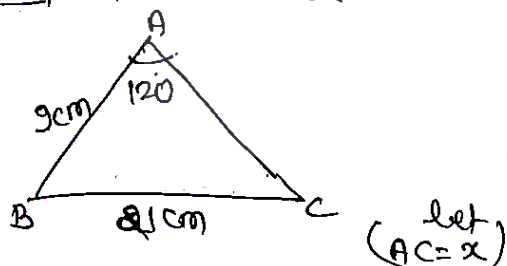
$$\Rightarrow 5120 - 8000 = -y^2$$

$$y^2 = 2880$$

$$y = \boxed{24\sqrt{5}\text{cm}} \text{ Ans}$$



Q.151 Find the Area



$$\text{Soln: } \cos 120^\circ = \frac{9^2 + x^2 - 21^2}{2 \cdot 9 \cdot x}$$

$$-\frac{1}{2} = \frac{81 + x^2 - 441}{2 \times 9 \times x}$$

$$x^2 + 9x - 360 = 0$$

$$x = \underline{15\text{cm}}$$

$$1 \text{ Area} = \frac{1}{2} \times 9 \times 15 \sin 120$$

$$1 = \frac{1}{2} \times 135 \times \frac{\sqrt{3}}{2}$$

$$1 = \boxed{\frac{135\sqrt{3}}{4}} \text{ Ans}$$

Regular Polygon

$n \rightarrow$ numbers of sides

i> Sum of Interior Angles = $(n-2) \times 180^\circ$

ii> Each angle of a Regular Polygon = $\frac{(n-2) \times 180^\circ}{n}$

iii> Sum of all the Exterior Angles = 360°

iv> Each exterior angle of a Regular polygon = $\frac{360^\circ}{n}$

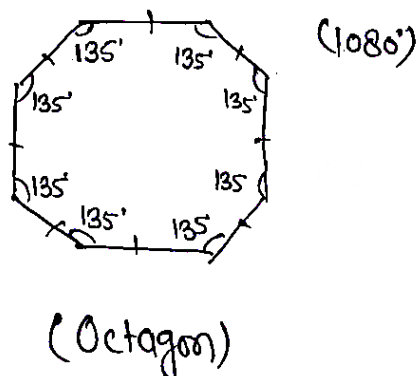
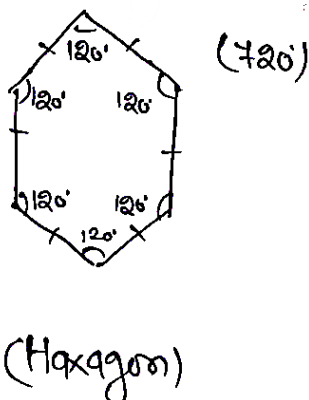
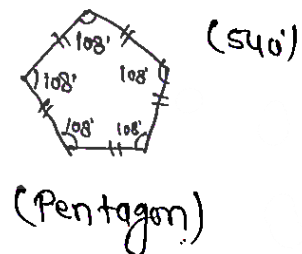
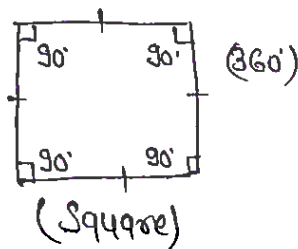
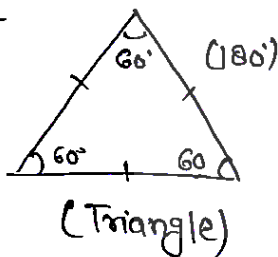
v> Each Interior Angle $\oplus$ each exterior angle = 180°

vi> No. of Diagonals = $\frac{n(n-3)}{2}$

Definition of Regular Polygon:-

All the interior angles and the sides are equal.

Ex:-



Q.152 If a regular polygon has 16 sides, then what is the measure of each interior angle?

Soln:- Each interior angle = $\frac{(n-2) \times 180}{n}$
 $= \frac{(16-2) \times 180}{16} = \boxed{157\frac{1}{2}^\circ}$ Ans

Q.153 The sum of the interior angles of a regular polygon is 1260° . What is the difference between an exterior angle and an interior angle of the polygon.

Soln:- $(n-2) \times 180^\circ = 1260$
 $n = 9$
 Exterior angle = $\frac{360^\circ}{9} = 40^\circ$
 Interior angle = $180^\circ - 40^\circ = 140^\circ$
 Difference = $140^\circ - 40^\circ = \boxed{100^\circ}$ Ans.

Q.154 The interior angle of a regular polygon exceeds its exterior angle by 90° . The number of sides of the Polygon is:

Soln:- **Q.154** Interior Angle = x
 Exterior Angle = $(180^\circ - x)$
 ($\because$ its sum should be 180°)
 $x = (180^\circ - x) + 90^\circ$
 $2x = 270^\circ$
 $x = 135^\circ$
 exterior = $180 - 135 = 45^\circ$
 $n = \frac{360^\circ}{45} = \boxed{8}$ Ans

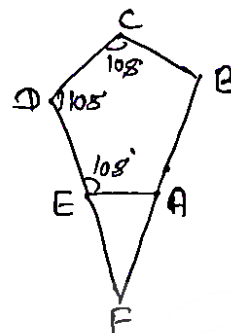
Q.154
 $\frac{(n-2) \times 180}{n} - \frac{360^\circ}{n} = 90^\circ$
 $180 - \frac{360^\circ}{n} - \frac{360^\circ}{n} = 90^\circ$
 $90 = \frac{720^\circ}{n}$
 $n = \boxed{8}$ Ans

Q.155 The sides BA and DE of a regular pentagon are produced to meet at F. What is the measure of $\angle EFA$?

Soln:- Interior angle = 108°

$$\angle AEF = 180^\circ - 108^\circ = 72^\circ = \angle EAF$$

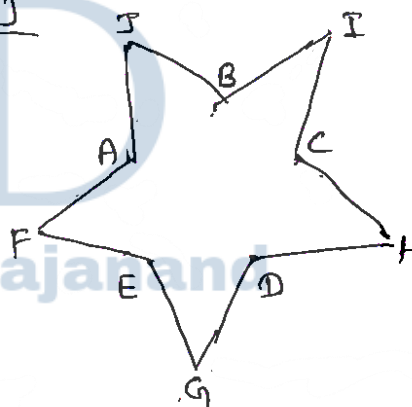
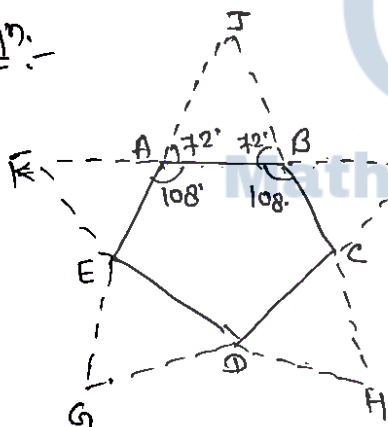
$$\begin{aligned}\angle EFA &= (180 - 72 - 72) \\ &= \boxed{36^\circ} \text{ Ans}\end{aligned}$$



Q.156 ABCDE is a regular pentagon its sides are extended as show in the figure. The value of

$$\frac{\angle ABC + 2\angle EGD + 3\angle BAJ}{6}$$

Soln:-



$$\angle EGD = (180 - 72 - 72)^\circ = 36^\circ$$

$$\angle BAJ = 72^\circ$$

$$\angle ABC = 108^\circ$$

$$\Rightarrow \frac{72^\circ + 2 \times 36^\circ + 3 \times 72^\circ}{6}$$

$$\Rightarrow \boxed{66^\circ} \text{ Ans}$$

Q.157 Find the number of diagonals of a regular polygon whose interior angles sum is 2700° ?

Soln

$$\begin{aligned} (n-2) \times 180^\circ &= 2700 \\ n-2 &= 15 \\ n &= 17 \end{aligned} \quad \left| \quad \begin{aligned} \text{diagonals} &= \frac{n(n-3)}{2} \\ &= \frac{17(17-3)}{2} \\ &= \boxed{119} \text{ Ans} \end{aligned}$$

Q.158 If each interior angle of a regular polygon is $(128\frac{4}{7})^\circ$, then what is the sum of the number of its diagonals and the number of its sides?

Soln:- Interior angle $= (128\frac{4}{7})^\circ$

Exterior angle $= 180^\circ - (128\frac{4}{7})^\circ = 51\frac{3}{7} = \frac{360}{7}$

$n = \frac{360^\circ}{\frac{360}{7}} = 7$

no. of diagonals $= \frac{7(7-3)}{2} = 14$

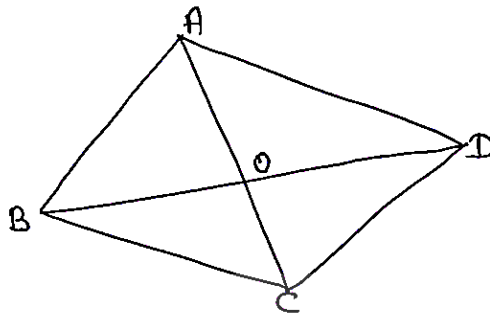
sum $= 14 + 7 = \boxed{21} \text{ Ans}$

Q.159 If one of the interior angle of a regular polygon is $\frac{15}{16}$ times of one of the interior angles of a regular decagon then find the number of diagonal of the polygon?

Soln

$$\begin{aligned} \frac{(n-2) \times 180}{n} &= \frac{15}{16} \left(\frac{(10-2) \times 180}{10} \right) \\ \frac{(n-2)}{n} &= \frac{3}{4} \\ 4n-8 &= 3n \\ n &= \boxed{8} \end{aligned} \quad \left| \quad \begin{aligned} \text{diagonal} &= \frac{n(n-3)}{2} \\ &= \frac{8(5)}{2} \\ &= \boxed{20} \text{ Ans} \end{aligned}$$

Quadrilateral



1. $\text{Area of } \triangle ABD : \text{Area of } \triangle BCD = AO : OC$

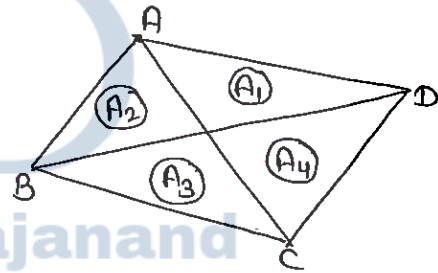
$\text{Area of } \triangle ABC : \text{Area of } \triangle ACD = BO : OD$

2.

$A_1 \times A_3 = A_2 \times A_4$

Proof:- $A_1 = \frac{1}{2} a \times b \sin \theta$
 $A_2 = \frac{1}{2} a \times d \sin(180 - \theta)$

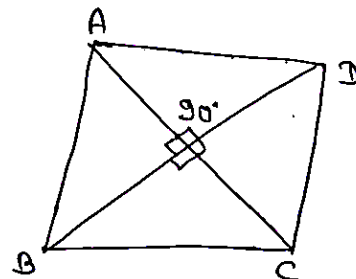
$A_3 = \frac{1}{2} d \times c \sin \theta$
 $A_4 = \frac{1}{2} c \times b \sin(180 - \theta)$



$A_1 \times A_3 = A_2 \times A_4$

3. If Diagonals makes 90° Angle / Diagonals are perpendicular

$AB^2 + CD^2 = AD^2 + BC^2$



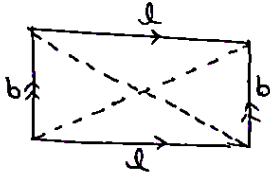
Rectangle

Square

Parallelogram

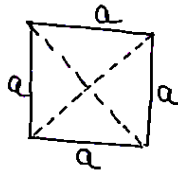
Rhombus

1



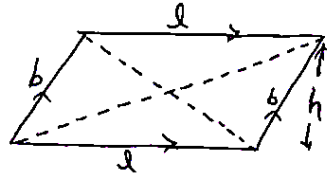
$$d_1 = d_2$$

$$d_1 \neq d_2$$



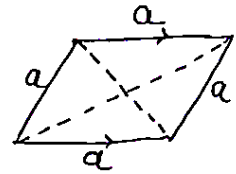
$$d_1 = d_2$$

$$d_1 \perp d_2$$



$$d_1 \neq d_2$$

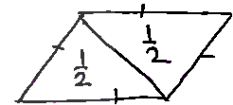
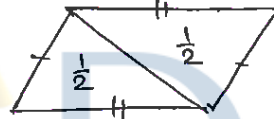
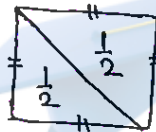
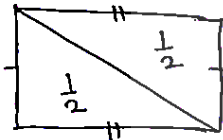
$$d_1 \neq d_2$$



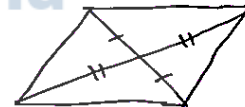
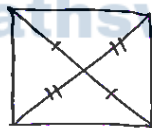
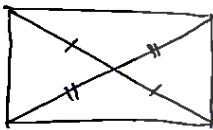
$$d_1 \neq d_2$$

$$d_1 \perp d_2$$

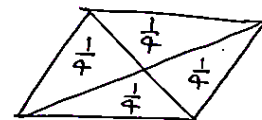
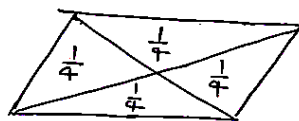
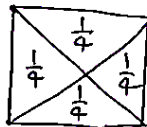
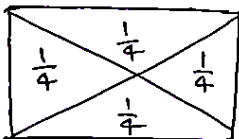
2 The diagonal bisects the Area of all four digram,



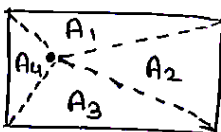
3 Two diagonals bisect each other in all digram.



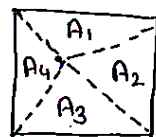
4 Both diagonals divides the Area in 4 equal parts,



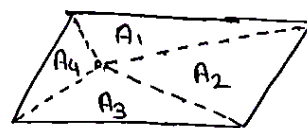
5



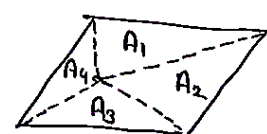
$$A_1 + A_3 = A_2 + A_4$$



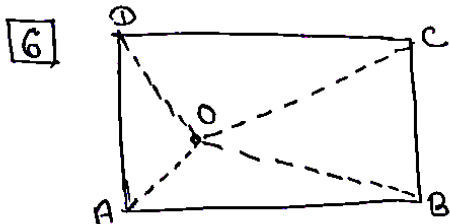
$$A_1 + A_3 = A_2 + A_4$$



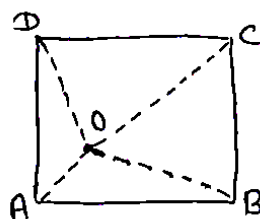
$$A_1 + A_3 = A_2 + A_4$$



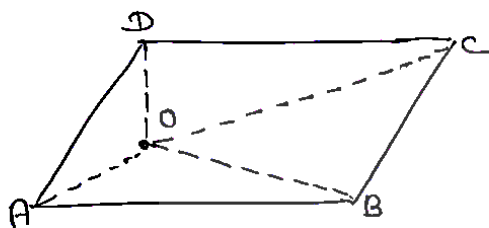
$$A_1 + A_3 = A_2 + A_4$$



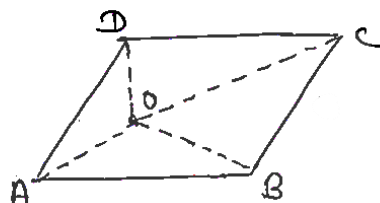
$$OA^2 + OC^2 = OB^2 + OD^2$$



$$OA^2 + OC^2 = OB^2 + OD^2$$

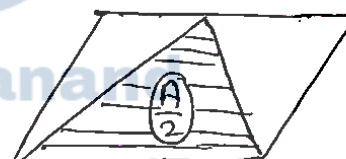
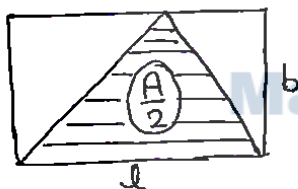


$$OA^2 + OC^2 \neq OB^2 + OD^2$$



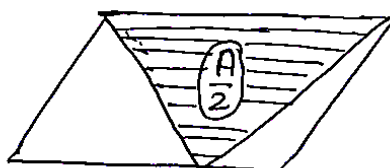
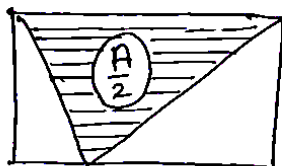
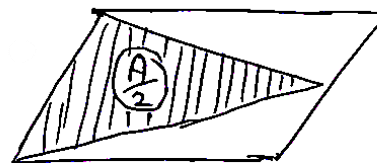
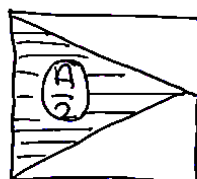
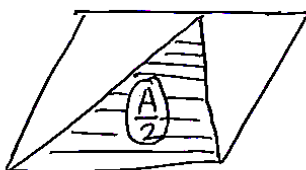
$$OA^2 + OC^2 \neq OB^2 + OD^2$$

7 If Total Area is (A)

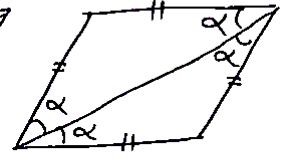
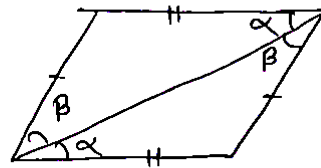
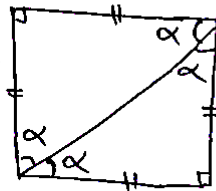
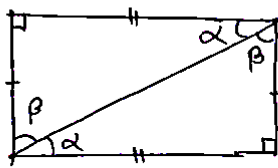


$$A = l \times b = (A)$$

$$A_1 = \frac{1}{2} \times l \times b = \left(\frac{A}{2}\right)$$

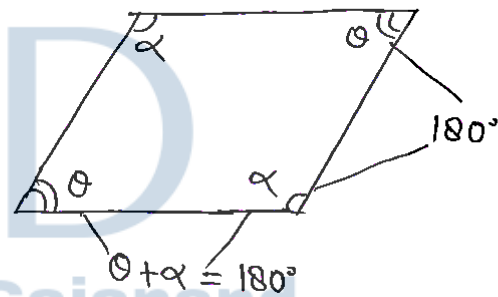
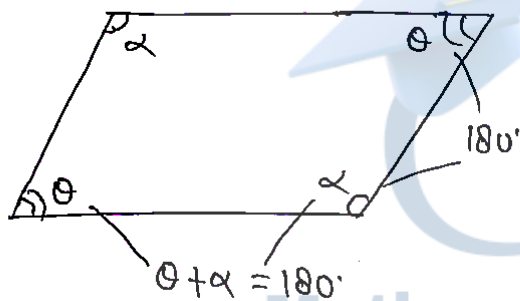


8 Diagonals are Angle Bisector in Square and Rhombus

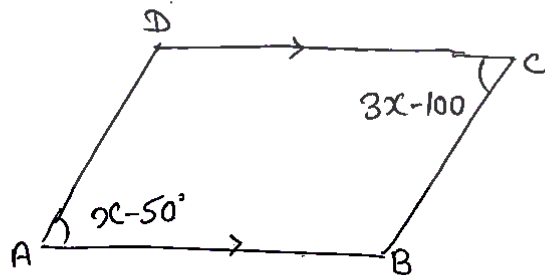


9 In Parallelogram and Rhombus

- (i) opposite Angle are same
- (ii) Sum of adjacent angles = 180°



Ex:-



$$x = ?$$

opposite Angle are equal

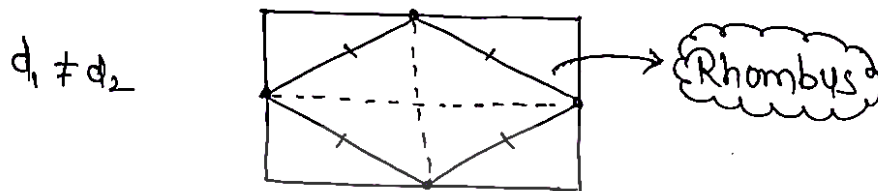
$$x - 50 = 3x - 100$$

$$2x = 50$$

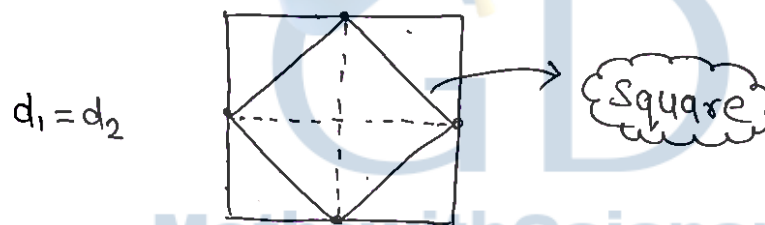
$$\boxed{x = 25} \text{ Ans}$$

Important Point

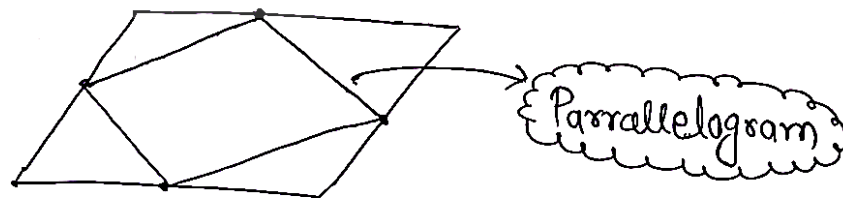
- (1) Shape formed by joining the midpoint of four sides of a rectangle



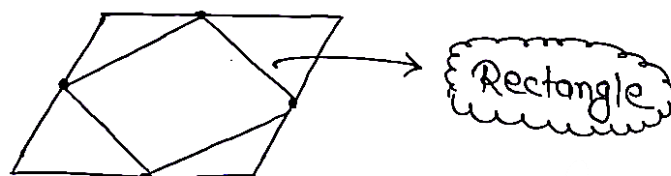
- (2) Shape formed by joining the mid-points of four sides of a Square



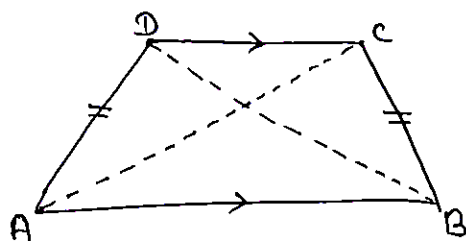
- (3) Shape formed by joining the mid-point of four sides of a parallelogram



- (4) Shape formed by joining the mid-point of four sides of a Rhombus



Isosceles Trapezium



- * $AB \parallel DC$
- * $AD = BC$
- * $AC = BD$

* Every isosceles trapezium is cyclic Quadrilateral.

Q.160 ABCD is a trapezium, such that $AB = CD$ and $AD \parallel BC$. $AD = 10\text{cm}$ and $BC = 18\text{cm}$. If the area of ABCD is 70cm^2 then what is the value (in cm) of CD?

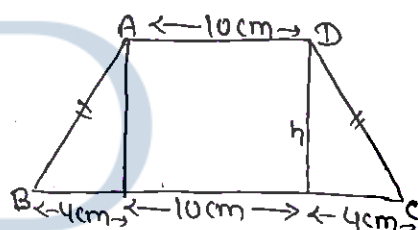
Soln:- Area = $\frac{1}{2}(AD + BC)h$

$$\Rightarrow 70 = \frac{1}{2}(10 + 18)h$$

$$\boxed{h = 5}$$

by Pythagoras theorem

$$CD = \sqrt{h^2 + (4)^2} = \sqrt{(5)^2 + (4)^2} = \boxed{\sqrt{41}\text{ cm}}$$



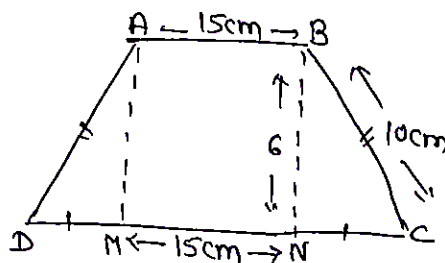
Q.161 In trapezium ABCD, AB and DC are parallel to each other with a perpendicular distance of 8m between them. Also, $AD = BC = 10\text{m}$ and $AB = 15\text{m} < (DC)$. What is the Perimeter (in m) of the trapezium ABCD?

Soln:- $DM = NC$

$$NC = \sqrt{(10)^2 - (6)^2} = 8\text{cm}$$

by Pythagoras theorem

$$\begin{aligned} CD &= DM + MN + NC \\ &= 8 + 15 + 8 = \boxed{31\text{cm}} \end{aligned}$$



$$\begin{aligned} \text{Perimeter of ABCD} &= AB + BC + CD + DA \\ &= 15 + 10 + 31 + 10 = \boxed{66\text{cm}} \text{ Ans} \end{aligned}$$

Trapezium

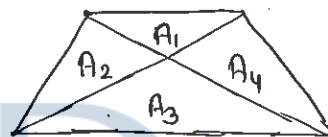
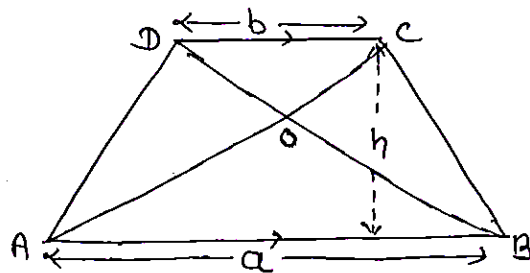
→ $\text{Area} = \frac{1}{2} (a+b)h$

→ $AB \parallel CD$

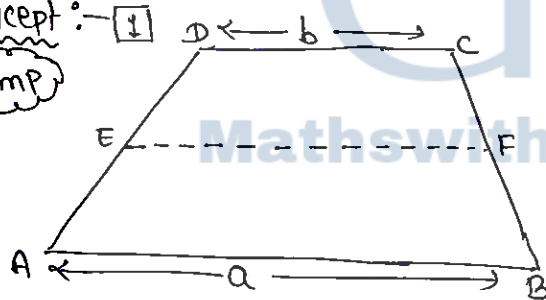
→ If $AD = BC$
then it is isocceles trapezium

→ $\text{Ar}(\triangle AOD) = \text{Ar}(\triangle BOC)$

→ $A_1 \times A_3 = A_2 \times A_4$

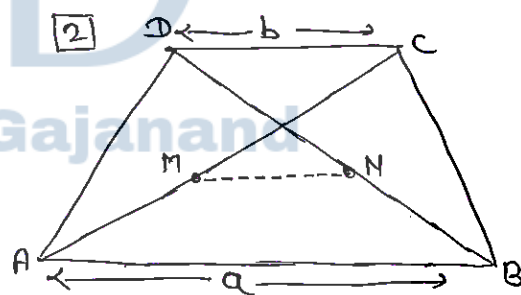


Concept:-
Imp



(E and F are mid-point of AD and BC)

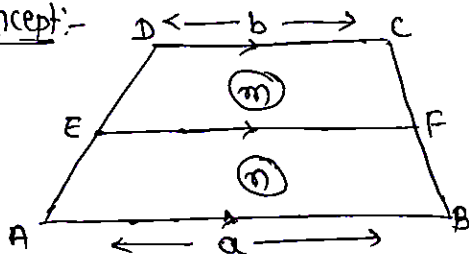
* $EF = \frac{a+b}{2}$



(M and N are mid-point of AC and BD)

* $MN = \frac{a-b}{2}$

Concept:-



$\frac{\text{ar } \square DCFE}{\text{ar } \square EFBA} = \frac{m}{n}$

* $EF = \sqrt{\frac{ma^2 + nb^2}{m+n}}$

Case → If $m:n = 1 \Rightarrow EF = \frac{a^2 + b^2}{2}$

162 The two parallel sides of a trapezium are 27cm and 13cm, respectively. If the height of the trapezium is 7cm, then what is the area (in m^2)?

Soln:-

$$\begin{aligned} \text{Area} &= \frac{1}{2}(27+13) \times 7 \\ &= \frac{1}{2} \times 40 \times 7 \\ &= 140 \text{ cm}^2 \longrightarrow \boxed{0.0140 \text{ m}^2} \text{ Ans.} \end{aligned}$$

163 The difference between the length of two parallel sides of a trapezium is 12cm. The perpendicular distance between these two sides is 60cm. If the area of the trapezium is 1380 cm^2 , then find the length (in cm) of each parallel side?

Soln:-

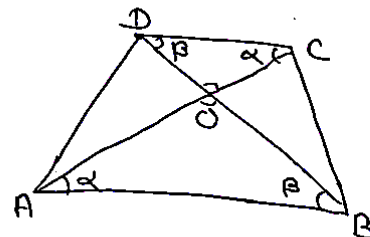
$$\begin{aligned} \frac{1}{2}(a+b) \times 60 &= 1380 \\ a+b &= 46 \\ a-b &= 12 \end{aligned} \quad \left\{ \begin{array}{l} a = \frac{46+12}{2} \\ b = \frac{46-12}{2} \end{array} \right.$$

$$a = \boxed{29} \quad b = \boxed{17}$$

Q.164 ABCD is a trapezium in which AB is parallel to CD and $AB = 4(CD)$. The diagonals of the trapezium intersect at O. What is the ratio of area of triangle DCO to the area of triangle ABO?

Soln:-

$$\begin{aligned} \triangle DOC &\sim \triangle BOA \\ \frac{\text{Area of } \triangle DOC}{\text{Area of } \triangle AOB} &= \frac{(DC)^2}{(AB)^2} \end{aligned}$$



$$= \frac{(1)^2}{(4)^2}$$

$$= \boxed{\frac{1}{16}}$$

$$\boxed{1:16} \text{ Ans}$$

165 ABCD is a trapezium in which $AB \parallel DC$ and its diagonals intersect at P. If $AP = (3x-1)\text{cm}$, $PC = (5x-3)$, $BP = (2x+1)\text{cm}$ and $PD = (6x-5)\text{cm}$, then the length of DB is:

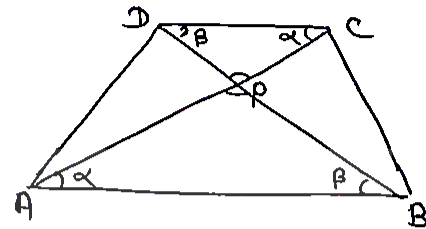
Soln:-

$$\triangle DPC \sim \triangle BPA$$

$$\frac{DP}{PC} = \frac{BP}{PA}$$

$$\frac{(6x-5)}{(5x-3)} = \frac{(2x+1)}{(3x-1)}$$

Put $x=2$ $\frac{7}{7} = \frac{5}{5}$ satisfy ✓



$$\begin{aligned} DB &= DP + PB \\ &= (6x-5) + (2x+1) \\ &= 8x-4 \\ &= 8 \times 2 - 4 = \boxed{12} \text{ Ans} \end{aligned}$$

166 ABCD is a quadrilateral in which $AB \parallel DC$. E and F are the midpoint of the diagonals AC and BD, respectively. If $AB = 18\text{cm}$ and $CD = 6\text{cm}$, then $EF = ?$

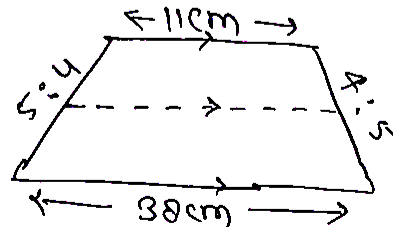
Soln:- $EF = \frac{a-b}{2} = \frac{18-6}{2} = \boxed{6\text{cm}} \text{ Ans}$

167 In the given trapezium ABCD, $DC \parallel AB$, P and Q are points on AD and BC respectively, $AP:PD = 5:4$, $BQ:QC = 5:4$ if $DC = 11\text{cm}$, $AB = 38\text{cm}$ then find the value of PQ?

Soln:-

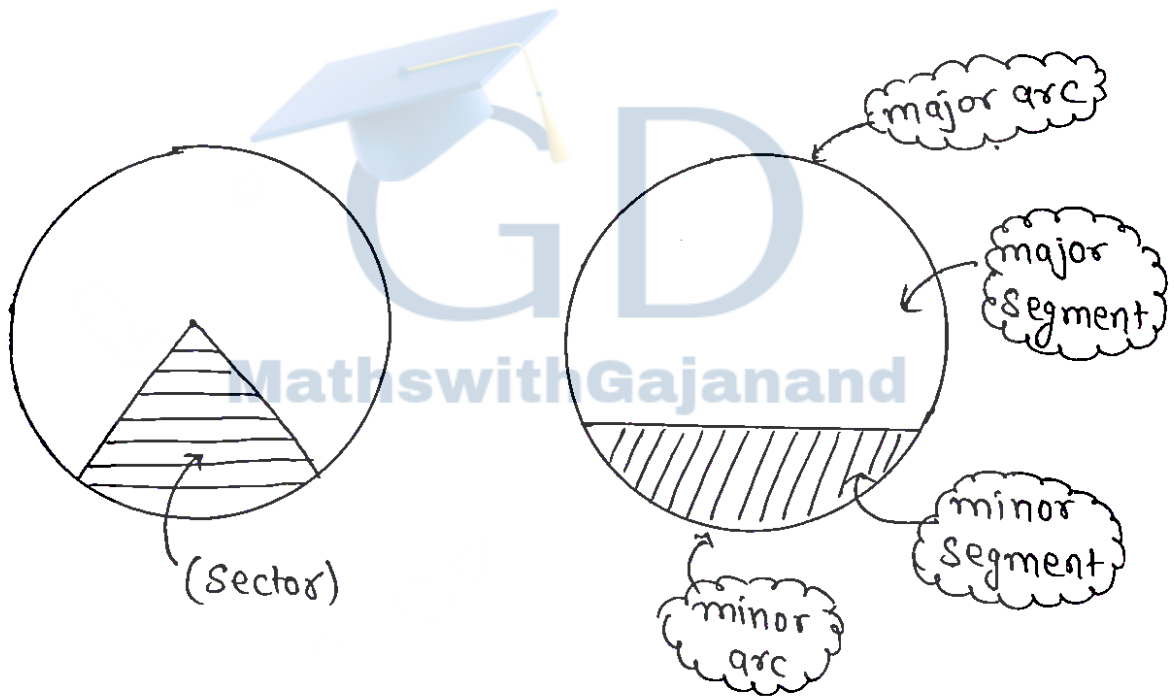
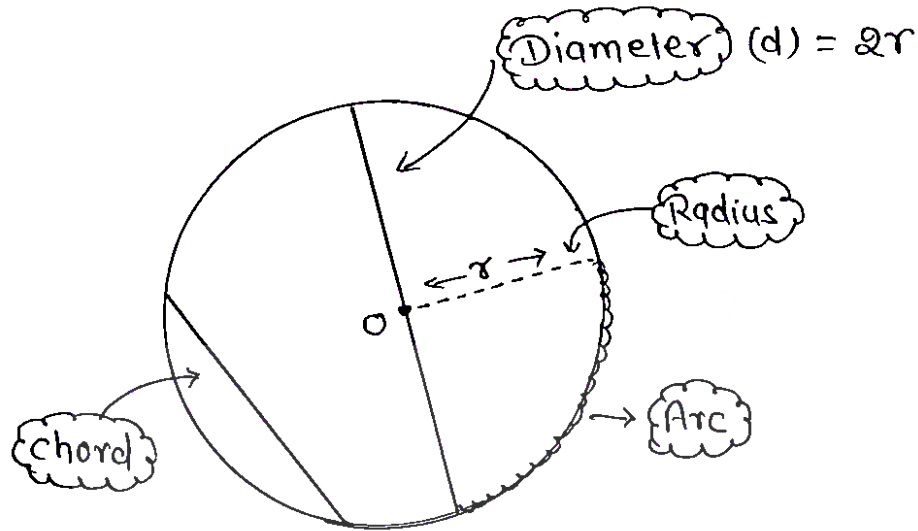
$$\begin{aligned} PQ &= \frac{5 \times 11 + 4 \times 38}{5+4} \\ &= \frac{152 + 55}{9} \end{aligned}$$

$$= \frac{207}{9} = \boxed{23\text{cm}} \text{ Ans}$$

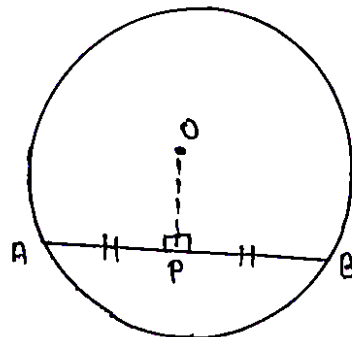


Circle

@mathswithgajanand



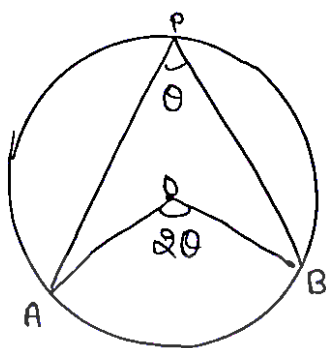
⇒ A perpendicular from the center of circle to the chord always bisect the chord.



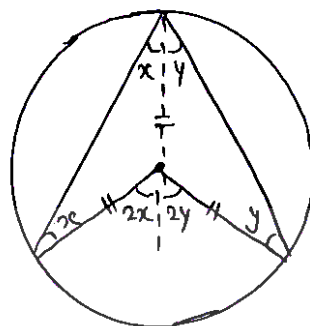
$$OP \perp AB$$

$$AP = PB$$

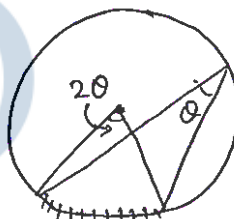
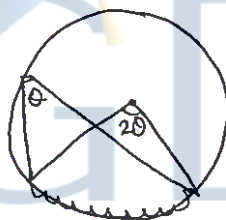
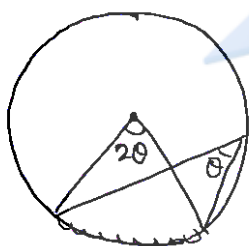
⇒ Angle formed at the center by any arc is twice the angle formed at the circumference.



Proof:-

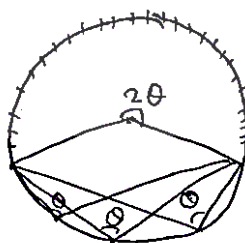
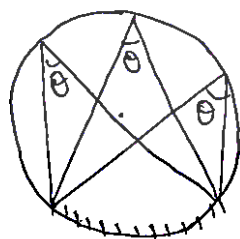


Imp * $\angle AOB = 2\angle APB$



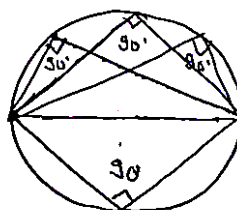
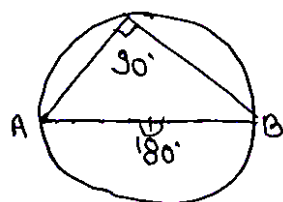
MathswithGajanand

⇒ Same arc make same angle on circumference.

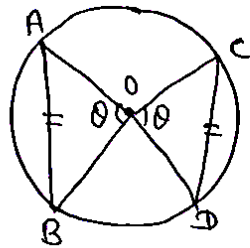


Imp

⇒ Angle formed by diameter at circumference is 90°



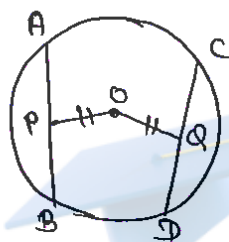
⇒ Equal chord subtended equal angle at the centre of circle,



$$AB = CD$$

$$\angle AOB = \angle COD$$

⇒ Equal chord are equidistance from the centre of circle,



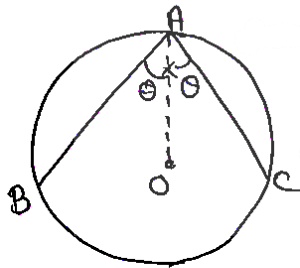
if chord

$$AB = CD$$

$$\text{then } \boxed{\text{arc}(AB) = \text{arc}(CD)}$$

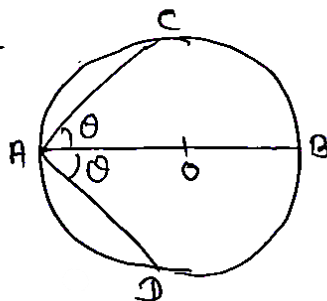
$$OP = OQ$$

⇒ If two chord $AB = AC$ then the line joining Point A to the center will be angle bisector of A



the reverse is also true

Example :-



(A) $AC < AD$

(B) $AC > AD$

(C) $AC \perp AD$

✓ (D) $AC = AD$

168 The radius of a circle is 15cm and the length of one chord of the circle is 20cm. What is the distance of the chord from the centre of the circle?

Soln:- radius (OB) = 15cm

chord (AB) = 20cm

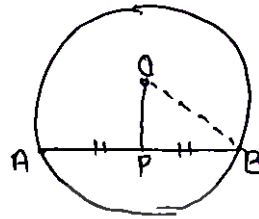
(PB) = 10cm

$$OP^2 = OB^2 - PB^2$$

$$OP^2 = (15)^2 - (10)^2$$

$$OP^2 = 225 - 100$$

$$OP = \sqrt{125} = \boxed{5\sqrt{5}\text{cm}} \text{ Ans}$$



169 A chord of length 24cm is at a distance of 5cm. from the centre of a circle. What is its area?

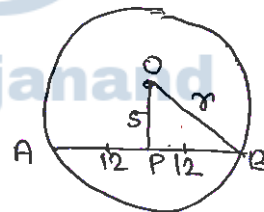
Soln:-

$$\text{radius} = \sqrt{(5)^2 + (12)^2}$$

$$r = 13\text{cm}$$

$$\text{Area} = \pi r^2$$

$$= \frac{22}{7} \times 169 = \boxed{531.14\text{cm}^2} \text{ Ans}$$



170 A circle with centre O has radius 15cm. D is a point on the circle such that a 24cm long chord AB is bisected by OD at point C. Find the length of CD (in cm).

Soln:-

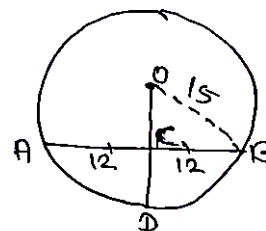
$$OC = \sqrt{(15)^2 - (12)^2}$$

$$= \sqrt{225 - 144} = \sqrt{81}$$

$$OC = 9$$

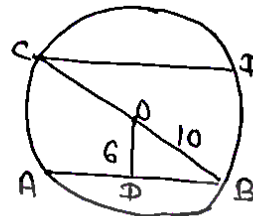
$$CD = OD - OC$$

$$= 15 - 9 = \boxed{6\text{cm}} \text{ Ans}$$



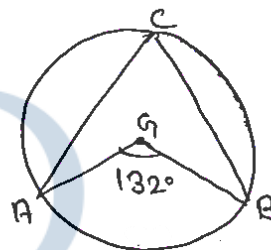
[171] In a circle of diameter 20cm, chord AB and CD are parallel to each other. BC is diameter. If AB is 6cm from the centre of the circle, what is the length (in cm) of chord CD?

Soln:- $BD = \sqrt{(10)^2 - (6)^2} = 8\text{cm}$
 $AB = 2 \times BD = 2 \times 8 = 16\text{cm}$
 $AB = CD = \boxed{16\text{cm}}$ Ans



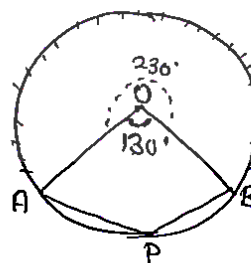
[172] In the given figure, 'G' is the centre of the circle. Find the angle ACB when $\angle AGB = 132^\circ$

Soln:- $\angle ACB = \frac{\angle AGB}{2}$
 $= \frac{132}{2} = \boxed{66^\circ}$ Ans



[173] In the given figure, O is the centre of the circle and $\angle AOB = 130^\circ$. Find the $\angle APB$.

Soln:- $\angle AOB (\text{Exterior}) = 360^\circ - 130^\circ$
 $= 230^\circ$
 $\angle APB = \frac{\angle AOB}{2} = \frac{230}{2}$
 $= \boxed{115^\circ}$ Ans



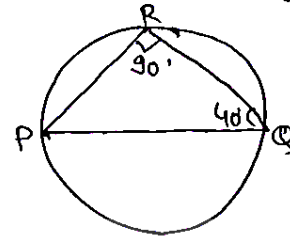
[174] Angle subtended by the largest chord of the circle to a point on the same circle measures:

- (a) $< 90^\circ$ (b) 180° (c) $> 90^\circ$ ~~(d) 90°~~

Ans: $\boxed{90^\circ}$

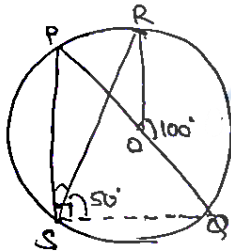
175 Triangle PQR is inscribed in a circle such that P, Q and R lie on the circumference. If PQ is the diameter of the circle and $\angle PQR = 40^\circ$, then what is the value (in degree) of $\angle QPR$?

Soln:- $\angle QPR = 90^\circ - 40^\circ = \boxed{50^\circ}$ Ans



176 In the figure given below, PQ is the diameter of the circle with centre O. If $\angle QOR = 100^\circ$ then the measure of $\angle PSR$ is:

Soln:-



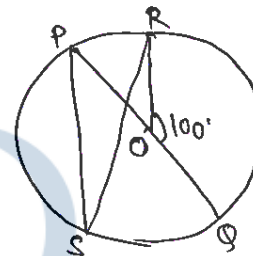
$$\angle RSQ = \frac{\angle ROQ}{2}$$

$$\angle RSQ = \frac{100}{2} = 50^\circ$$

$$\angle PSQ = 90^\circ$$

$$\angle PSR = \angle PSQ - \angle RSQ$$

$$= 90 - 50 = \boxed{40^\circ}$$
 Ans



177 A, B and C are three points on the circle. if $AB = AC = 7\sqrt{2}$ cm and $\angle BAC = 90^\circ$ then the radius is equal to:

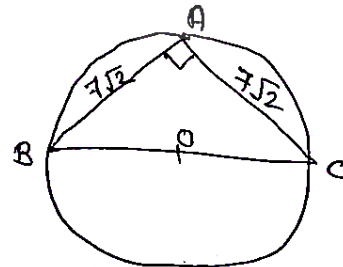
Soln:- $BC^2 = AB^2 + AC^2$

$$BC = \sqrt{(7\sqrt{2})^2 + (7\sqrt{2})^2}$$

$$BC = 7\sqrt{2} \times \sqrt{2}$$

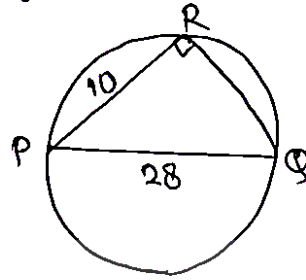
$$BC = 14 \text{ cm}$$

$$\text{radius} = \frac{\text{Hypotenuse}}{2} = \frac{14}{2} = \boxed{7 \text{ cm}}$$
 Ans



178 Triangle PQR is inscribed in the circle whose radius is 14cm. If PQ is the diameter of the circle and PR = 10cm then what is the area of the triangle?

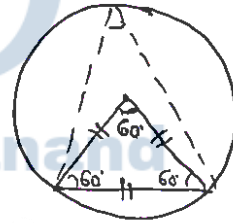
Soln:- $RQ = \sqrt{(28)^2 - (10)^2}$
 $= \sqrt{784 - 100} = \sqrt{684}$
 $= 6\sqrt{19} \text{ cm}$



Area = $\frac{1}{2} \times b \times h$
 $= \frac{1}{2} \times 6\sqrt{19} \times 28 = \boxed{30\sqrt{19} \text{ cm}^2} \text{ Ans}$

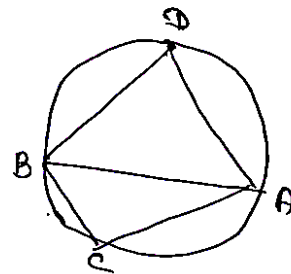
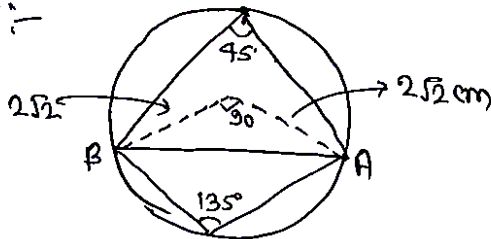
179 A chord of a circle is equal to its radius. The angle subtended by this chord at a point on the circumference is:

Soln:- Angle = $\frac{60^\circ}{2} = \boxed{30^\circ} \text{ Ans}$



180 If in the following figure (not to the scale), $\angle ACB = 135^\circ$ and the radius of the circle is $2\sqrt{2} \text{ cm}$, then the length of the chord AB is:

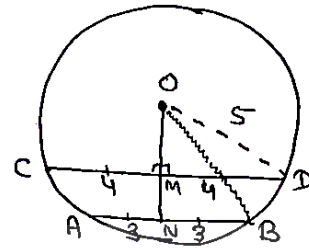
Soln:-



$AB^2 = (2\sqrt{2})^2 + (2\sqrt{2})^2$
 $AB = 2\sqrt{2} \times \sqrt{2} = \boxed{4 \text{ cm}} \text{ Ans}$

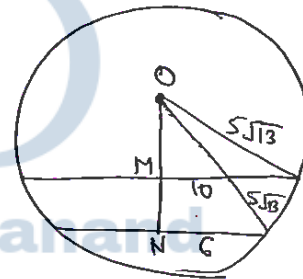
- 181** The length of two parallel chords of a circle of radius 5cm are 6cm and 8cm in the same side of the centre. The distance between them is:

Soln:- by Triplet (3,4,5)
 $OM = 3\text{cm}$ (In $\triangle OMD$)
 $ON = 4\text{cm}$ (In $\triangle ONB$)
 $MN = ON - OM$
 $= 4 - 3 = \boxed{1\text{cm}}$ Ans



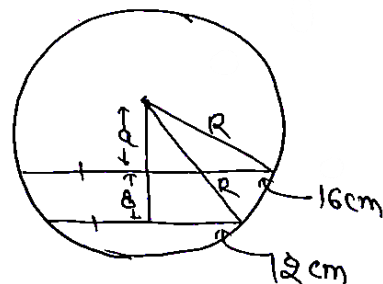
- 182** Two parallel chords on the same side of the centre of a circle are 12cm and 20cm long and the radius of the circle is $5\sqrt{13}$ cm. What is the distance (in cm) between the chords?

Soln:-
 $OM = \sqrt{(5\sqrt{13})^2 - (10)^2}$
 $OM = 15\text{cm}$
 $ON = \sqrt{(5\sqrt{13})^2 - (6)^2}$
 $ON = 17\text{cm}$
 $MN = 17 - 15 = \boxed{2\text{cm}}$ Ans



- 183** Two parallel chords are on one side of the circle. The length of the two chords is 24cm. and 32cm. If the distance between of the two chords is 8cm, then what is area of the circle?

Soln:-
 $R^2 = a^2 + 16^2 \text{ --- (1)}$
 $R^2 = (a+8)^2 + 12^2 \text{ --- (2)}$
Solve
 $a^2 + 16^2 = (a+8)^2 + 12^2$
 $a^2 + 256 = a^2 + 16a + 64 + 144$
 $a = \boxed{3}$
 $R^2 = 3^2 + 16^2 = \boxed{265}$



$$\text{Area of circle} = \pi r^2$$

$$= \frac{22}{7} \times 265 = \boxed{832.86 \text{ cm}^2} \text{ Ans}$$

184 AB = 28 cm and CD = 22 cm are two parallel chords on the same side of the centre of a circle. The distance between them is 4 cm. The radius of the circle is --?

Soln:- $R^2 = a^2 + 14^2$ — (1)

$$R^2 = (a+4)^2 + 11^2$$
 — (2)

by (1) & (2)

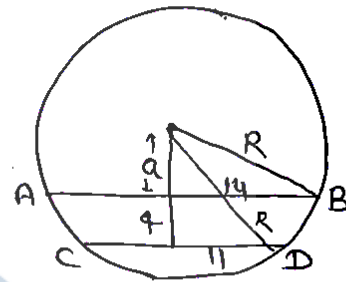
$$a^2 + 14^2 = (a+4)^2 + 11^2$$

$$a^2 + 196 = a^2 + 16 + 8a + 121$$

$$a = \frac{59}{8} = 7.375$$

$$R^2 = (7.375)^2 + (14)^2$$

$$R = \boxed{15.82 \text{ cm}} \text{ Ans}$$

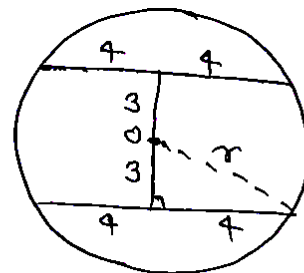


185 In a circle two equal and parallel chords are 6 cm apart and lie on the opposite sides of the center of the circle. If the length of each chord is 8 cm, then the radius of the circle is:

Soln:-

$$\text{radius} = \sqrt{(3)^2 + (4)^2}$$

$$= \boxed{5 \text{ cm}} \text{ Ans}$$



186 Two parallel chords are drawn in a circle of diameter 20cm. The length of one chord is 16cm. and the distance between the two chords is 12cm. The length of the other chord is:

Soln :- diameter = 20cm

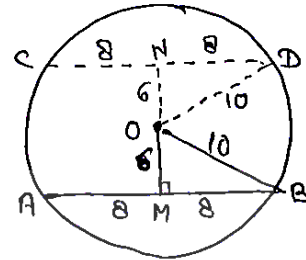
radius = 10cm

$$OM = \sqrt{(10)^2 - (8)^2} \\ = 6\text{cm}$$

$$ON = MN - OM \\ = 12 - 6 = 6\text{cm}$$

$$ND = \sqrt{(10)^2 - (6)^2} = 8\text{cm}$$

$$CD = \boxed{16\text{cm}} \text{ Ans}$$

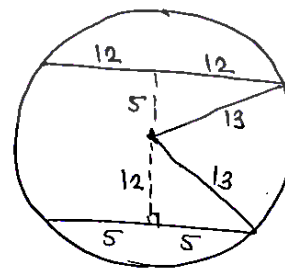


187 AB and CD are two parallel chords of a circle lying on the opposite side of the centre and the distance between them is 17cm. The length of AB and CD are 10cm and 24cm respectively. The radius of the circle is:

Soln :- Triplet $\rightarrow (5, 12, 13)$

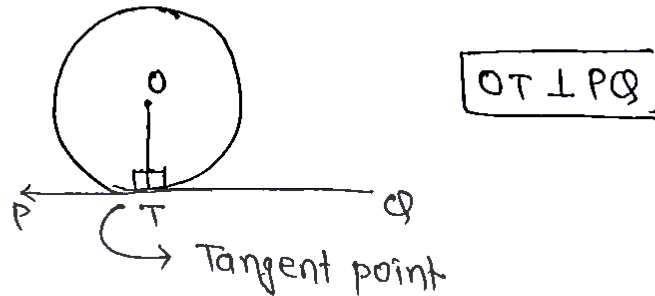
satisfy $\checkmark$

$$\text{radius} = \boxed{13\text{cm}} \text{ Ans}$$

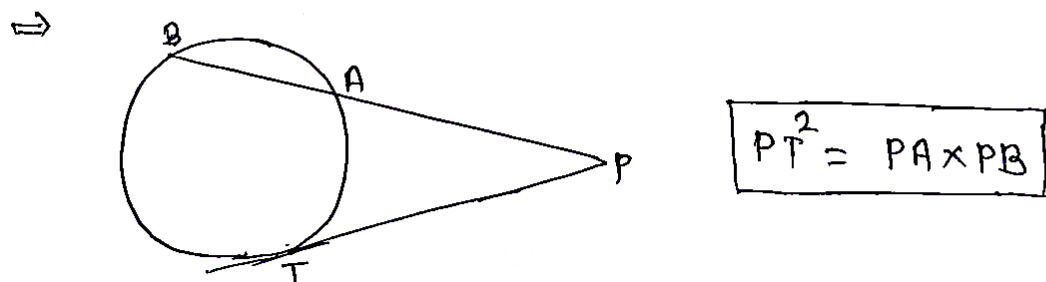
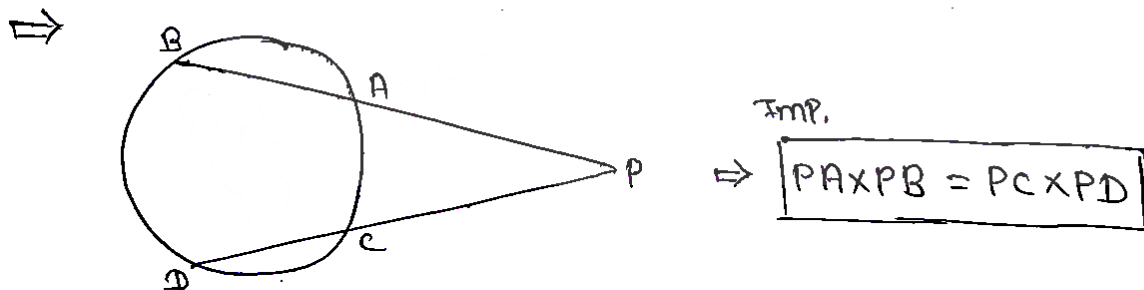
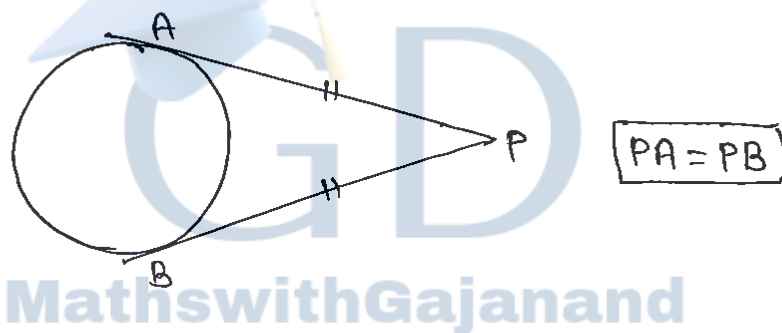


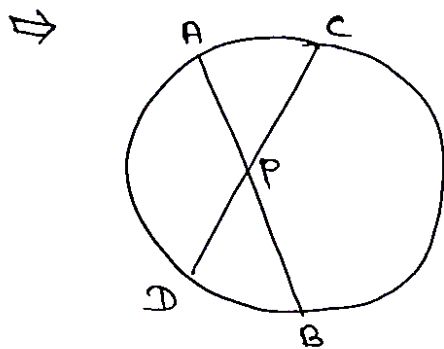
v.v. Imp Tangent theorems * *

⇒ The radius is perpendicular to the tangent (at the point)

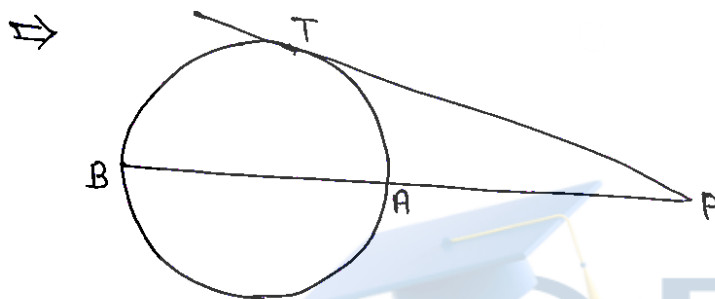


⇒ The Tangent drawn from the outside point are equal.





Imp
 $\Rightarrow PA \times PB = PC \times PD$



$$PT^2 = PA \times PB$$

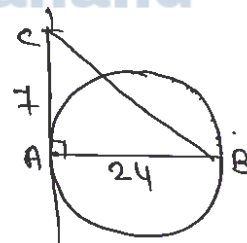
188 AB is the diameter of a circle C is a point on a tangent A. If AB = 24cm and AC = 7cm then find the length of BC.

Soln:-

$$BC = \sqrt{(7)^2 + (24)^2}$$

$$= \boxed{25\text{cm}}$$

(Triplet $\rightarrow 7, 24, 25$)



189 In the given figure, AB, AC and EF are tangents to a circle. If AC = 15 and DE = 3cm. then the length of AE is:

Soln:-

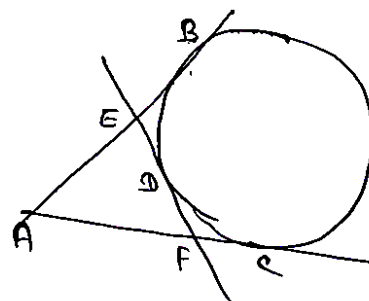
$$AB = AC = 15\text{cm}$$

$$DE = BE = 3\text{cm}$$

$$AE = AB - BE$$

$$= 15 - 3 = \boxed{12\text{cm}}$$

Ans

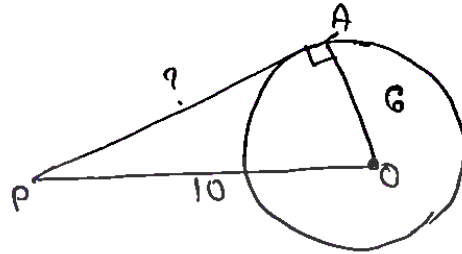


- 190** The radius of a circle is 6cm. The distance of a point lying outside the circle from the centre is 10cm. The length of the tangent drawn from the outside point to the circle is:

Soln:-

$$PA = \boxed{8\text{cm}} \text{ Ans.}$$

(Triplet $\rightarrow 6, 8, 10$)



- 191** Two chords AB and CD of a circle when produced, meet at a point P outside the circle. If $AB = 6\text{cm}$, $PB = 5\text{cm}$, $PD = 4\text{cm}$, then CD is equal to:

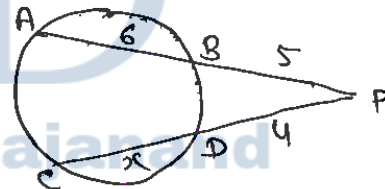
Soln:-

$$PA \times PB = PC \times PD$$

$$11 \times 5 = 4(4+x)$$

$$\frac{55}{4} - x = 4$$

$$x = \boxed{9.75} \text{ Ans}$$



- 192** chords AB and CD of a circle intersect externally at P. If $AB = 7\text{cm}$, $CD = 1\text{cm}$ and $PD = 5\text{cm}$, then the length of PB is:

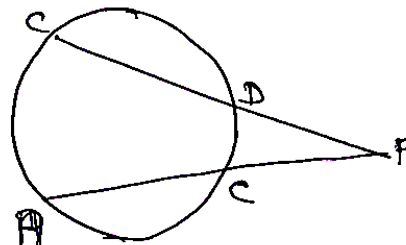
Soln:-

$$PD \times PC = PB \times PA$$

$$\Rightarrow 5 \times 6 = PB(PB + 7)$$

$$PB = \boxed{3\text{cm}} \text{ Ans}$$

(satisfy)



[193] In the figure, chords AB and CD of a circle intersect externally at P. If AB = 4cm, CD = 11cm and PD = 15cm, then the length of PB is:

Soln:-

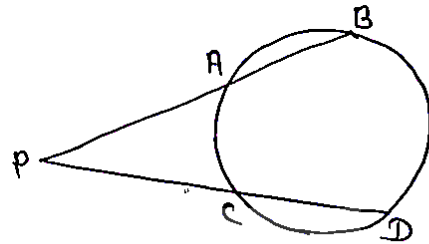
$$PA \times PB = PC \times PD$$

$$x(x+4) = 4 \times 15$$

$$x^2 + 4x - 60 = 0$$

$$\text{Solve } x = 6\text{cm}$$

$$PB = x + 4 = 6 + 4 = \boxed{10\text{cm}} \text{ Ans}$$



[194] Chord AB of a circle is produced to a point P, and C is a point on the circle such that PC is a tangent to the circle. If PC = 18cm, and BP = 15cm, then AB is equal to

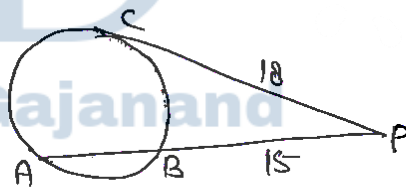
Soln:-

$$PC^2 = PB \times PA$$

$$18^2 = 15 \times (15 + AB)$$

$$\frac{324}{15} = 15 + AB$$

$$AB = 21.6 - 15 = \boxed{6.6\text{cm}} \text{ Ans}$$



[195] Find the value of x in the given figure when TP (tangent on point T) = 15cm, PB = $2x + 3$, PA = 9cm.

Soln:-

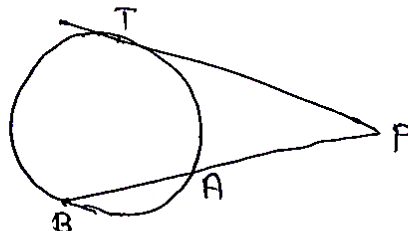
$$PT^2 = PA \times PB$$

$$(15)^2 = 9 \times (2x + 3)$$

$$\frac{225}{9} = 2x + 3$$

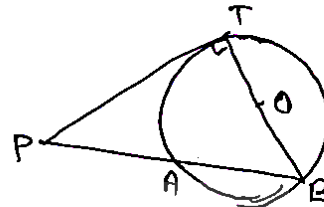
$$25 = 2x + 3$$

$$x = \boxed{11\text{cm}} \text{ Ans}$$



[196] In the given figure, TB is a chord which passes through the centre of the circle. PT is a tangent to the circle at the point T on the circle. If $PT = 10\text{ cm}$, $PA = 5\text{ cm}$ and $AB = x\text{ cm}$, then the radius of the circle is:

Soln:- $PT^2 = PA \times PB$
 $(10)^2 = 5(5+x)$
 $x = 15\text{ cm}$



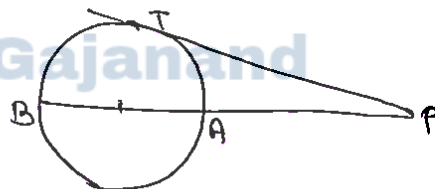
In ΔPTB $TB^2 = (20)^2 - (10)^2$

$TB = 10\sqrt{3}$

$OT = \frac{10\sqrt{3}}{2} = \boxed{5\sqrt{3}} \text{ Ans}$

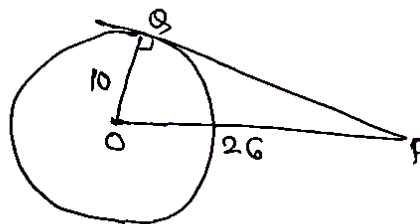
[197] In the figure PT is a tangent. If $TP = 12\text{ cm}$, $PB = x+7$ and $PA = x\text{ cm}$, then find the value of x .

Soln:- $PT^2 = PA \times PB$
 $(12)^2 = (x+7)x$
 $x^2 + 7x - 144 = 0$
 solve - $\boxed{x = 9\text{ cm}} \text{ Ans}$



[198] O is the centre of a circle of radius 10 cm . P is a point outside the circle and PQ is a tangent to the circle. What is the length of PQ if the length OP is 26 cm ?

Soln:- $PQ = 24\text{ cm}$
 (Triplet $\rightarrow 10, 24, 26$)



199 The Centre of circle is O and PT is a tangent at T. BC is the diameter of the circle. If BC is extended, then it meets the tangent PT at P. It is given that PC = 4cm and PT = 8cm. Find the radius of the circle.

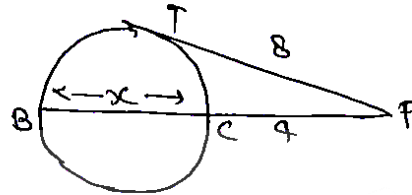
Soln:-

$$PT^2 = PC \times PB$$

$$64 = 4 \times (4+x)$$

$$x = 12 \text{ cm}$$

$$\text{radius} = \frac{x}{2} = \frac{12}{2} = \boxed{6 \text{ cm}} \text{ Ans.}$$



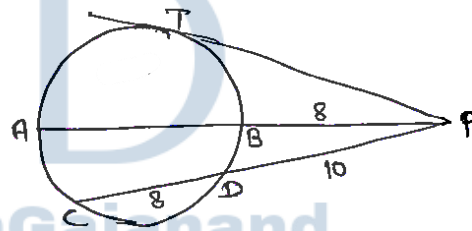
200 In the given figure, diameter AB and chord CD of a circle meet at P. PT is a tangent to the circle at T. If CD = 8cm, PD = 10cm and PB = 8cm, find AB.

Soln:-

$$PA \times PB = PC \times PD$$

$$8(8+x) = 10 \times 8$$

$$x = \boxed{14.5 \text{ cm}} \text{ Ans}$$



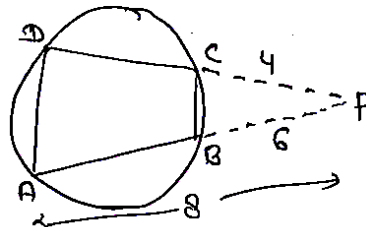
201 ABCD is a cyclic quadrilateral. AB and DC when produced meet at P, if PA = 8cm, PB = 6, PC = 4cm, then the length of PD is:

Soln:-

$$PA \times PB = PC \times PD$$

$$\Rightarrow 8 \times 6 = 4 \times PD$$

$$\Rightarrow PD = \boxed{12 \text{ cm}} \text{ Ans}$$



202 Two chords AB and CD of a circle meet inside the circle at point P. If $AP = 12\text{cm}$, $AB = 20\text{cm}$, $CP = 16\text{cm}$, then $CD = ?$

Soln:-

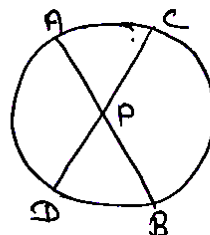
$$PA \times PB = PC \times PD$$

$$12 \times (20 - 12) = 16 \times PD$$

$$\frac{12 \times 8}{16} = PD$$

$$PD = 6\text{cm}$$

$$CD = PC + PD = 16 + 6 = \boxed{22\text{cm}} \text{ Ans}$$



203 In the figure $AB = AD = 9\text{cm}$ and $AC = AE = 13\text{cm}$ and $BC = 15\text{cm}$. Find $ED = ?$

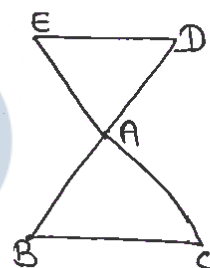
Soln:-

$$AB = AD = 9\text{cm}$$

$$AC = AE = 13\text{cm}$$

$$\triangle ABC \cong \triangle ADE$$

$$\text{so } BC = ED = \boxed{15\text{cm}} \text{ Ans}$$



204 AB is a chord of a circle with O as centre. C is a point on the circle such that, $OC \perp AB$ and OC intersects AB at P. If $PC = 2\text{cm}$ and $AB = 6\text{cm}$ then the diameter of the circle is:

Soln:-

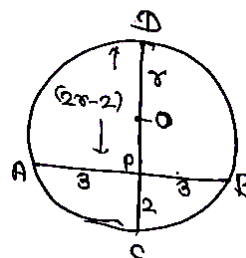
$$PA \times PB = PC \times PD$$

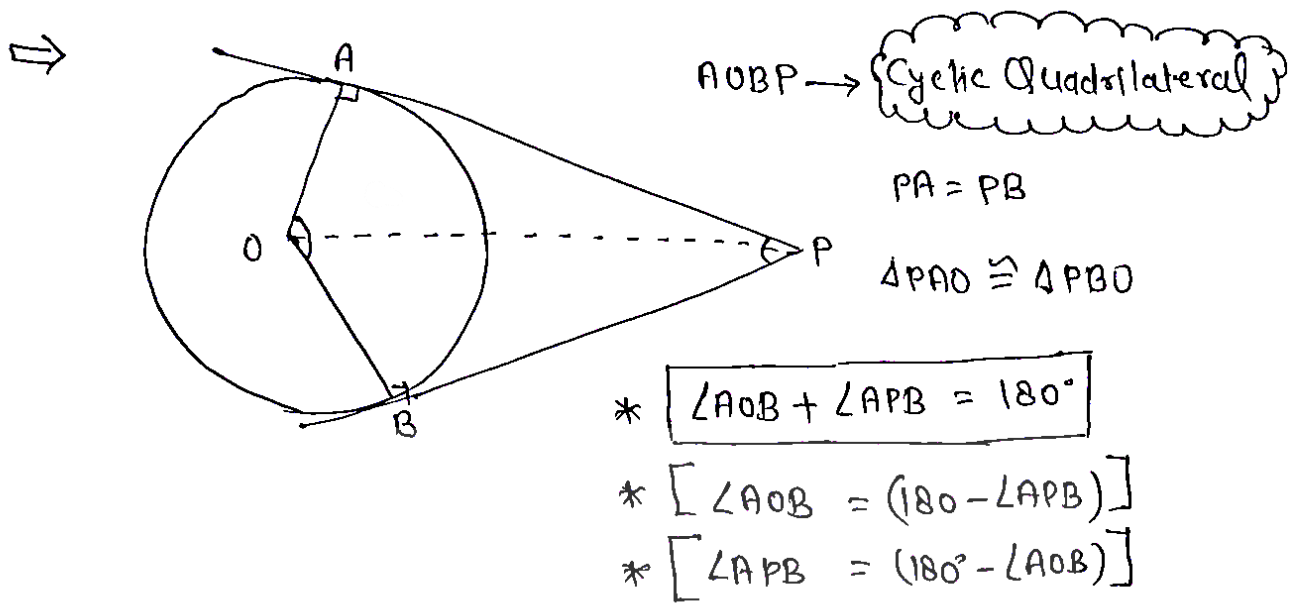
$$3 \times 3 = (2r - 2) \times 2$$

$$9 = 4r - 4$$

$$r = \frac{13}{4}$$

$$\text{diameter} = 2r = \frac{13}{2} = \boxed{6.5\text{cm}} \text{ Ans}$$



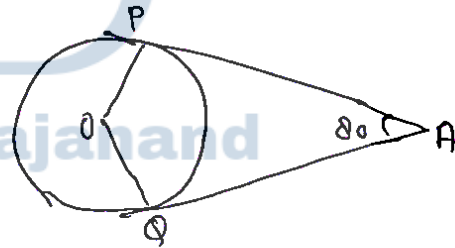


Q05 To a circle with centre at O, two tangents AP and AQ are drawn from an external point A. If the $\angle PAQ = 80^\circ$ then the $\angle POQ$ is:

Soln:-

$$\angle POQ = 180^\circ - 80^\circ$$

$$= \boxed{100^\circ} \text{ Ans.}$$

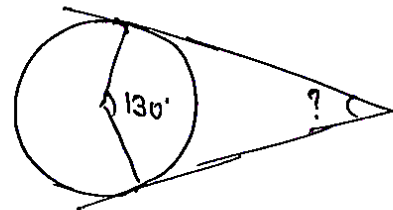


Q06 If the angle between two radii of a circle be 130° , then the angle between the tangents at the end of these radii (in degree) is:

Soln:-

$$\text{Angle} = 180 - 130^\circ$$

$$= \boxed{50^\circ} \text{ Ans}$$



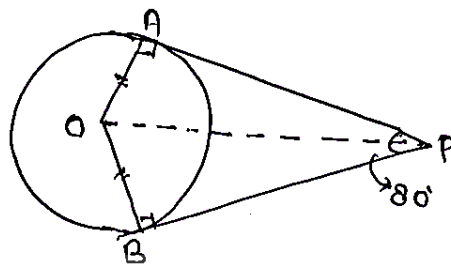
Q07 If PA and PB are two tangents to a circle with centre O such that $\angle APB = 80^\circ$. Then, $\angle AOP = ?$

Soln:-

$$\angle AOB = 180^\circ - 80^\circ$$

$$= 100^\circ$$

$$\angle AOP = \frac{100}{2} = \boxed{50^\circ} \text{ Ans}$$



Q08 O is the center of the circle and two tangents are drawn from a point P to this circle at points A and B. If $\angle AOP = 50^\circ$, then what is the value of $\angle APB = ?$

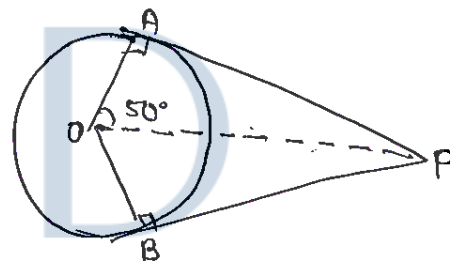
Soln:-

$$\angle AOP = 50^\circ$$

$$\angle AOB = 50^\circ \times 2 = 100^\circ$$

$$\angle APB = 180 - 100$$

$$= \boxed{80^\circ} \text{ Ans.}$$



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Q09 Two tangent PA and PB are drawn from an external Point P to a circle with centre O at the point A and B respectively on it. such that $\angle APB = 120^\circ$ and $AP = 12.5\text{cm}$. The length of OP is:

Soln:-

$$\angle APB = 120^\circ$$

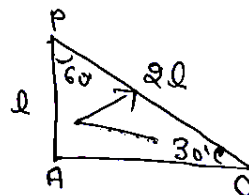
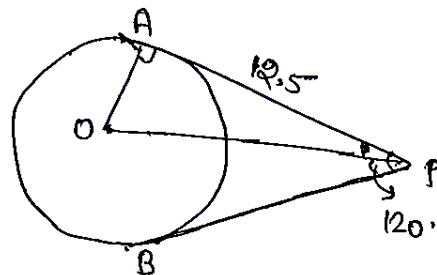
$$\angle APO = 60^\circ$$

$$\text{then } \angle POA = 90 - 60 = 30^\circ$$

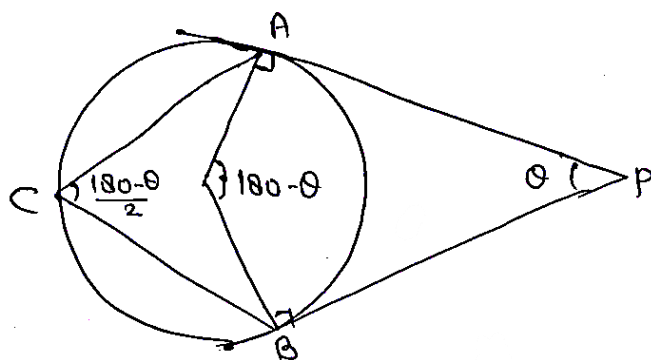
$$AP = 12.5\text{cm}$$

$$OP = 2 \times 12.5$$

$$= \boxed{25\text{cm}} \text{ Ans}$$



⇒

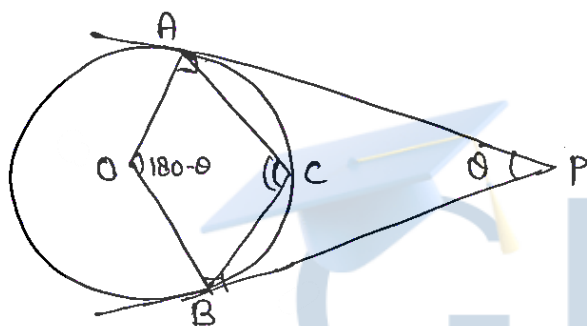


IF $\angle APB = \theta$

then

* $\angle ACB = 90 - \frac{\theta}{2}$ *

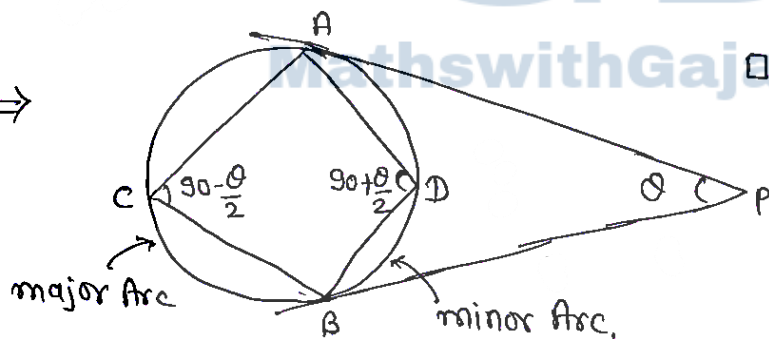
⇒



IF $\angle APB = \theta$

* $\angle ACB = 90 + \frac{\theta}{2}$ *

⇒

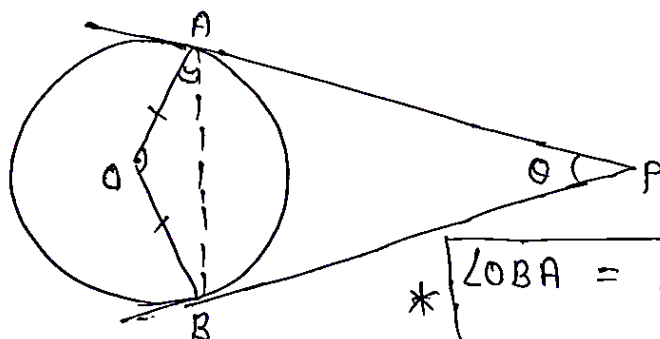


□ ACBD will be

cyclic Quadrilateral

Imp.

⇒



IF $\angle APB = \theta$

then

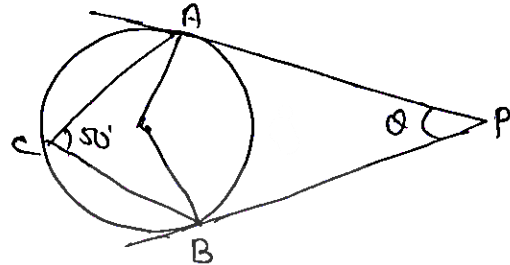
* $\angle OBA = \angle OAB = \frac{\theta}{2}$ *

Q10. In a circle with centre O, PA and PB are tangents to the circle at point A and B, respectively. C is a point on the major arc AB. If $\angle ACB = 50^\circ$, then find the measure of $\angle APB$.

Soln:-

$$\angle ACB = 90 - \frac{\theta}{2} = 50^\circ$$

$$\theta = 80^\circ \quad \underline{\underline{\text{Ans}}}$$

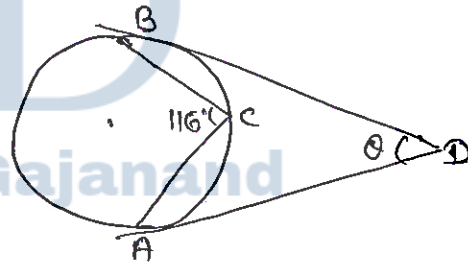


Q11. AB is a chord in the minor segment of a circle with centre O. C is a point between A and B on the minor arc AB. The tangents to the circle at A and B meet at the point D. If $\angle ACB = 116^\circ$, then the measure of $\angle ADB$ is.

Soln:-

$$90 + \frac{\theta}{2} = 116^\circ$$

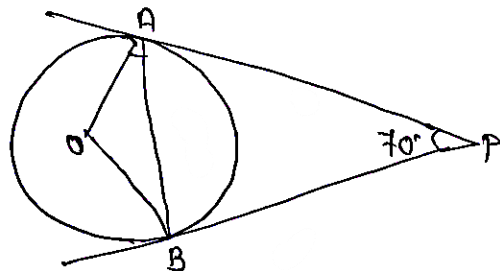
$$\theta = 52^\circ \quad \underline{\underline{\text{Ans}}}$$



Q12. PA and PB are tangents to a circle with centre O. A and B are points on the circle. If $\angle APB = 70^\circ$, then value of $\angle OAB$ is:

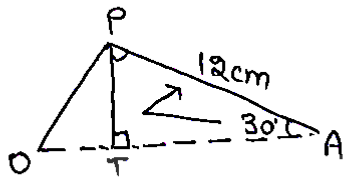
Soln:-

$$\begin{aligned} \angle OAB &= \frac{\theta}{2} \\ &= \frac{70^\circ}{2} \\ &= 35^\circ \quad \underline{\underline{\text{Ans}}} \end{aligned}$$



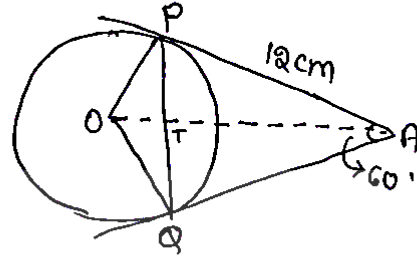
- Q13** Two tangents AP and AQ are drawn to a circle with centre O from an external point A, where P and Q are points on the circle. If AP = 12cm and $\angle PAQ = 60^\circ$, then the length of chord PQ is:

Soln:-



$$PT = \frac{12}{2} = 6\text{cm}$$

$$PQ = 2 \times PT \\ = 2 \times 6 = \boxed{12\text{cm}} \text{ Ans.}$$



- Q14** PA and PB are tangents to the circle and O is the centre of the circle. The radius is 5cm and PO is 13cm. If the area of the triangle PAB is M, then the value of $\sqrt{\frac{M}{15}}$ is:

Soln:- PA = 12cm

$$\sin \theta = \frac{AC}{AP} = \frac{AO}{OP}$$

$$\frac{AC}{12} = \frac{5}{13} \Rightarrow AC = \frac{60}{13}$$

$$\tan \theta = \frac{AC}{PC} = \frac{AO}{AP}$$

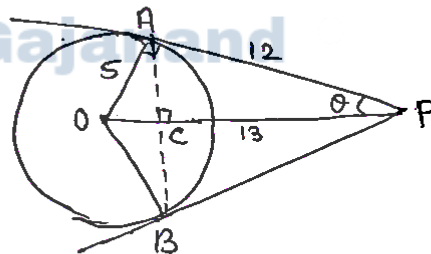
$$\frac{\frac{60}{13}}{PC} = \frac{5}{12} \Rightarrow PC = \frac{144}{13}$$

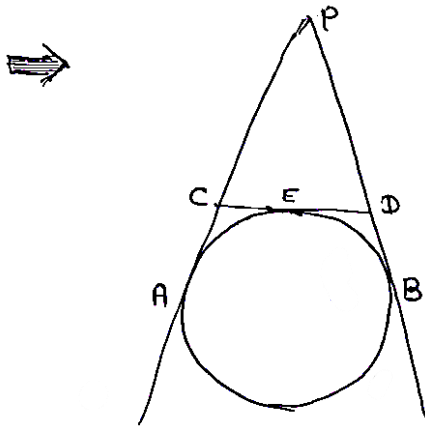
$$\text{Area of } \triangle PCB = \frac{1}{2} \times PC \times AC$$

$$\text{Area of } \triangle PAB = 2(\triangle PCB) = PC \times AC$$

$$M = \frac{144}{13} \times \frac{60}{13}$$

$$\sqrt{\frac{M}{15}} = \sqrt{\frac{144 \times 60}{13 \times 13 \times 15}} = \boxed{\frac{24}{13}} \text{ Ans}$$





$$* PA = \frac{\text{Perimeter of } \triangle PCD}{2}$$

- Q15** A circle touches the side PQ of a $\triangle APQ$ at the point R and sides AP and AQ produced at the points B and C, respectively. If the perimeter of $\triangle APQ = 30\text{cm}$, then the length of AB is:

Soln:- $AB = \frac{30}{2} = \boxed{15\text{cm}}$ Ans

- Q16** A circle touches the side BC of $\triangle ABC$ at D and AB and AC are produced to E and F, respectively. If $AB = 10\text{cm}$, $AC = 8.6\text{cm}$ and $BC = 6.4\text{cm}$, then $BE = ?$

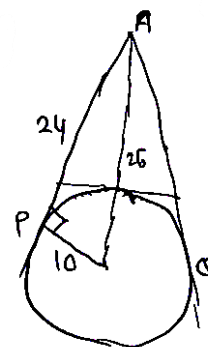
Soln:- $AE = \frac{(10 + 6.4 + 8.6)}{2} = \frac{25}{2} = 12.5\text{cm}$

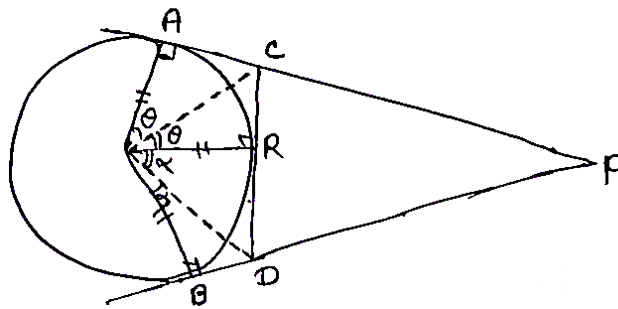
$BE = 12.5 - 10 = \boxed{2.5\text{cm}}$ Ans



- Q17** A is a point at a distance 26cm from the centre O of a circle of radius 10cm . AP and AQ are the tangents to the circle at the points of contacts P and Q. If a tangent BC is drawn at a point R lying on the minor arc PQ to intersect AP at B, AQ at C, then the perimeter of $\triangle ABC$ is:

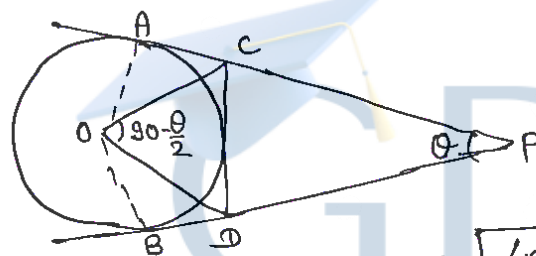
Soln:- $\text{Perimeter of } \triangle ABC = 2 \times 24$
 $= \boxed{48\text{cm}}$ Ans





$$* \quad \angle COD = \frac{\angle AOB}{2}$$

$$* \quad \begin{aligned} \angle AOB &= 2\theta + 2\alpha \\ &= 2(\angle COD) \end{aligned}$$



$$\angle COD = 90 - \frac{\theta}{2}$$

$$* \quad \angle AOB = 180 - \theta$$

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Q18 In the given figure, O is centre of the circle. Circle has 3 tangents. If $\angle QPR = 45^\circ$, then what is the value of $\angle QOR$?

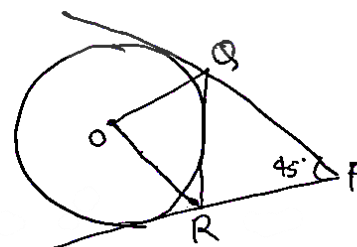
Soln:-

$$\angle QOR = 90^\circ - \frac{\angle P}{2}$$

$$= 90 - \frac{45^\circ}{2}$$

$$= 90 - 22.5$$

$$= \boxed{67.5^\circ} \text{ Ans.}$$



Q19 If $PE = 18\text{cm}$ and radius of circle is 7cm , then find tangent $CD = ?$

Soln:- {Triplet $\rightarrow 7, 24, 25$ }

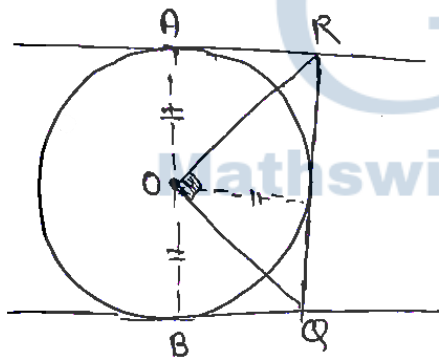
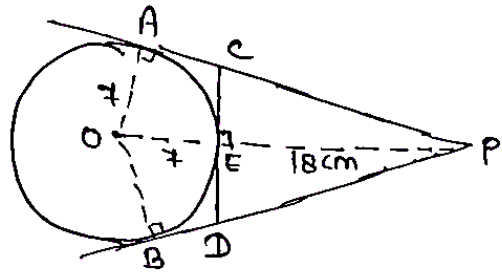
$$PA = 24\text{cm}$$

$$\triangle AOP \sim \triangle PCE$$

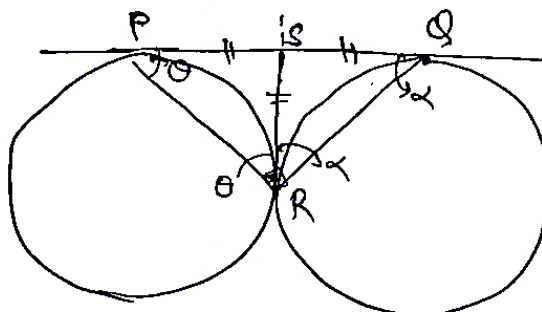
$$\frac{CE}{AO} = \frac{EP}{AP} \Rightarrow \frac{CE}{7} = \frac{18}{24}$$

$$CE = \frac{21}{4}$$

$$CD = 2 \times CE = \frac{21}{4} \times 2 = \boxed{10.5\text{cm}} \text{ Ans//}$$



$$* \angle RPO = 90^\circ *$$



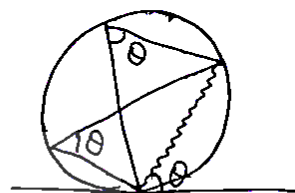
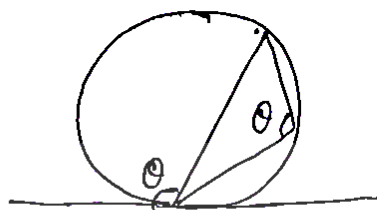
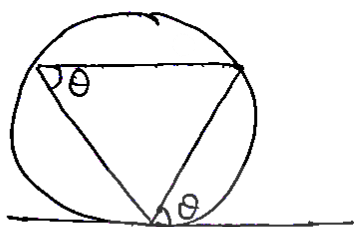
$$* \angle PRQ = 90^\circ *$$

$$\text{Proof:- } 2\theta + 2\alpha = 180^\circ$$

$$\theta + \alpha = \underline{90^\circ}$$

⇒ Alternate Segment Theorem:-

The angle between the tangent and the side of the triangle is equal to the opposite interior angle.



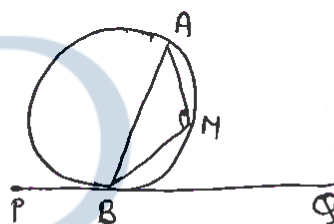
Q20 In the given figure, $\angle AMB = 130^\circ$, then what is the value of $\angle ABQ$..

Soln

$$\angle AMB = 130^\circ$$

$$\text{So } \angle ABP = 130^\circ$$

$$\angle ABQ = 180 - 130 = \boxed{50^\circ} \text{ Ans}$$

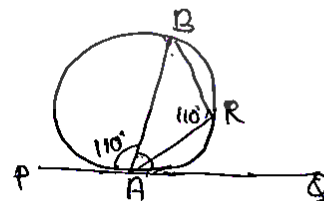


Q21 AB is a chord of a circle with centre O, while PAQ is the tangent at A. R is a point on the minor arc AB. If $\angle BAQ = 70^\circ$ then find the measure of $\angle ARB$.

Soln:- $\angle BAQ = 70^\circ$

$$\angle BAP = 180^\circ - 70^\circ = 110^\circ$$

$$\angle ARB = \boxed{110^\circ} \text{ Ans}$$

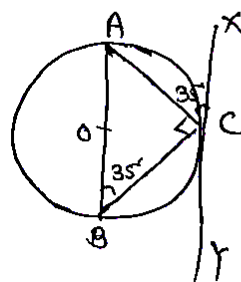


Q22 In the given figure, AB is a diameter of the circle with centre O and XY is the tangent at a point C. If $\angle ACX = 35^\circ$ then what is the value of $\angle CAB$?

Soln:-

$$\angle CBA = 35^\circ$$

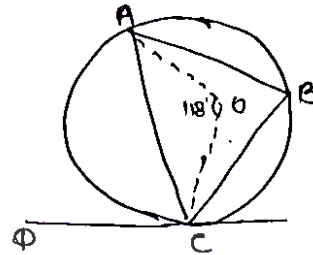
$$\angle CAB = 90 - 35^\circ = \boxed{55^\circ} \text{ Ans}$$



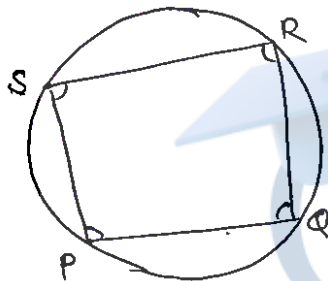
Q23 In the given figure, BC is a chord and CD is a tangent through the point C. If $\angle AOC = 118^\circ$, then find the $\angle ACD$.

Soln:- $\angle B = \frac{\angle AOC}{2} = \frac{118}{2} = 59^\circ$

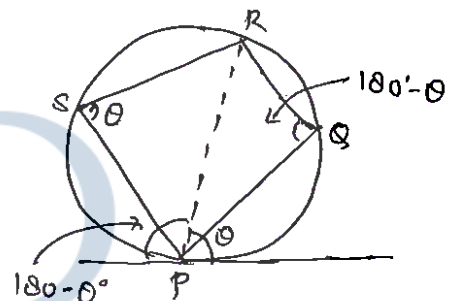
$\angle ACD = \boxed{59^\circ}$ Ans



$\Rightarrow$ Cyclic Quadrilateral :-



Proof



* $\angle P + \angle R = 180^\circ$
 $\angle Q + \angle S = 180^\circ$

$\angle S + \angle Q = \theta + 180 - \theta$
 $= \boxed{180^\circ}$

Q24 In the given figure, $\angle QRU = 72^\circ$, $\angle TRS = 15^\circ$ and $\angle PSR = 95^\circ$, then what is the value of $\angle PQR$

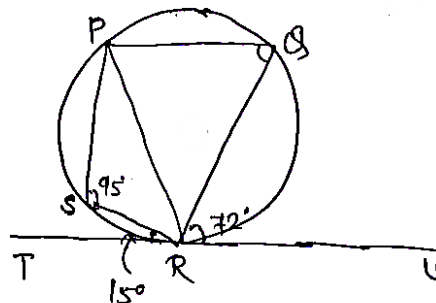
Soln:-

$\angle S + \angle Q = 180^\circ$

$95^\circ + \angle Q = 180^\circ$

$\angle Q = 180^\circ - 95^\circ$

$\angle Q = \boxed{85^\circ}$ Ans



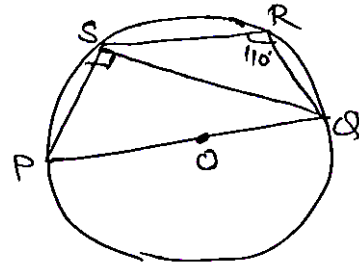
Q25 In the given figure, PQ is a chord passing through the centre 'O' of the circle, calculate $\angle PQS$.

Soln:-

$$\angle R = 110^\circ \text{ (Given)}$$

$$\angle P = 180^\circ - 110^\circ = 70^\circ$$

$$\angle PQS = 90^\circ - 70^\circ = \boxed{20^\circ} \text{ Ans}$$

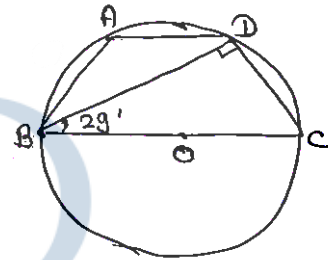


Q26 ABCD is a cyclic quadrilateral and BC is a diameter of the circle. If $\angle PBC = 29^\circ$, then $\angle BAD = ?$

Soln:-

$$\begin{aligned} \angle C &= 90^\circ - \angle B \\ &= 90^\circ - 29^\circ = 61^\circ \end{aligned}$$

$$\begin{aligned} \angle A &= 180^\circ - \angle C \\ &= 180^\circ - 61^\circ = \boxed{119^\circ} \text{ Ans} \end{aligned}$$



Q27 In a circle with centre O, AB is a diameter and CD is a chord such that ABCD is a trapezium. If $\angle BAC = 28^\circ$, then $\angle CAD$ is equal to:

Soln:-

$$\begin{aligned} \angle B &= 90^\circ - 28^\circ \\ &= 62^\circ \end{aligned}$$

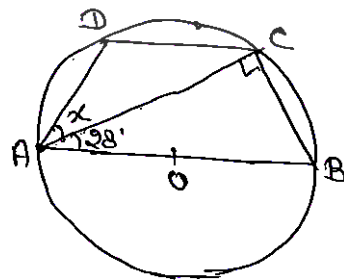
$$\begin{aligned} \angle D &= 180^\circ - 62^\circ \\ &= 118^\circ \end{aligned}$$

$$\angle ACD = 28^\circ = \angle CAB$$

$$\angle A + \angle C = 180^\circ$$

$$x + 28^\circ + 28^\circ + 90^\circ = 180^\circ$$

$$\boxed{x = 34^\circ} \text{ Ans}$$



Q28 ABCD is a cyclic quadrilateral. Side AB and DC, when produced, meet at E and sides AD and BC when produced, meet at F. If $\angle ADC = 76^\circ$ and $\angle AED = 55^\circ$ then $\angle AFB$ is equal to:

Soln:- In $\triangle AED$

$$\angle A = 180 - (76 + 55)$$

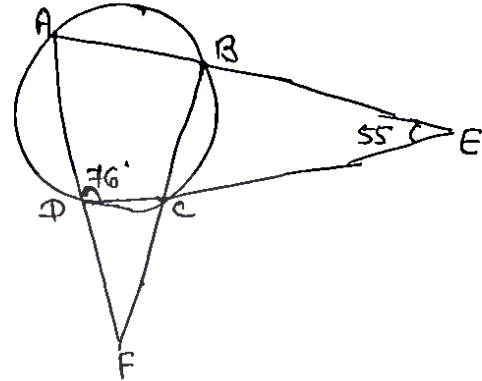
$$\angle A = 49^\circ$$

$$\angle B = 180 - 76$$

$$\angle B = 104^\circ$$

In $\triangle ABF$: $\angle F = 180 - 104 - 49$

$$\angle AFB = 27^\circ \text{ Ans}$$



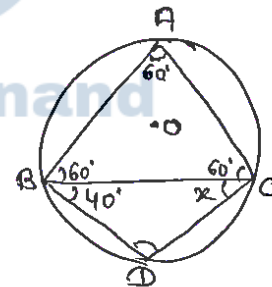
Q29 An equilateral triangle ABC is inscribed in a circle with centre O. D is point on the minor arc BC and $\angle CBD = 40^\circ$. Find the measure of $\angle BCD$.

Soln:-

$$\angle B + \angle C = 180^\circ$$

$$120^\circ + 60^\circ + x = 180^\circ$$

$$x = 20^\circ \text{ Ans}$$



Q30 A cyclic quadrilateral ABCD is drawn in a circle with centre O. A and C are joined to O. If $\angle ABC = 2p$ and $\angle ADC = 3p$, what is the measure of the $\angle AOC$ reflex?

Soln:-

$$\angle B + \angle D = 180^\circ$$

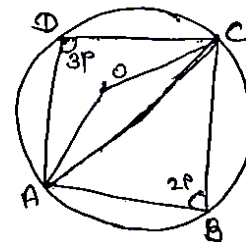
$$3p + 2p = 180^\circ$$

$$p = 36^\circ$$

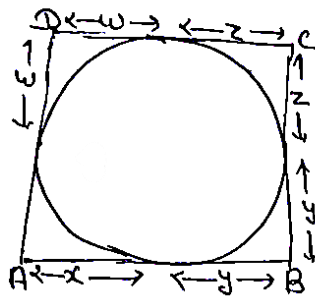
$$\angle ADC = 3p = 3 \times 36^\circ = 108^\circ$$

$$\angle AOC = \frac{1}{2} \times \angle ADC = \frac{108}{2} = 54^\circ$$

$$\text{Reflex } \angle AOC = 360^\circ - 54^\circ = 306^\circ \text{ Ans}$$



Concept :-



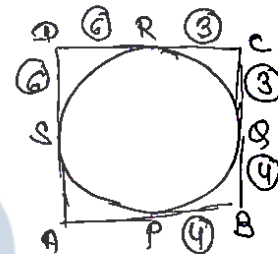
Imp.

$$AB + CD = BC + AD$$

$$[x + y + z + w = x + y + z + w]$$

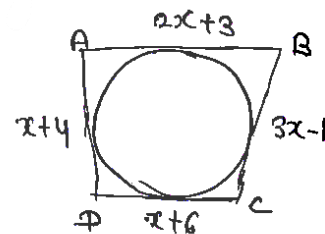
Q31 A circle is inscribed in a quadrilateral ABCD touching sides AB, BC, CD and AD at the points P, Q, R and S respectively. If BP = 4cm, SD = 6cm and BC = 7cm, then the length of DC is:

Solⁿ:- $DC = 6 + 3 = \boxed{9\text{cm}}$ Ans



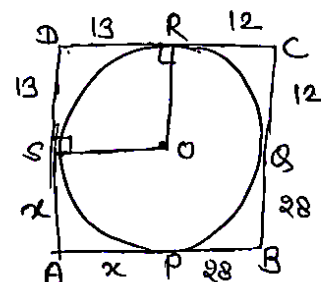
Q32 In the given figure, a circle touches quadrilateral ABCD. If $AB = 2x + 3$, $BC = 3x - 1$, $CD = x + 6$ and $DA = x + 4$, then what is the value of x ?

Solⁿ:- $AB + CD = BC + AD$
 $2x + 3 + x + 6 = 3x - 1 + x + 4$
 $\boxed{x = 6}$ Ans

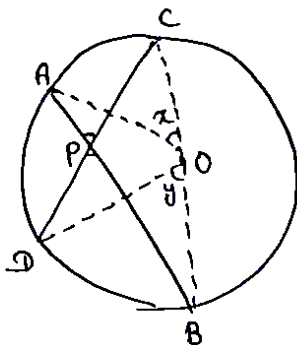


Q33 ABCD is a quadrilateral such that $\angle D = 90^\circ$. A circle with centre O touches the sides AB, BC, CD and DA at P, Q, R and S respectively. If $BC = 40\text{cm}$, $BP = 28\text{cm}$ and $CD = 25\text{cm}$, then what is the radius of the circle?

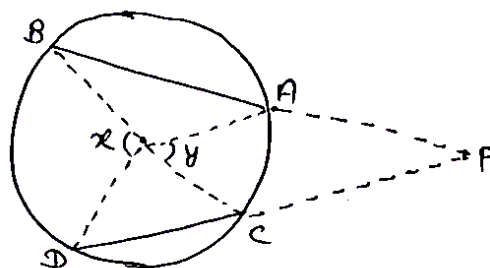
Solⁿ $QC = 40 - 28 = 12\text{cm}$
 $SO = OR = RD = DS = \boxed{13\text{cm}}$ Ans



Concept :-



$$* \angle APC = \angle BPD = \frac{1}{2}(\angle x + \angle y)$$



$$* \angle APC = \angle BPD = \frac{1}{2}(\angle x - \angle y)$$

Q34 In a circle, O is the centre of the circle. Chords AB and CD intersect at P. If $\angle AOD = 32^\circ$ and $\angle COB = 26^\circ$ then the measure of $\angle APD$ lies between:

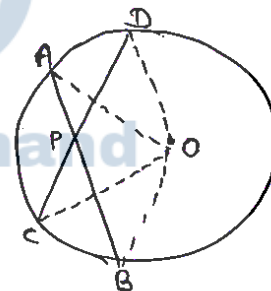
- (a) 18° and 22° (b) 26° and 30° (c) 30° and 34° (d) 22° and 26°

Soln:-

$$\angle APD = \frac{1}{2}(\angle 32^\circ + \angle 26^\circ)$$

$$= \frac{58}{2}$$

$$= \boxed{29^\circ} \text{ Ans}$$



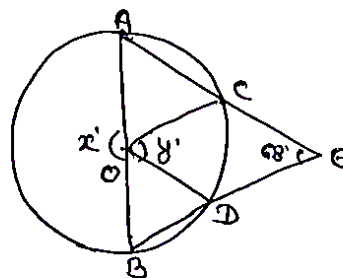
Q35 In a circle with centre O, AC and BD are two chords. AC and BD meet at E when produced. If AB is the diameter and $\angle AEB = 68^\circ$, then the measure of $\angle DOC$ is:

Soln:-

$$\frac{x-y}{2} = 68^\circ$$

$$180 - y = 136^\circ$$

$$y = \boxed{44^\circ} \text{ Ans}$$



Q35 AB and CD are two chords in a circle with centre O and AD is the diameter. When produced, AB and CD meet at the point P. If $\angle DAP = 27^\circ$, $\angle APD = 35^\circ$, then what is the measure of $\angle OBC$?

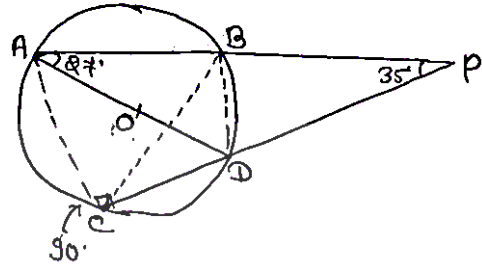
Soln:-

$$\angle CDA = \angle DAP + \angle APD$$

$$\angle CDA = 27^\circ + 35^\circ = 62^\circ$$

$$\text{In } \triangle ACD \quad \angle DAC = 90^\circ - 62^\circ = 28^\circ$$

$$\angle OBC = \angle DAC = 28^\circ \text{ (same arc)}$$



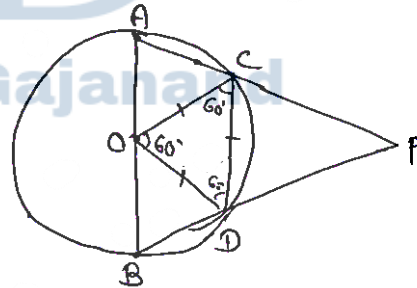
Q36 In a circle with centre O, AB is a diameter and CD is a chord which is equal to the radius OC. AC and BD are extended in such a way that they intersect each other at a point P, exterior to the circle. The measure of $\angle APB$ is:

Soln

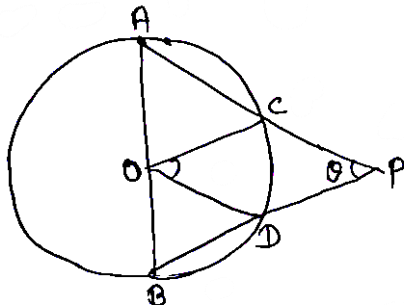
$$\angle APB = 90 - \frac{\angle COD}{2}$$

$$= 90^\circ - \frac{60}{2}$$

$$= \boxed{60^\circ} \text{ Ans}$$

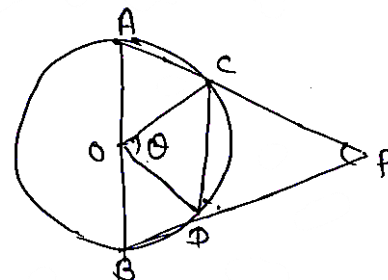


Imp. Concept:-



$$* \quad \angle COD = 180^\circ - 2\theta$$

OR

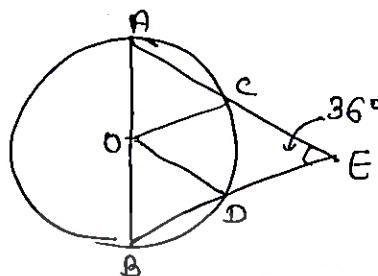


$$* \quad \angle APB = 90^\circ - \frac{\theta}{2}$$

Q37 In a circle with centre O, AC and BD are two chords, AC and BD meet at E, when produced. If AB is a diameter and $\angle AEB = 36^\circ$, then the measure of $\angle DOC$ is:

Soln:-

$$\begin{aligned}\angle DOC &= 180 - 2\theta \\ &= 180 - 2 \times 36^\circ \\ &= \boxed{108^\circ} \text{ Ans}\end{aligned}$$



Q38 In a $\triangle ABC$, points P, Q and R are taken on AB, BC and CA, respectively, such that $BQ = PQ$ and $QC = QR$. If $\angle BAC = 75^\circ$, what is the measure of $\angle PQR$?

Soln:- In $\triangle ABC$

$$\angle A + \angle B + \angle C = 180^\circ$$

$$75 + \theta + \alpha = 180$$

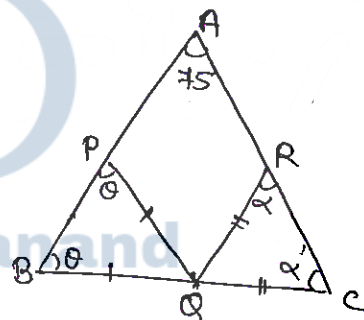
$$\theta + \alpha = 105^\circ$$

$$\angle PQB + \angle PQR + \angle RQC = 180^\circ$$

$$180 - 2\theta + \angle PQR + 180 - 2\alpha = 180$$

$$\angle PQR = -180^\circ + 2(\alpha + \theta)$$

$$= -180^\circ + 2 \times 105^\circ = \boxed{30^\circ} \text{ Ans.}$$



Q39 In a circle with centre O, ABCD is a cyclic quadrilateral and AC is the diameter. Chords AB and DC are produced to meet at E. If $\angle CAE = 34^\circ$ and $\angle E = 30^\circ$, then $\angle CBD$ is equal to:

Soln:-

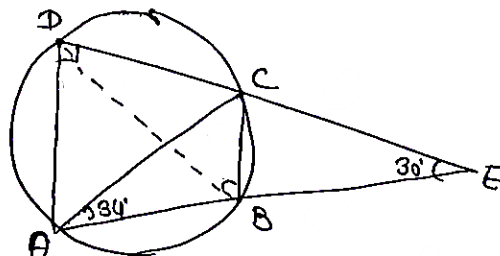
$$\text{In } \triangle ADE \quad \angle D = 90^\circ$$

$$\angle A = 90 - 30 = 60^\circ$$

$$\angle DAC = 60 - 34 = 26^\circ$$

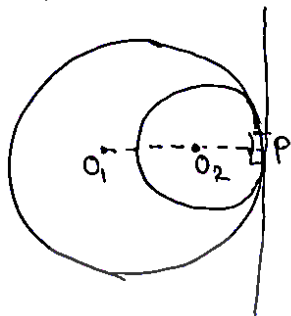
$$\angle DAC = \angle DBC = 26^\circ \text{ (same Arc DC)}$$

$$\boxed{\angle CBD = 26^\circ} \text{ Ans}$$



Common Tangents

Case-1

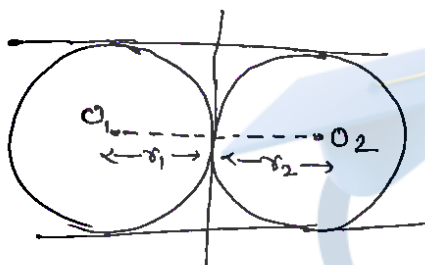


* maximum common tangent = **1**

* $O_1P = r_1$
 $O_2P = r_2$

$$O_1O_2 = r_1 - r_2$$

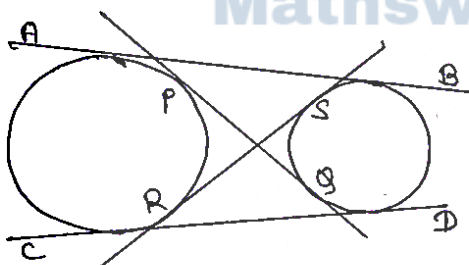
Case-2



* maximum common tangent = **3**

$$O_1O_2 = r_1 + r_2$$

Case-3

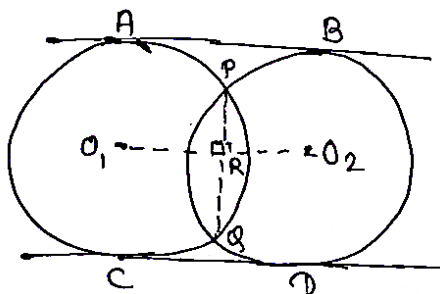


* maximum common tangent = **4**

$AB = CD =$ Direct Common Tangent

$PQ = RS =$ Transverse Common Tangent

Case-4



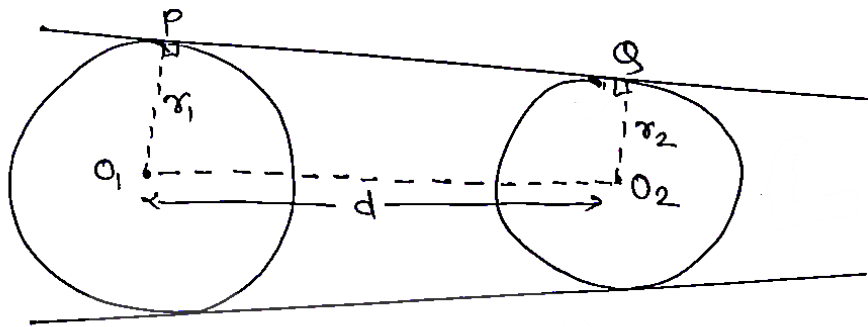
* maximum common tangent = **2**

$AB = CD =$ Direct Common tangent

* $PQ \Rightarrow$ Common chord

$$PR = QR = \frac{PQ}{2}$$

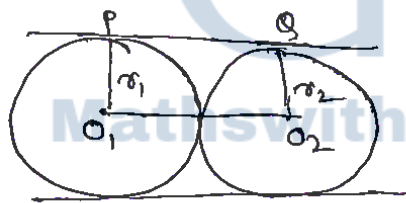
Direct Common Tangent



$$PQ = \sqrt{(\text{Distance between centres})^2 - (r_1 - r_2)^2}$$

Imp. * $PQ = \sqrt{d^2 - (r_1 - r_2)^2}$

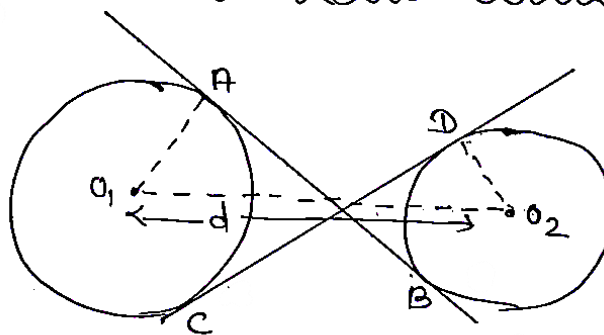
Special Case :- If $d = r_1 + r_2$



$$PQ = \sqrt{(r_1 + r_2)^2 - (r_1 - r_2)^2}$$

* $PQ = 2\sqrt{r_1 r_2}$

Transverse Common Tangent



Imp. * $AB = CD = \sqrt{d^2 - (r_1 + r_2)^2}$

Q40 The distance between the centres of two circles having radii 16cm and 18cm, is 26cm. The length of the direct common tangent of the two circles is:

Soln:-

$$\begin{aligned} DCT &= \sqrt{(d)^2 - (r_1 - r_2)^2} \\ &= \sqrt{(26)^2 - (18 - 16)^2} \\ &= \sqrt{676 - 4} \\ &= \sqrt{672} = \boxed{4\sqrt{42}} \text{ Ans} \end{aligned}$$

Q41 Observe the given figure. The distance between the two centres AB is:

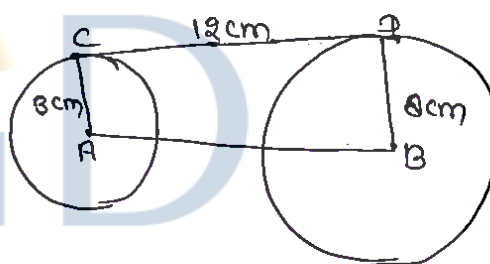
Soln:-

$$12 = \sqrt{(AB)^2 - (8-3)^2}$$

$$144 = AB^2 - 25$$

$$AB^2 = 169$$

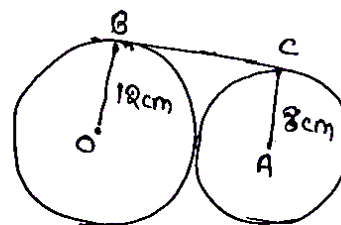
$$AB = \sqrt{169} = \boxed{13\text{cm}} \text{ Ans}$$



Q48 There are two circles that touch each other externally. Radius of the first circle with centre O is 12cm, Radius of the second circle with centre A is 8cm. Find the length of their common tangent BC.

Soln:-

$$\begin{aligned} \text{Common tangent} &= 2\sqrt{r_1 r_2} \\ &= 2\sqrt{8 \times 12} \\ &= \boxed{8\sqrt{6}\text{ cm}} \text{ Ans} \end{aligned}$$



Q43 The distance between the centres of two equal circles each of radius 4cm is 17cm. The length of a transverse tangent is:

Soln:-

$$\begin{aligned} \text{TCT} &= \sqrt{d^2 - (r_1 + r_2)^2} \\ &= \sqrt{(17)^2 - (4+4)^2} \\ &= \sqrt{289 - 64} = \sqrt{225} = \boxed{15\text{cm}} \text{ Ans} \end{aligned}$$

Q44 The radius of two circles is 3cm and 4cm. The distance between the centres of the circles is 10cm. What is the ratio of the length of direct common tangent to the length of the transverse common tangent?

Soln:-

$$\begin{aligned} \sqrt{(10)^2 - (4-3)^2} &: \sqrt{(10)^2 - (4+3)^2} \\ \sqrt{99} &: \sqrt{51} \\ \boxed{\sqrt{33} : \sqrt{17}} & \text{ Ans} \end{aligned}$$

Q45 There are two identical circles of radius 10cm. each. If the length of the direct common tangent is 26cm, then what is the length of the transverse common tangents?

Soln:-

$$\begin{aligned} \text{Direct Common tangent} &= \sqrt{d^2 - (r_1 - r_2)^2} \\ 26 &= \sqrt{d^2 - 0} \\ d &= 26 \\ \text{Transverse Common Tangent} &= \sqrt{(26)^2 - (10+10)^2} \\ &= \sqrt{676 - 400} \\ &= \boxed{2\sqrt{69}} \text{ Ans} \end{aligned}$$

246 Two common tangents AC and BD touch two equal circles each of radius 7cm, at points A, C, B and D, respectively, as shown in the figure. If the length of BD is 48cm, what is the length of AC?

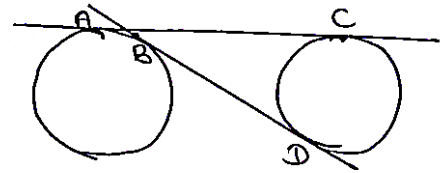
Soln:-

$$BD = \sqrt{d^2 - (r_1 + r_2)^2}$$

$$(48)^2 = d^2 - (7+7)^2$$

$$d^2 = 2304 + 196$$

$$d^2 = 2500$$

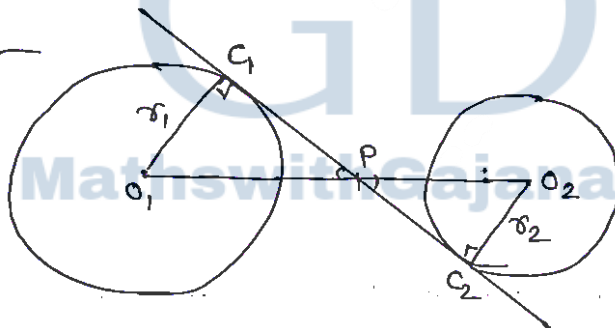


AC is direct Common Tangent

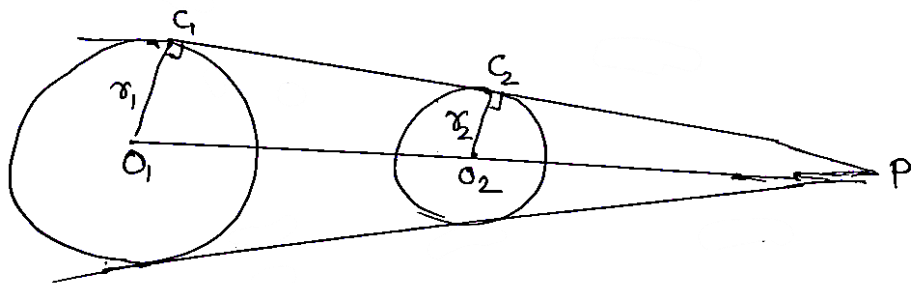
$$AC = \sqrt{d^2 - (r_1 - r_2)^2}$$

$$= \sqrt{2500 - (7-7)^2} = \sqrt{2500} = \boxed{50\text{cm}} \text{ Ans}$$

Concept :-



$$* \frac{O_1P}{O_2P} = \frac{O_1C_1}{O_2C_2} = \frac{r_1}{r_2}$$



$$* \frac{O_1P}{O_2P} = \frac{O_1C_1}{O_2C_2} = \frac{r_1}{r_2}$$

Q47 Two circles of radii 5cm and 3cm touch externally, then the ratio in which the direct common tangent to the circles divides externally the line joining the centres of the circles is:

Soln:- $r_1 : r_2 = \boxed{5 : 3}$ Ans

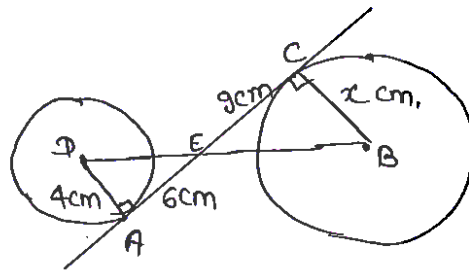
Q48 In the following figure, the circles with centres B and D have radii 4cm and xcm, respectively. AC is a tangent to both the circles. Find the value of x.

Soln:- by similarity

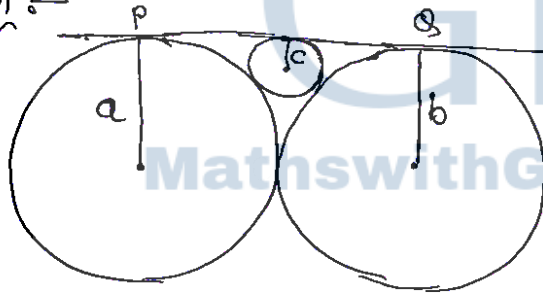
$$\frac{AE}{EC} = \frac{r_1}{r_2}$$

$$\frac{6}{9} = \frac{4}{x}$$

$$x = \boxed{6\text{cm}}$$
 Ans

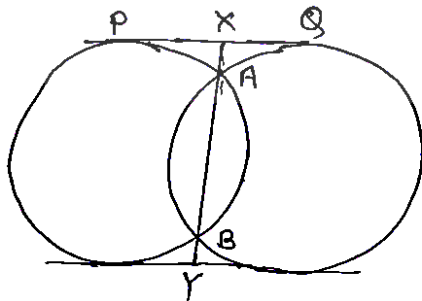


Concept:-



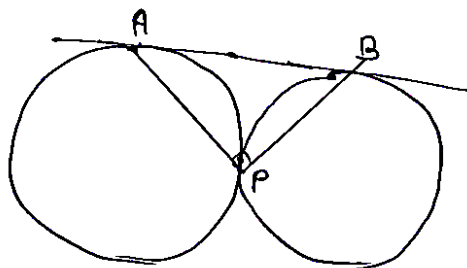
$$* \frac{1}{\sqrt{c}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}}$$

*



$$* PQ = \sqrt{(xy)^2 - (AB)^2}$$

*

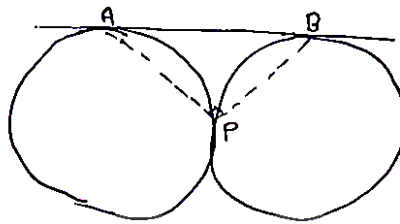


$\triangle APB$ is Right Angle $\triangle$

$$* \boxed{\angle P = 90^\circ}$$

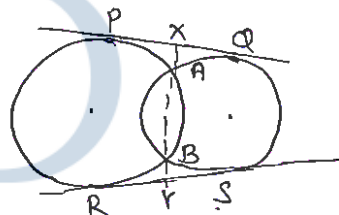
Q49 Two circles touch each other externally at P, AB is a direct common tangent to the two circles. If A and B are points of contact and $\angle PAB = 65^\circ$, then $\angle ABP$ is:

Soln:- $\angle ABP = 90^\circ - 65^\circ$
 $= 25^\circ$ Ans



Q50 PQ and RS are common tangents to two circles intersecting at A and B. AB when produced both sides, meet the tangents PQ and RS at X and Y, respectively. If $AB = 3\text{cm}$, $XY = 5\text{cm}$, then PQ will be?

Soln:- $PQ = \sqrt{(XY)^2 - (AB)^2}$
 $= \sqrt{(5)^2 - (3)^2}$
 $= \sqrt{16} = 4\text{cm}$ Ans



Q51 PP' and QQ' are two direct common tangents to two circles intersecting at A and B. The common chord on produced intersect PP' at R and QQ' at S. Which of the following is true?

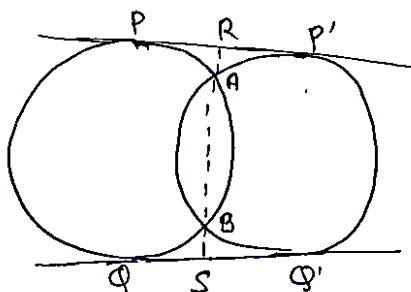
(a) $RA^2 + BS^2 = AB^2$

(b) $RS^2 = PP'^2 + AB^2$

(c) $RS^2 = PP'^2 + QQ'^2$

(d) $RS^2 = BS^2 + PP'^2$

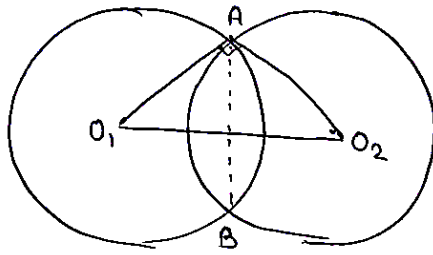
Soln:-



$RS^2 = PP'^2 + AB^2$ Ans

Common CHORD

Case-1 :-



$\Delta O_1 A O_2$ is Right Angle Δ

$$\angle A = 90^\circ$$

$$r_1 = 3\text{cm}$$

$$r_2 = 4\text{cm}$$

$$O_1 O_2 = 5\text{cm}$$

} Triplet

$$* \text{ Common chord} = \frac{r_1 \times r_2}{O_1 O_2} \times 2$$

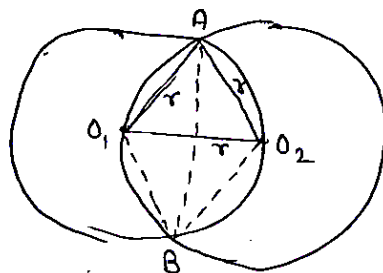
$$AB = 2 \times \frac{3 \times 4}{5} = \boxed{4.8\text{cm}}$$

Q52 Two circles are having radii 9cm and 12cm. the distance between their centres is 15cm. What is the length of their common chord?

Soln

$$\text{Common chord} = 2 \times \frac{9 \times 12}{15} = \boxed{14.4\text{cm}} \text{ Ans}$$

Case-2 :- Two circle having radii r unit intersect each other in such a way that each of them passes through the centres of the other then the length of their common chord is:



$O_1 A O_2$ is Equilateral Δ

$$\text{Common chord} = 2 \times \text{height of } \Delta O_1 A O_2$$

$$= 2 \times \frac{\sqrt{3}}{2} r$$

$$= \boxed{\sqrt{3} r}$$

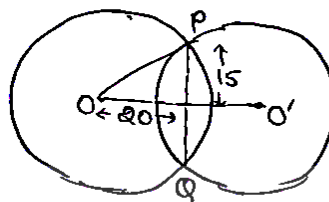
$$* \text{ Common chord} = \boxed{\sqrt{3} r}$$

Q53 Two circles of same radius intersect each other at P and Q. If the length of the common chord is 30 cm and distance between the centres of the two circles is 40 cm, then what is the radius of the circles?

Soln:- $OO' = 40\text{cm}$
 $PQ = 30\text{cm}$

$$\text{radius} = OP \Rightarrow \sqrt{20^2 + 15^2}$$

$$OP \Rightarrow \sqrt{625} = \boxed{25\text{cm}}$$



Case-3:- $\triangle AOO_2$ is Scalene Triangle

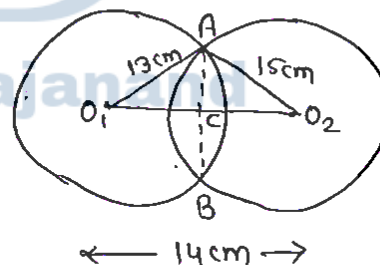
Q54 The length of the common chord of two circles of radii 15cm and 13cm, whose centres are 14cm apart is:

Soln:- $S = \frac{(13+15+14)}{2} = 21\text{cm}$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\Delta = \sqrt{21 \times 8 \times 6 \times 7}$$

$$\Delta = 84\text{cm}^2$$



In $\triangle AOO_2$

$$\Delta = \frac{1}{2} \times 14 \times AC$$

$$84 = \frac{1}{2} \times 14 \times AC$$

$$AC = 12\text{cm}$$

$$\text{Common chord} = AB = 2 \times 12\text{cm}$$

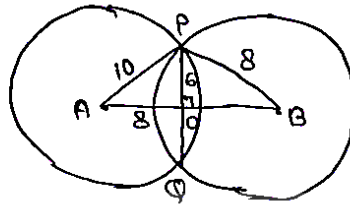
$$= \boxed{24\text{cm}} \text{ Ans}$$

Q55 Two circles of radii 10cm and 8cm intersect at the points P and Q. If $PQ = 12$, and the distance between the centres of the circles is x cm. The value of x (Correct to one decimal place) is:

Soln:- $OB = \sqrt{(8)^2 - (6)^2} = \sqrt{28}$
 $= 5.3$

Triplet (6, 8, 10) $\rightarrow AO = 8$

$AB = 8 + 5.3 = \boxed{13.3\text{cm}}$ Ans

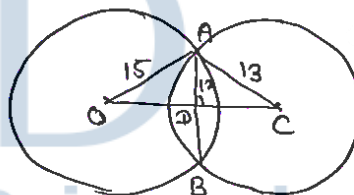


Q56 Two circles of radius 13cm and 15cm intersect each other at points A and B. If the length of the common chord is 24cm, then what is the distance between their centres?

Soln:- $OD = 9\text{cm}$ (Triplet $\rightarrow 9, 12, 15$)

$OC = 5\text{cm}$ (Triplet $\rightarrow 5, 12, 13$)

$AC = 9 + 5 = \boxed{14\text{cm}}$ Ans



Q57 Two circles of same radius 6cm, intersect each other at P and Q. If $PQ = 10\text{cm}$, then what is the distance between the centres of the two circles?

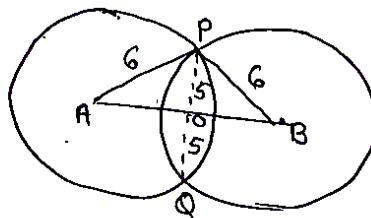
Soln:-

$AO = \sqrt{(6)^2 - (5)^2}$

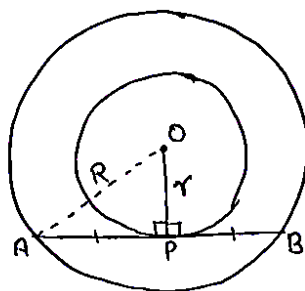
$AO = \sqrt{11}$

$AB = 2 \times AO$

$= \boxed{2\sqrt{11}}$ Ans



Concentric Circle



* $OP \perp AB$

* $AP = PB$

* $OP = r$ (Small) Circle radius)

* $OA = R$ (Big circle radius)

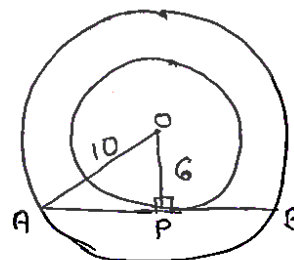
Q58 C_1 and C_2 are two concentric circles such that the radius of C_2 is less than the radius of C_1 . If AB is the Chord of C_1 of length 12cm, touching C_2 at P, find the length of AP.

Soln:- $AP = \frac{AB}{2} = \frac{12}{2} = \boxed{6\text{cm}}$ Ans

Q59 Two concentric circles are of radii 10cm and 6cm. Find the length of the chord of the larger circle which touches the smaller circles.

Soln:- $AP = 8\text{cm}$ (Triplet $\rightarrow 6, 8, 10$)

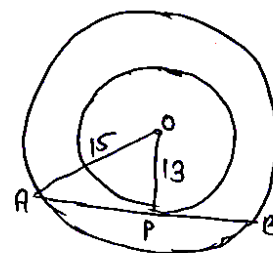
$AB = 2 \times AP$
 $= 2 \times 8 = \boxed{16\text{cm}}$ Ans



Q60 Two concentric circles of radii 15cm and 13cm are given. Find the length of the chord of the larger circle which touches the smaller circles

Soln:- $AP = \sqrt{(15)^2 - (13)^2} = 2\sqrt{14}$

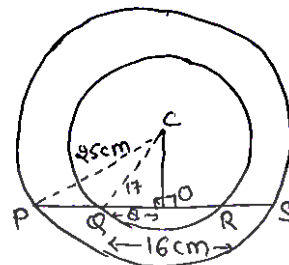
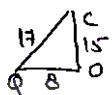
$AB = 2 \times 2\sqrt{14} = \boxed{4\sqrt{14}}$ Ans



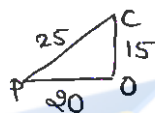
Q61 The radii of two concentric circles are 17cm and 25cm. a straight line PQRS intersects the larger circle at the points P and S and intersects the smaller circle at the points Q and R. If QR = 16cm then the length of PS is:

Soln:-

$$OC = 15\text{cm} \text{ (Triplet } \rightarrow 8, 15, 17)$$



$$PO = 20\text{cm} \text{ (Triplet } \rightarrow 15, 20, 25)$$



$$PS = 2 \times PO$$

$$= 2 \times 20 = \boxed{40\text{cm}} \text{ Ans}$$

Q62 A line cuts two concentric circles. The lengths of chords formed by that line on the two circles are 4cm and 16cm. What is the difference in squares of radii of two circles?

Soln:- Given:- $BC = 4\text{cm}$
 $AD = 16\text{cm}$

In $\triangle OMA$

$$OM^2 = OA^2 - AM^2 \text{ --- (1)}$$

In $\triangle OMB$

$$OM^2 = OB^2 - BM^2 \text{ --- (2)}$$

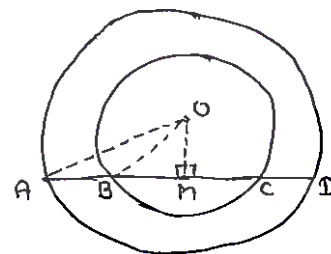
from (1) and (2)

$$OA^2 - AM^2 = OB^2 - BM^2$$

$$OA^2 - OB^2 = AM^2 - BM^2$$

$$= (8)^2 - (2)^2$$

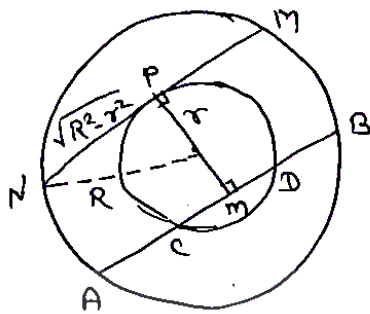
$$= 64 - 4 = \boxed{60\text{cm}} \text{ Ans}$$



$$\left[\begin{aligned} AM &= \frac{AD}{2} = \frac{16}{2} = 8\text{cm} \\ BM &= \frac{BC}{2} = \frac{4}{2} = 2\text{cm} \end{aligned} \right]$$

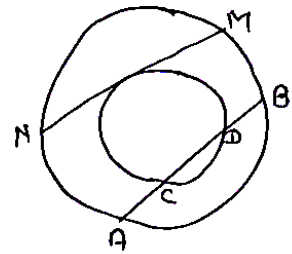
Q63 In the given figure, $AB = 30\text{cm}$ and $CD = 24\text{cm}$. What is the value of $MN = ?$

Soln:-



$$AM = 15\text{cm}$$

$$CM = 12\text{cm}$$



Let bigger circle radius = R

Smaller circle radius = r

A.T.Q $R^2 - 15^2 = r^2 - 12^2$

$$R^2 - r^2 = 15^2 - 12^2$$

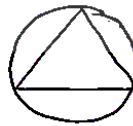
$$R^2 - r^2 = 81$$

$$\sqrt{R^2 - r^2} = 9 \Rightarrow NP$$

$$MN = 2 \times NP = 2 \times 9 = \boxed{18\text{cm}} \text{ Ans}$$

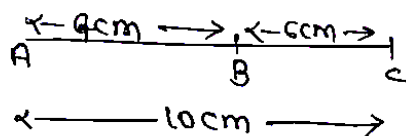
Q64 Number of circles that can be drawn through three non-collinear points are:

Soln:- One



Q65 A, B, C are three points so that $AB = 4\text{cm}$, $BC = 6\text{cm}$ and $AC = 10\text{cm}$. The number of circles passing through the points A, B, C is:.

Soln:- Zero



Type

[SSC CHSL-20]

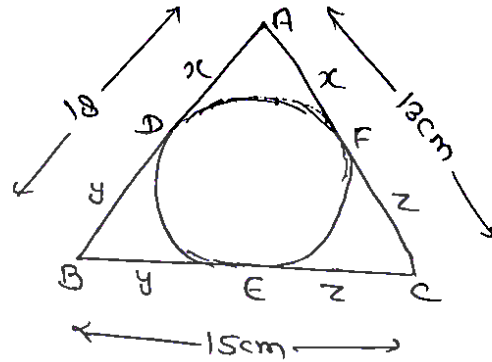
- Q66** A circle inscribed in a triangle ABC touches its sides AB, BC and AC at the points D, E and F respectively. If $AB = 18\text{cm}$, $BC = 15\text{cm}$ and $AC = 13\text{cm}$, then the value of $AD + BE + CF$ is:

Soln:-

$$2(x + y + z) = 18 + 13 + 15$$

$$x + y + z = 23$$

$$AD + BE + CF = \boxed{23} \text{ Ans}$$



- Q67** Triangle ABC is circumscribed around circle D. Segments AQ, BR and SC measure 13, 10.5 and 6cm, respectively. The perimeter of triangle ABC is:

[SSC CGL PRE-2023]

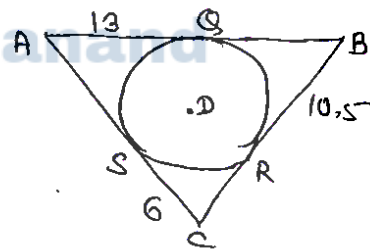
Soln:-

$$AQ + BR + SC = 13 + 10.5 + 6$$

$$= 29.5\text{cm}$$

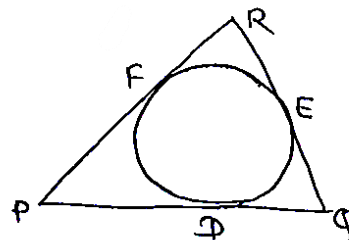
$$\text{Perimeter} = 2 \times 29.5$$

$$= \boxed{59\text{cm}} \text{ Ans}$$



- Q68** In the given figure, a circle is inscribed in $\triangle PQR$, such that it touches the sides PQ, QR and RP at points D, E, F, respectively. If the lengths of the sides $PQ = 18\text{cm}$, $QR = 13\text{cm}$ and $RP = 15\text{cm}$, then find the length of PD.

[SSC CGL-23]



Soln:-

$$2(x+y+z) = 18+13+15$$

$$x+y+z = 23\text{cm}$$

$$PQ = y+z = 18\text{cm}$$

$$QR = x+z = 13\text{cm}$$

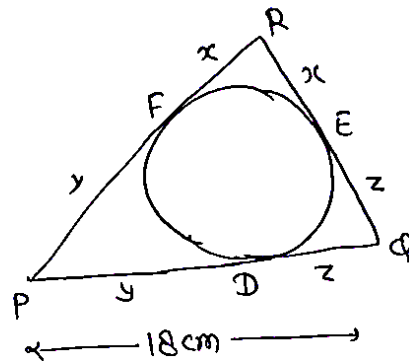
$$PR = x+y = 15\text{cm}$$

$$x = 23-18 = 5\text{cm}$$

$$y = 23-13 = 10\text{cm}$$

$$z = 23-15 = 8\text{cm}$$

$$PD = y = \boxed{10\text{cm}} \text{ Ans}$$



[269] A circle is inscribed in the triangle ABC whose sides are given as $AB = 10$, $BC = 8$, $CA = 12$ units as shown in the figure. The value of $AD \times BF$ is:

Soln:- Let $EB = x$

$$AC = 10-x+8-x$$

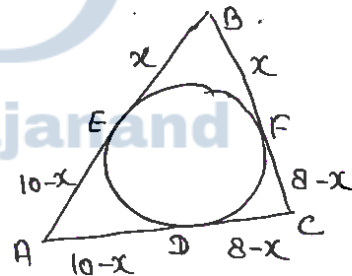
$$12 = 18-x$$

$$\boxed{x = 3}$$

$$AD = 10-x = 7\text{cm}$$

$$BF = 3$$

$$AC \times BF = 7 \times 3 = \boxed{21} \text{ Ans}$$

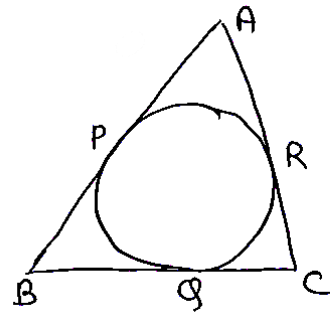


[270] A Circle is inscribed in $\triangle ABC$, touching AB , BC and AC at the points P , Q and R , respectively. If $AB-BC=4\text{cm}$, $AB-AC=2\text{cm}$, and the perimeter of $\triangle ABC = 32\text{cm}$, then AC (in cm) = ?

[SSC CGL Mains - 2019]

[SSC CGL PRE - 2022]

Soln:- $AB - BC = 4\text{cm}$
 $\Rightarrow BC = AB - 4\text{cm}$
 $AB - AC = 2\text{cm}$
 $\Rightarrow AC = AB - 2\text{cm}$



Perimeter of $\triangle ABC = 32\text{cm}$
 $\Rightarrow AB + BC + AC = 32$
 $\Rightarrow AB + AB - 4 + AB - 2 = 32$
 $\Rightarrow 3AB = 32 + 6$
 $AB = \frac{38}{3}$

$AC = AB - 2 \Rightarrow \frac{38}{3} - 2 \Rightarrow \boxed{\frac{32}{3}\text{cm}}$ Ans

Q71 ABC is an isosceles triangle where $AB = AC$ which is circumscribed about a circle. If P is the point where the circle touches the side BC, then which of the following is true?

Soln (a) $BP = PC$ (b) $BP > PC$ (c) $BP < PC$ (d) $BP = \frac{1}{2} PC$

$AB = AC$
 $AQ + BQ = AR + RC$
 we know that

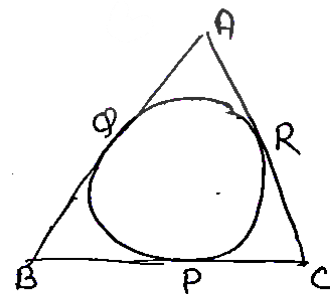
$BQ = PB$ & $PC = RC$

$AQ + PB = AR + PC$

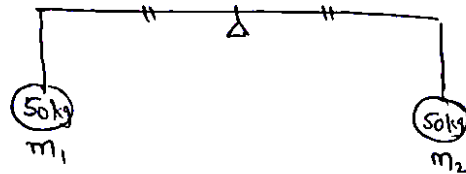
also $AQ = AR$

$AR + PB = AR + PC$

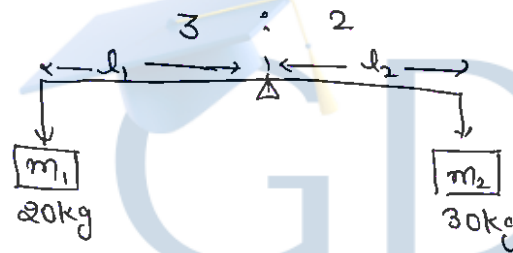
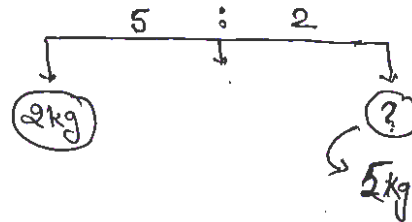
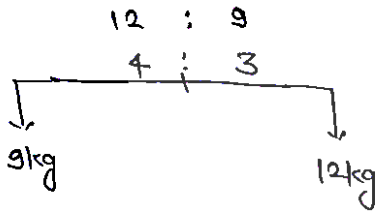
$\boxed{PB = PC}$ Ans



Mass Point Geometry (MPG)



$$m_1 = m_2$$

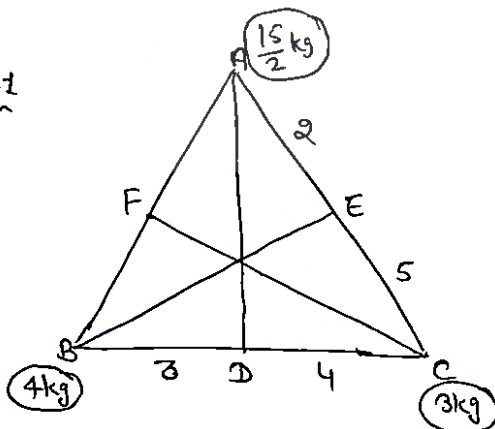


$$* m_1 l_1 = m_2 l_2$$

$$20 \times 3 = 30 \times 2$$

$$* \frac{l_1}{l_2} = \frac{m_2}{m_1}$$

Case-1



Given that

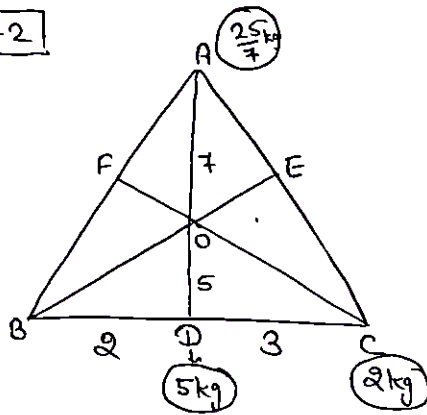
$$BD : DC = 3 : 4$$

$$CE : EA = 5 : 2$$

$$\text{then } AF : FB = ??$$

$$AF : FB = 4 : \frac{15}{2} = \boxed{8 : 15} \text{ Ans}$$

Case-2



Given that

$$BD : DC = 2 : 3$$

$$AO : OD = 7 : 5$$

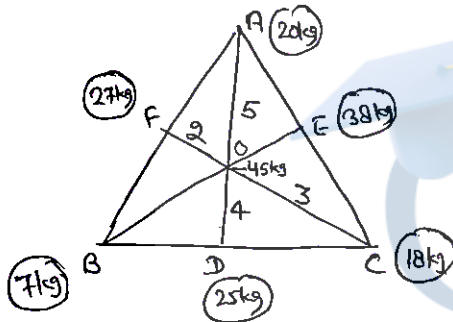
$$AE : EC = ??$$

$$AE : EC = 2 : \frac{25}{7} = \boxed{14 : 25} \text{ Ans}$$

Case-3

In the given triangle, $AO : OD = 5 : 4$, $CO : OF = 3 : 2$

find $BO : OE = ?$

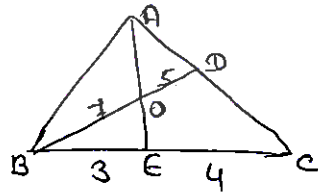


$$BO : OE = \boxed{38 : 7} \text{ Ans}$$

MathswithGajanand

Q12

In the given triangle ABC, if $AE = 96\text{cm}$, $BO : OD = 7 : 5$ & $BE : EC = 3 : 4$ then find $OE = ?$

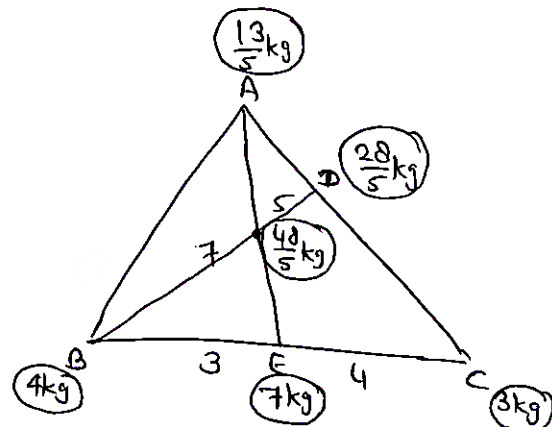


Soln :- $AO : OE = 7 : \frac{13}{5}$

$$AO : OE = 35 : 13$$

$$OE = \frac{96}{(35+13)} \times 13$$

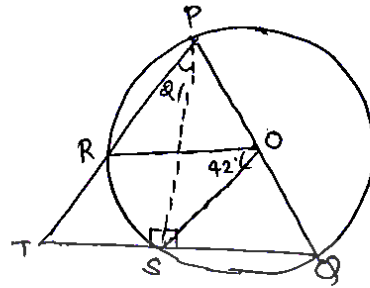
$$= \boxed{26\text{cm}} \text{ Ans}$$



Q73 In a circle with centre O, chords PR and QS meet at the point T, when produced and PQ is a diameter. If $\angle ROS = 42^\circ$ then the measure of $\angle PTQ$?

Soln :- $\angle SPT = \frac{1}{2} \angle ROS$
 $= \frac{1}{2} \times 42^\circ = 21^\circ$

In $\triangle PST$ $\angle PTQ = 90^\circ - 21^\circ$
 $= \boxed{69^\circ} \text{ Ans}$



Q74 In the following figure, P and Q are centres of two circles, The circles are intersecting at point A and B. PA produced on both the sides meets the circles at C and D. If $\angle CPB = 100^\circ$, then find the value of x.

Soln :- $PA = PB = \text{Radius}$

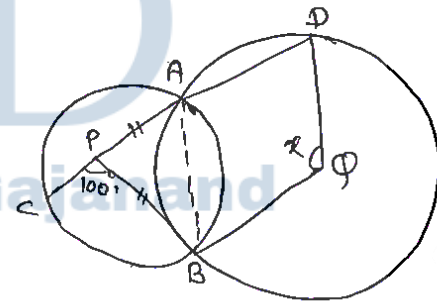
$\angle PAB = \angle PBA$

$\angle PAB = \frac{100}{2} = 50^\circ$

$\angle BAD = 180^\circ - 50^\circ = 130^\circ$

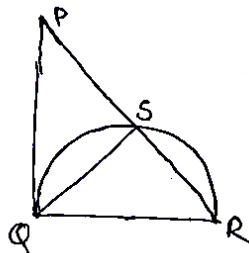
Reflex $\angle BQD = 2 \times 130^\circ = 260^\circ$

$x = 360^\circ - 260^\circ = \boxed{100^\circ} \text{ Ans}$



Q75 In the given figure, triangle PQR is a right angled triangle at Q. If PQ = 35 cm and QS = 28 cm, then what is the value of SR?

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Soln:-

$$PS = 21 \text{ cm (Triplet :- } 21, 28, 35)$$

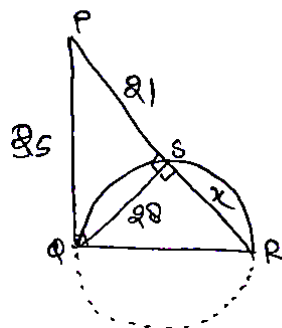
Tangent He theorem:

$$PQ^2 = PS \times PR$$

$$(35)^2 = 21 \times (21 + x)$$

$$21 + x = 58.33$$

$$x = \boxed{37.33 \text{ cm}} \text{ Ans}$$



Q76 In the given, figure, CD and AB are diameters of circle and AB and CD are perpendicular to each other, LQ and SR are Perpendiculars to AB and CD respectively. Radius of circle is 5cm, PB : PA = 2 : 3. What is the length of SM ?

Soln:- PB : PA = 2 : 3 diameter = 10cm
(AB)

$$PA = \frac{10}{2+3} \times 3 = 6 \text{ cm}$$

$$PB = 4 \text{ cm}$$

$$OP = PA - OA = 6 - 5 = 1 \text{ cm}$$

Similarly $\rightarrow ON = 1 \text{ cm}$

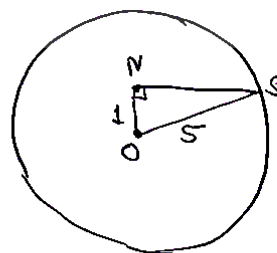
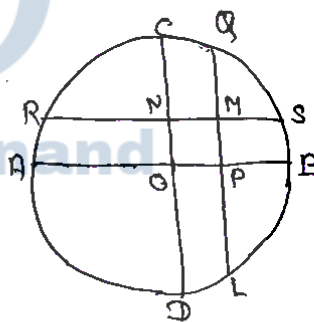
$$NS = \sqrt{(5)^2 - (1)^2} = \sqrt{24} \text{ cm}$$

$$SM = NS - MN$$

$$\therefore MN = OP = 1 \text{ cm}$$

$$SM = \sqrt{24} - 1$$

$$SM = \boxed{(2\sqrt{6} - 1) \text{ cm}} \text{ Ans}$$



Q77 In the given figure, PQRS is a square inscribed in a circle of radius 4cm. PQ is produced till point Y. From Y a tangent is drawn to the circle at point R. What is the length of SY?

Soln:- $\angle PRQ = 45^\circ$

$\angle ORY = 90^\circ$

$\angle QRY = 45^\circ$

$RQ = QY$

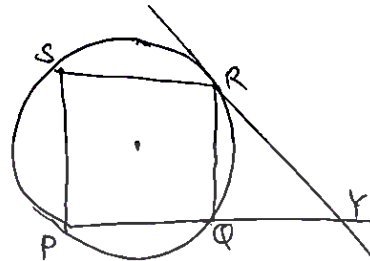
Diagonal = 8 = Diameter

Side = $\frac{8}{\sqrt{2}} = 4\sqrt{2}$

In $\triangle SPY$

$$\begin{aligned} SY^2 &= SP^2 + PY^2 \\ &= (8\sqrt{2})^2 + (4\sqrt{2})^2 \\ &= 128 + 32 = 160 \end{aligned}$$

$SY = \boxed{4\sqrt{10}}$ Ans



Q78 If the given figure, E and F are the centers of two identical circles. What is the ratio of area of triangle AOB to the area of triangle DOC?

Soln:- $\triangle DEM \sim \triangle DBN$

$\therefore \frac{EM}{BN} = \frac{DM}{DN}$

$\frac{r}{2r} = \frac{DM}{DN}$

$\Rightarrow DM : MN = 1 : 1$

Similarly $MN : NC = 1 : 1$

$MN = AB = 1$

$\frac{\text{ar. } \triangle AOB}{\text{ar. } \triangle DOC} = \left(\frac{1}{3}\right)^2 = \frac{1}{9} \Rightarrow \boxed{1 : 9}$ Ans

